Waiting for Closure: Deep Networks Still Do Not Complete Contours

Benjamin ten Wolde *bendotten@gmail.com*
University of California San Diego

Abstract

Whether deep networks exhibit the Gestalt law of closure has been answered both ways, using stimuli that are rarely sized to the network's sampling grid and measures that are rarely checked against observers that cannot close contours. We built a patch-aware test battery in which every gap, inducer and dash is specified in units of the model's patch size and rendered natively, with matched controls that rotate, scramble or re-orient the same pieces. We trained and tested 24 classifiers, frozen and fine-tuned CNNs and vision transformers, looped (weight-tied, recurrent-depth) transformers, and fGRU-H, γ-net and deep fGRU-H/fGRU circuits built from the published fGRU cell, including stacks of up to eight recurrent layers behind a CNN stem; probed contour-tracing networks and masked autoencoders; reimplemented cued segmentation circuits; and trained 34 further cued segmentation networks that loop the last blocks of frozen pretrained vision transformers with learned feedback. Illusory contours are at best weakly represented: ViT-B/16 variants show a small but consistent advantage of Kanizsa figures over rotated-inducer controls (+0.08 to +0.14 across three independently trained copies, +0.10 frozen), with Kanizsa accuracy of at most 0.39 (chance 0.10) against 1.00 on complete outlines; DINOv2-S shows an inconsistent advantage (+0.01 to +0.15 across seeds), and in the other networks it stays at or below +0.07, far from the fragment effects. Fine-tuned transformers do integrate fragmented outlines, and the arrangement carries the effect: scrambling piece positions while keeping orientations lowers accuracy by +0.46 to +0.65, and pieces placed correctly but misoriented are not recognized. Neither looping nor recurrence consistently adds closure: in five of six recurrent circuits a control trained with a single iteration matches or exceeds the eight-iteration network, although the recurrent states in the gap become more similar to those for the complete outline over iterations. Finally, we audited published closure measures with null observers. A blur reproduces the illusory-brightness signature reported for a predictive-coding network; a pixel-growth observer, which only sees how far contour pixels lie from their centroid, scores +0.289 on the closure measure of Kim et al. (2021), several times any CNN, and rotating each corner in place about the centroid of its own pixels, which removes the radial difference this observer detects, reduces the measure to +0.005 and +0.005 in the two networks previously reported to show closure most consistently. The configural-effect (CE) measure survives null observers and is specific to closing arrangements, but it generally grows as the outline approaches completion, its advantage over a non-closing arrangement gains nothing further, on average, from a fully closed square, and changes with inducer shape and size at a fixed ratio of luminance-specified to total edge length, which predicts the clarity of human illusory boundaries across display sizes (Shipley & Kellman, 1992). For CNN and recurrent networks the mid-gap closure index halves at gaps of 6–50 px, inside the network's measured effective receptive field in 9 of 10. Networks trained specifically to trace contours, including weight-tied looped transformers, link aligned pieces only across gaps a few pixels wider than in training and label almost none of the missing contour as contour. In the transformers whose classifications favour Kanizsa figures over rotated inducers, activation patching locates that evidence in the inducers' tokens, not in the tokens of the gaps. Recurrent circuits trained to spread a grouping label from a cue across objects in natural images, reimplemented after Mollard et al. (2026), label the surface of a Kanizsa figure only when the gaps between inducers are a few pixels wide;

single-step controls do as much, and the limit does not move with the circuits' kernel size. Putting the recurrence inside pretrained vision transformers instead, by looping the last blocks of frozen V-JEPA 2 and 2.1, DINOv3 and SAM 3 encoders, and of a LeVJEPA encoder trained through, with learned feedback and training them for cued segmentation, gives a selective Kanizsa label in two encoders, the SAM 3 trunk trained with object masks and DINOv3 (one of whose two seeds also labels the inside of scrambled outlines, and so fails our selection rule), once the training clips contain outlines and hidden-layer labels; in both the single pass through the looped blocks already produces the label, on controls as well, and the recurrence removes it from the configurations that form no figure, so cutting the feedback raises the Kanizsa fill in SAM 3, and a network trained without the loop passes the same battery. Deep networks group nearby, consistently oriented contour pieces; they do not close gaps, and illusory contours are at most weakly represented. Across these architectures and training regimes the search for closure is, we argue, exhausted: what moved the result was the training signal, and models supervised for amodal structure, tested against the graded and late profile of human and cortical illusory contours on the same displays, are where to look next.

1 Introduction

Closure, the tendency to perceive an incomplete figure as a whole, is among the oldest Gestalt laws (Wagemans et al., 2012; Kanizsa, 1979) and a standard test of whether artificial vision organizes the world as people do (Kim et al., 2021; Zhang et al., 2025). Studies of deep networks disagree. Similarity-based analyses report closure in ImageNet-trained CNNs (Kim et al., 2021; Zhang et al., 2025); accuracy-based protocols report its absence (Zhang et al., 2024); classification images suggest networks rely on inducer orientation rather than illusory edges (Baker et al., 2018); and recurrent predictive-coding networks have been reported to perceive Kanizsa squares (Pang et al., 2021). For vision-language models, whose image encoders are vision transformers, answers to classic illusions persist when the illusion-inducing elements are inverted, which Sun et al. (2026) trace to different causes in different models, from recall of the canonical answer to limits of visual processing.

Two problems run through this literature. First, stimuli are drawn in pixels at arbitrary scales and then resized, so the size of a gap relative to a vision transformer's patches or a CNN's receptive field is uncontrolled. A six-pixel gap need not leave any 16-pixel patch empty, although it remains detectable within its patch; a forty-pixel gap may exceed every receptive field below the last stage. Second, measures are checked against some null controls, such as untrained networks (Kim et al., 2021), but not against simple observers that *cannot* close contours yet share the stimuli's low-level statistics, such as the radial distribution of contour pixels. If a blur, a filter bank or an untrained network produces the same signature, the signature is not evidence of closure.

We address both problems. Our contributions are: (i) a patch-aware battery with geometry in patch units and matched controls (rotated, position-scrambled and orientation-swapped pieces, Kanizsa inducers, and dashed, noisy linking segments); (ii) a comparison of 24 networks spanning feed-forward, looped-transformer and horizontal/top-down recurrent architectures built from the published fGRU cell, each recurrent circuit but one with a matched single-iteration control; (iii) an analysis of the gap itself, relating each network's critical gap to its patch size or effective receptive field and measuring how far the empty part of the gap comes to resemble the drawn contour, and whether attention or recurrence increases this (Section 7.1); (iv) a null-observer audit of four published closure measures; (v) a faithfulness check of the saliency overlays commonly used to argue that networks "see" a shape; and (vi) a family of cued segmentation networks that loop the last blocks of frozen self-supervised and segmentation-trained vision transformers with learned feedback, read step by step, which locates the selective Kanizsa labels we found in two encoders under one training recipe and the recurrence's role in a veto on non-figures (Section 6.6). The results are organized first by network family (Section 6), then by probe across families (Section 7); the step-by-step and layer-by-layer maps of every family are animated in the supplementary material.

2 Related work

Closure and illusory contours in biological vision. Closure and good continuation are among the Gestalt grouping principles (Wagemans et al., 2012), and Kanizsa figures, in which aligned notches in separate inducers produce the percept of an occluding surface with contours where there is no luminance edge, are their best-known demonstration (Kanizsa, 1979). Neurons in macaque V2 respond to illusory contours (von der Heydt et al., 1984), and in V1 such responses appear later than responses to real contours, consistent with feedback from higher areas (Lee & Nguyen, 2001). Psychophysics of contour integration describes an association field that links elements by their relative position and orientation (Field et al., 1993), and boundary-completion models complete a contour only when aligned edge signals are present on both sides of the gap (Grossberg & Mingolla, 1985). These accounts differ in what they predict for a gap: linking of present elements in the first case, active completion of the absent contour in the second.

Closure in deep networks. Evidence that deep networks show closure comes mainly from representational similarity: fragmented shapes evoke representations closer to complete shapes than disordered controls do in ImageNet-trained CNNs (Kim et al., 2021; Zhang et al., 2025), an effect also reported with configural measures that compare discrimination with and without context (Biscione & Bowers, 2023; Zhang et al., 2025). Self-supervised vision transformers have been reported to show emergent Gestalt organization (Li et al., 2025), and masked autoencoders to complete the edges of Kanizsa figures in their reconstructions (Li et al., 2026). Behavioural tests with valid and invalid Kanizsa triangles (Ehrensperger et al., 2019) and with fragmented polygons (Zhang et al., 2024) have reached mixed conclusions, the latter reporting an absence of closure; classification images suggest that networks use inducer orientation rather than illusory edges (Baker et al., 2018), and Toosi & Miller (2024) propose that illusory contours emerge in deep networks from the accumulation of implicit priors. Tang & Ley (2023) degraded regular polygons at their corners or at the middle of each edge, following the vertex and midsegment deletions of Biederman (1987), and found that ResNet classifiers trained on this task suffer more from edge than from corner degradation, the opposite of human observers, while a ViT is affected equally by both. A predictive-coding network with feedback was reported to produce Kanizsa-like brightness in its reconstructions (Pang et al., 2021). Most of these studies resize stimuli drawn at arbitrary scales, and their null controls (untrained, noise-trained or shuffled networks in Kim et al., 2021) do not include observers that share the stimuli's low-level statistics but cannot close contours; these are the two issues we address.

Recurrence and iterative computation. The horizontal gated recurrent unit (hGRU) was introduced to solve long-range contour tasks such as Pathfinder with far fewer parameters than feed-forward networks (Linsley et al., 2018); the feedback gated recurrent unit (fGRU) and the γ-net built from it add top-down connections and learn contour detection with better sample efficiency (Linsley et al., 2020b), and recurrent grouping mechanisms have been compared with feed-forward ones on segmentation tasks that require them (Kim et al., 2020). Weight-tied transformers apply the same block repeatedly (Dehghani et al., 2019) and can be run for more iterations at test time than in training (Geiping et al., 2025); recirculation turns a frozen feed-forward transformer into a dynamical system by feeding a deep layer's state back into a shallow one (Mozer et al., 2026). If closure requires iterative completion, these are the architectures in which it should appear, which is why we compare the recurrent circuits with single-iteration controls (trained with $T{=}1$) and run the looped transformers for different numbers of loops.

Pretrained encoders as fronts. The encoders we loop in Section 5.4 were chosen to differ in what their pretraining rewards. V-JEPA 2 predicts the features of masked video (Assran et al., 2025), and V-JEPA 2.1 adds dense image features to it (Mur-Labadia et al., 2026); LeVJEPA is a video encoder trained without the usual heuristics (Kuhn et al., 2026); DINOv3 is a self-distilled image encoder (Siméoni et al., 2026). SAM 3's vision trunk, from the Perception Encoder family (Bolya et al., 2025), is trained on image–text pairs and then inside a model that segments and tracks every instance of a concept from a prompt (Carion et al., 2026), continuing the Segment Anything line (Kirillov et al., 2023; Ravi et al., 2025); it is the only encoder among them trained with object masks, which matters for what it carries into a line drawing.

Receptive fields and attribution. The effective receptive field of a CNN unit is much smaller than its theoretical one (Luo et al., 2016), which matters for whether a gap can be bridged at all. Some gradient-based saliency maps are insensitive to the model's parameters (Adebayo et al., 2018), and deletion tests check

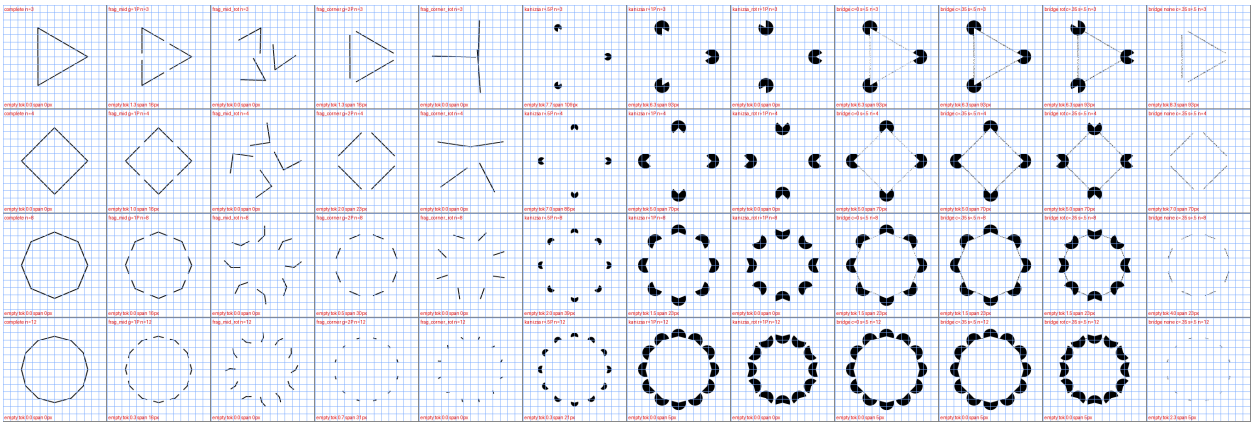


Figure 1: Patch-aware stimuli at 224 px, shown with the 16 px patch grid. Columns: complete outline; gaps in the middle of every edge and their rotated control; corners removed and their rotated control; Kanizsa figures with inducer radius 0.5 and 1 patch and the rotated-inducer control; dashed linking segments in the Kanizsa gap with inward, rotated and no inducers. Rows: 3, 4, 8 and 12 sides. Red text gives the number of empty patches each gap crosses.

whether a map ranks the pixels the model relies on (Petsiuk et al., 2018); we apply such a test to the Grad-CAM and transformer relevance maps (Selvaraju et al., 2017; Chefer et al., 2021a) that are commonly used to argue that a network "sees" a shape.

3 Stimuli

Networks classify regular polygons with 3 to 12 sides, the ten-way task of Zhang et al. (2024). All lengths are set in units of the patch size P (16 px for ViT-B/16 and the CNNs, 14 px for DINOv2) and rendered natively at 224 px with 4× supersampling; images are never resized. The circumradius is $4.5P$ for every class, so overall size carries no class information. Training uses complete outlines with random rotation, scale (±15%), position ($\pm P$) and background polarity. The test battery (Fig. 1; about 38,000 images per patch size) contains: complete outlines; *mid-gap fragments*, with a gap of $g \in \{0.25, \ldots, 4\}P$ removed from the middle of each edge (the manipulation of Zhang et al. (2024), but in absolute patch units; midsegment deletion in Biederman, 1987; edge degradation in Tang & Ley, 2023); *corner-removed fragments*, in which contour within $g/2$ of every vertex is removed so that corner angles cannot be counted (vertex deletion in Biederman, 1987; corner degradation in Tang & Ley, 2023); *Kanizsa figures* with inducer radius $r \in \{0.5, \ldots, 2\}P$; and *dashed links*, in which the Kanizsa gap (inducer radius P) contains a real dashed segment (dash length $0.25P$, spacing 0–$2P$, contrast 0–1, pixel noise of s.d. 0, 0.25 or 0.5 only on the segment). Every fragment condition has a matched control, a disordered control in the sense of Kim et al. (2021), in which each piece is rotated about a point inside it (a straight piece about its midpoint; a corner piece of a mid-gap fragment about the centroid of the triangle formed by its vertex and its two ends) by a fixed pseudo-random angle (the rotated control). *Contour pixels* are the pixels of the physically specified contour (Shipley & Kellman, 1992); a patch, token or feature-map cell is *empty* if it contains none. Each image is shifted diagonally (equally in x and y) by 0, $\frac{1}{4}$, $\frac{1}{2}$ or $\frac{3}{4}$ of a patch relative to the patch grid, and its geometry, gap masks and the number of empty patches each gap crosses are recorded.

Two further test-only controls separate closure from orientation statistics (Fig. 5B,C): *scrambled* stimuli move each piece to a position inside the shape's disk, chosen among 200 random candidates to keep the pieces apart, while keeping its orientation and length; *cue conflict* stimuli, a position–orientation analogue of the shape–texture cue conflict of Geirhos et al. (2019), keep each corner-removed piece's midpoint but give it the edge orientation of the polygon with one more side, so positions imply n sides and the orientations are n of the $n{+}1$ edge orientations of the polygon with one more side (for 12-sided polygons, the 11-sided one).

4 Measures and protocols

Measures. The *closure index* is accuracy on a stimulus minus accuracy on its rotated control. We also report the closure measure $\bar{C}$ of Kim et al. (2021) on our stimuli (cosine similarity to the matched complete image, stimulus minus control), a linear probe (a linear classifier probe; Alain & Bengio, 2017) trained at every layer on z-scored features with the frozen-backbone protocol of Section 5.1, the effective receptive field (Luo et al., 2016), and, for recurrent models, the *gap similarity*, $\bar{C}$ restricted to the gap region of the recurrent state (for Fig. 12B: one-patch corner-removed fragments, all cells the missing contour crosses, the first recurrent layer, over the T=8 training iterations; Section 7.1.3 uses mid-gap fragments, cells empty in both images and the layer the classifier reads): $\cos(h_t^{\text{stim}}, h_t^{\text{complete}}) - \cos(h_t^{\text{ctrl}}, h_t^{\text{complete}})$ at each iteration t.

Measures for the feedback networks (Section 5.4). These networks label the object at a cued location rather than classify, so their unit of measurement is the *interior fill*: the share of a figure's interior pixels (the ink-free pixels inside it, beyond one patch from the cue) that the network labels as the cued object, meaning that its read-out's most probable class for that pixel is not background, with the cue at the figure's centre and read at the last recurrent step (80). The cue is placed by us, at the centre of the figure, in every display alike: the network is asked what object lies at that point, and the measure is what it labels in answer. It does not have to find the Kanizsa figure; the question is whether, told where to look, it labels the figure between aligned inducers and not the same region between rotated inducers, closed discs or full discs. The cue selects the location, it does not create the label: run with no cue at all, the same networks label the line square completely, nothing between rotated inducers, and the Kanizsa square at 0.15–0.61 of its interior (Appendix L), so the selectivity reported below is in the response to the image. Fill is therefore a proportion between 0 and 1: a fill of 0.09 means that 9% of the figure's interior area is labelled, 0.90 that 90% is, and a complete, clean fill reads 1.00; the same quantity on the strip just outside the figure is the *spill*. The maps in the figures show, pixel by pixel, the read-out's probability that the pixel belongs to the cued object, from which the label is the pixels above one half. The displays are a square of side 64 px at 224 px, drawn at two *support ratios*, the share of each side that is bounded by contour pixels (Shipley & Kellman, 1992) (0.5 and 0.7), and four orientations: the *Kanizsa* square; *rotated*, the same inducers turned by 180° in place; *closed discs*, the inducers with their notches closed by hairlines; *full discs*, solid discs at the same positions; *line*, the gaps between the inducers drawn as real lines; *outline*, the whole square drawn; *corners*, only the notches' straight sides, and the same corners turned by 180°; and clutter versions that add stray inducers, discs and crosses or a second figure (Fig. 2). The *false fill* is the fill of a scrambled contour in a separate validation set of real contours; checkpoints are selected on that set (Appendix L). The pass criteria, fixed before the runs, were: line or outline fill of at least 0.85, rotated fill of at most 0.05, and Kanizsa fill exceeding closed-disc fill by at least 0.2, each on two of two seeds. Two further read-outs are reported, not used as gates: the fill at every recurrent step (the time course), and the fill with the feedback cut from the first step, which shows what the looped blocks produce in a single pass. Two linear probes are used alongside: the *separation probe*, one logistic read-out per token of an untouched encoder, fitted on clip frames to mark the object with no cue and read on the line drawings as the probability inside the square minus outside it; and the *contour probe* of Section 7.2, a per-layer logistic regression fitted on complete outlines that labels a feature-map cell as holding contour, which is also what the layer-by-layer figures of Section 6 show for the classifiers and circuits. Appendix L adds the fill as a function of support ratio from 0.22 to 0.95 and the fill on the inducers' own bodies. The cued circuits of Section 6.5 are measured on a different display, a large hexagon at support ratios 0.50–0.95, and their fill is defined there (the label after 40 steps, or 1 for the single-step controls).

5 Models

5.1 Feed-forward classifiers, looped transformers and recurrent circuits

Frozen backbones follow the protocol of Zhang et al. (2024), who replace the last layer of a frozen network by a fully connected layer (Adam, 10^{-3}, 25 epochs): VGG16, ResNet50, and, not tested by them, ViT-B/16 and DINOv2-S/14. **Fine-tuned** ResNet50, ViT-B/16 and DINOv2-S/14 are trained end to end. **Masked autoencoders** (MAE; He et al., 2022) start from the official ViT-B/16 MAE with its decoder, continued for 20 epochs on ImageNet-100 with one of three masks: random patches (75%, the MAE default), blocks

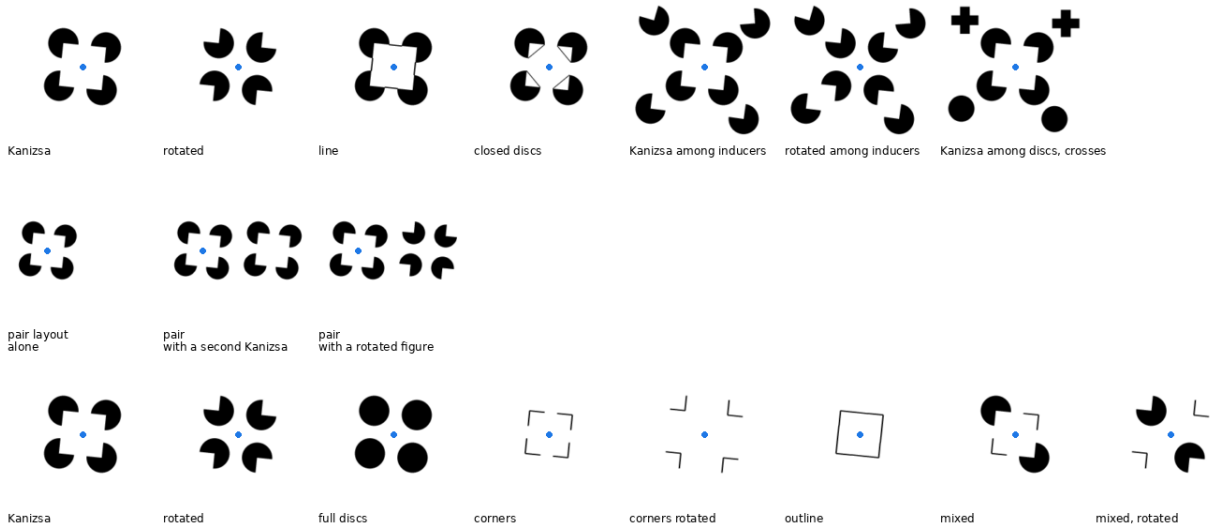


Figure 2: The displays of the feedback networks' battery, one item each (support ratio 0.7, first orientation), with the cue (blue dot) at the centre. Top row: the single displays and the clutter versions; middle: the pair layouts (the cued square alone, with a second Kanizsa figure, with a rotated figure); bottom: the corners probe's displays (Kanizsa, rotated, full discs, drawn corners, corners rotated, outline, inducers mixed with drawn corners, the same rotated). Every display is drawn at two support ratios and four orientations.

of 1–4 × 1–4 patches (75%), or the same blocks painted into the image as visible occluders (50%; the loss is the hidden pixels). Their encoders, and the official one, are fine-tuned like the other networks. **Looped transformers** split a pretrained ViT into prelude (blocks 0–3), a weight-tied core (blocks 4–7) applied T times, and coda (blocks 8–11), re-injecting the prelude output through a zero-initialized linear map so that T=1 is exactly the pretrained model (prelude, core and coda are the terms of Geiping et al., 2025; iterating a weight-tied block follows Dehghani et al., 2019; Giannou et al., 2023); T is sampled from $\{1, \ldots, 4\}$ during training and swept to 12 at test. **Horizontal recurrence** uses the fGRU cell (Linsley et al., 2018; 2020b) as specified in the appendix of Linsley et al. (2020b): attention-modulated suppression gate, a gate that updates the recurrent state, channel-symmetric horizontal kernels, instance normalization of the gate inputs and the published initialization. Our adaptations are a VGG16 front end (conv2 features, stride 4), a convolutional readout (three stride-2 3 × 3 convolutions with ReLUs, global average pooling and a linear layer; a 1 × 1 convolution in the deep hierarchies and stacks below), unrolling for T=8, and, for the γ-net, two levels with max-pooled bottom-up connections. The models are a horizontal-only fGRU circuit (fGRU-H; the hGRU of Linsley et al. (2018) is a different cell), an fGRU-H with a 1 × 1 readout that adds no spatial convolution after the recurrence, and a two-level γ-net whose top-down fGRUs update the lower layer from a projection of the upper one, read out from the lowest layer. **Deep fGRU-H/fGRU** hierarchies place fGRU layers at the four VGG stages: an eight-layer fGRU-H (two per stage, bottom-up only) and a four-layer fGRU with top-down connections, unrolled for T=8. **Divisive-normalization (DN) stacks** keep a CNN only at the input (VGG16 conv1–2, stride 4) and then stack 4 or 8 fGRU layers at that single resolution, each preceded by divisive normalization (Heeger, 1992) with learned channel weights, as in generalized divisive normalization (Ballé et al., 2016), over a fixed 5 × 5 average pool of our choosing; layers are linked by 1 × 1 convolutions, so spatial integration beyond the stem comes from the recurrent horizontal kernels, the 3 × 3 attention kernel of the suppression gate and the divisive-normalization pool, and top-down fGRUs update each layer through a zero-initialized gate.

The fGRU cell. For input (feed-forward drive) X and state H at one layer, each iteration computes

$$G^S = \sigma\big(\mathrm{IN}(U^A * H \odot U^M * H) + b^S\big), \quad C^S = \mathrm{IN}\big(W^S * (H \odot G^S)\big), \quad S = \big[X - [(\alpha H + \mu) \odot C^S]_+\big]_+,$$
$$G^F = \sigma\big(\mathrm{IN}(U^F * S) + b^F\big), \quad C^F = \mathrm{IN}\big(W^F * S\big), \quad \tilde{H} = \big[\nu(C^F + S) + \omega(C^F \odot S)\big]_+,$$

and $H \leftarrow (1 - G^F) \odot H + G^F \odot \tilde{H}$ (Linsley et al., 2020b). U^A is a 1×1 convolution and U^M a 3×3 convolution to one channel (spatial attention on the suppression gate only); U^F is 1×1; W^S and W^F are the $k \times k$ horizontal kernels, constrained to be symmetric between channels; IN is instance normalization (Ulyanov et al., 2016) with learned scale (initialized at 0.1); α, μ, ν, ω are per-channel parameters initialized at $0.1, 0, 0, 0.1$, and H starts at 0. A top-down fGRU uses the same cell with 1×1 horizontal kernels (the 3×3 attention kernel of the suppression gate is kept), the lower layer's state as X and a projection of the upper layer's state (instance normalization, resizing and two 1×1 convolutions with ReLU) as H, and adds its output to the lower layer. Gate biases use the chrono initialization (Tallec & Ollivier, 2018) and the horizontal kernels are orthogonally initialized; making them symmetric afterwards breaks exact orthogonality. Other departures from the published specification: α is kept positive by a softplus, the front ends are ImageNet VGG16 layers (conv1–2 for the single-resolution circuits, the four VGG stages for the deep hierarchies, with the first convolutional block frozen in the deep hierarchies and the DN stacks), and the DN stacks gate the top-down update by $\tanh(g)$ with g initialized at zero.

Every recurrent circuit (fGRU-H, γ-net, deep hierarchies and DN stacks) except the fGRU-H with a 1×1 readout has a single-iteration control (trained with T=1); the looped transformers are instead run for different numbers of loops. All networks except the frozen ones are trained for 15–20 epochs on 6,000 complete polygons and reach 1.00 validation accuracy; frozen backbones reach 0.97–1.00. Where a network was trained more than once, each training run started from a different random seed, the number that sets the random initialization of the new layers and the order of the training data; throughout the paper a *seed* is such an independently trained copy of the same network, so "seed 0" and "seed 1" in tables and figures are two trainings of one architecture on the same data, not different inputs, cue positions or starting points in the image. Transformer inputs are normalized with the statistics of each checkpoint's own pretraining. Training hyperparameters are listed in Table 7.

5.2 Contour tracers from a companion project

The contour-tracing networks come from a companion project and are described with their probe in Section 6.4 and Appendix E.

5.3 Reimplemented cued segmentation circuits

The reimplementation of the circuit of Mollard et al. (2026) and its variants are described with their results in Section 6.5 and Appendix J.

5.4 Feedback networks on frozen fronts: architecture and training

The cued circuits (Section 6.5) put recurrence on top of a frozen convolutional backbone; this family puts it inside a vision transformer. Each network keeps the early blocks of a pretrained encoder frozen (the *front*, run once) and loops the last blocks (six, in every network but the deeper cuts below) with feedback: the loop's output is passed through a layer norm and a linear map, initialized at zero so that the first step is the pretrained encoder, and added to the front's output before the looped blocks run again. A cue map, a disc of radius 8 px at the cued location, enters through a 1×1 convolution added to the front's tokens (Fig. 3). Two read-outs of 1×1 convolutions follow the looped blocks: the cued-object label, and a *layer label* that classifies every pixel as a visible part of the cued object, a hidden part, an occluder, or background. The number of recurrent steps is sampled from 1 to 8 during training and the networks are run for 80 at test.

Encoders. V-JEPA 2 ViT-L (Assran et al., 2025), a video encoder trained to predict the features of masked video, with the image repeated as a two-frame tubelet at 512 px (32×32 tokens), blocks 1–18 frozen and 19–24 looped; V-JEPA 2.1 ViT-L and ViT-B (Mur-Labadia et al., 2026) (hereafter 2.1 L and 2.1 B), which add an image tokenizer (512 px, the same split for the ViT-L; blocks 1–6 frozen and 7–12 looped for the ViT-B); LeVJEPA ViT-L (Kuhn et al., 2026), at its native 224 px (14×14 tokens), with the same split and, in a second variant, with the front trained as well (the *whole encoder*, blocks 1–18, the patch embedding and the class token at the looped blocks' learning rate); DINOv3 ViT-B/16 (Siméoni et al., 2026) at 448 px (28×28 tokens), blocks 1–6 frozen and 7–12 looped, and DINOv2 ViT-B/14 (Oquab et al., 2024) with the same split,

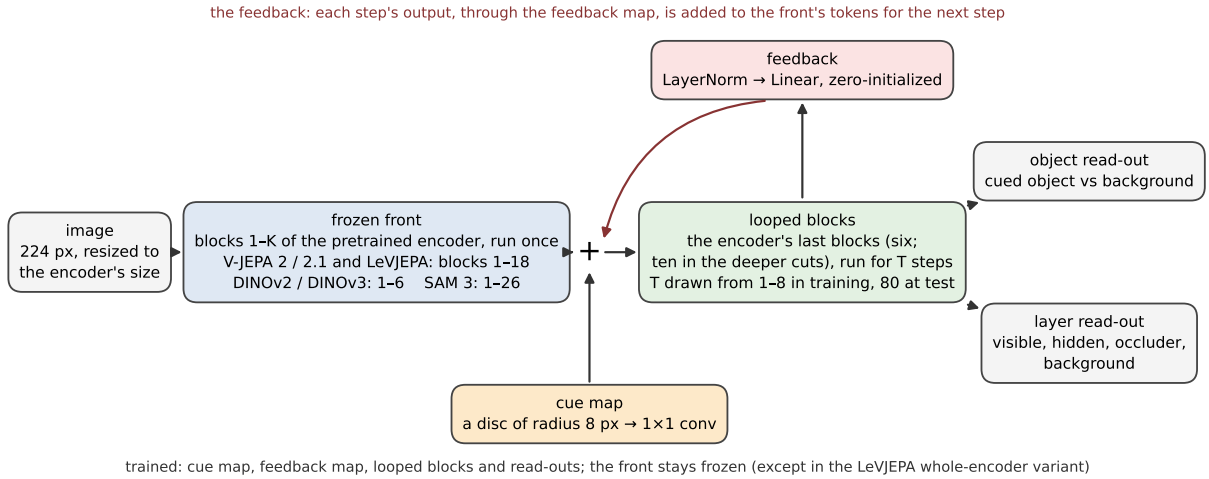


Figure 3: The feedback networks. The pretrained encoder's early blocks (the front) are frozen and run once; the cue map is added; the encoder's last blocks, copied, run for T steps with a zero-initialized feedback map that adds each step's output to the front's tokens for the next step; two read-outs give the cued-object label and the layer label. In training T is sampled from 1 to 8; at test the networks run for 80 steps.

the first encoder of this family; and the vision trunk of SAM 3 (Carion et al., 2026), a 32-block, 1024-wide ViT with 14-px patches from the Perception Encoder family (Bolya et al., 2025), pretrained on image–text pairs at 336 px and then trained inside SAM 3 to delineate objects in images and video from concept prompts, run at 336 px (24×24 tokens, so that its windowed and global attention blocks attend over the same grid), blocks 1–26 frozen and 27–32 looped. Among these, SAM 3's trunk is the only encoder trained with object masks; the others are self-supervised.

Two hybrids swap halves between V-JEPA 2.1 L and SAM 3 through a linear adapter initialized at the identity: SAM 3's front (blocks 1–26) with 2.1's looped blocks (19–24), and 2.1's front (blocks 1–18, at 384 px so that the grids match) with SAM 3's looped blocks (27–32).

Training has two stages, 4,000 iterations each (batch 24; Adam, Kingma & Ba, 2015; learning rate 10^{-4} for the cue map, the feedback map and the read-outs and 10^{-5} for the looped blocks), back-propagated through the sampled steps. Stage 1 is cued segmentation of COCO objects (Lin et al., 2014), the task of the cued circuits. Stage 2 is cued segmentation on eight-frame synthetic clips in which a COCO cut-out moves behind flat-coloured covers (the *unity* clips), with the layer labels as a second target; it is this stage whose variants the family was built to compare. The clips used for the main results draw 30% of the objects as their outline only (1–3 px lines; the *outline clips*); a *texture* variant fills the object with its own background texture instead, and a third uses both. Three controls change one thing each: the *no-loop* network is trained with the feedback zeroed throughout, so that it is the same architecture run as a feed-forward network; the *no layer labels* network drops the layer-label loss; and the *faded covers* network trains on clips in which covers may be blended with the background at a random opacity (0.15–0.7 per clip). Two seeds were trained for every cell reported with a verdict; earlier cells, trained on plain unity clips without outlines, have two to three seeds. Before this recipe was settled, the same architecture on DINOv2 was trained with several other objectives (static layered COCO pastes, moving occluders, a slit-reveal objective, one-step training, and variants of the unity clips with background-coloured or evidence-limited covers), each with one to eight seeds; their outcomes are summarized in Appendix L.

5.5 Null observers

The null observers used in the audit of published measures are defined in Section 8.

6 Results by family

6.1 Feed-forward classifiers

The frozen and fine-tuned CNNs and vision transformers, and the four masked autoencoders, are the family on which most of the battery was first run; Fig. 4 shows the contour probe moving through two of them layer by layer, and the subsections below give their behavioural, probe and patching results.

6.1.1 Illusory contours are at most weakly represented

Across all networks that learned the training task (first training seed of each), accuracy on Kanizsa figures lies between 0.04 and 0.39 (chance 0.10), and the Kanizsa closure index reaches at most +0.10 (ViT-B/16 (frozen); Fig. 5A). Turning every inducer in place by 15° halves the transformers' Kanizsa advantage, but part of it remains at 45°, so it does not require the inducer edges to line up (Appendix G). Across three to five training seeds of three networks (Table 5), fragment closure indices and arrangement effects (accuracy in place minus accuracy with scrambled positions; Section 6.1.2) replicate (largest s.d. for these two measures 0.03 and 0.05); the Kanizsa index is small but consistent in fine-tuned ViT-B/16 (+0.08 to +0.14, Kanizsa accuracy 0.18–0.35), inconsistent in DINOv2-S (+0.01 to +0.15) and near zero in ResNet50 (-0.05 to +0.07), always against 1.00 accuracy on complete outlines (Table 5). The frozen and looped ViT-B/16 show indices of the same size (Table 3), so ViT-B/16 carries a weak illusory-contour signal that is consistent across seeds only in this network. Class-specific relevance maps (Chefer et al., 2021a) for the top-1 class illustrate this: in Fig. 7 Kanizsa triangles and squares are labelled as hexagons, and their rotated inducers receive the same label. Dashed linking segments raise accuracy about as much with aligned inducers as with no inducers: from contrast 0 to 1 the median gain over 22 networks (including the six single-iteration controls) is +0.14 with aligned inducers and +0.11 without, the median of the per-network differences is -0.01 (above +0.05 in 6 networks, below −0.05 in 8), and rotated inducers suppress the gain (+0.01). Noise was drawn independently for each image.

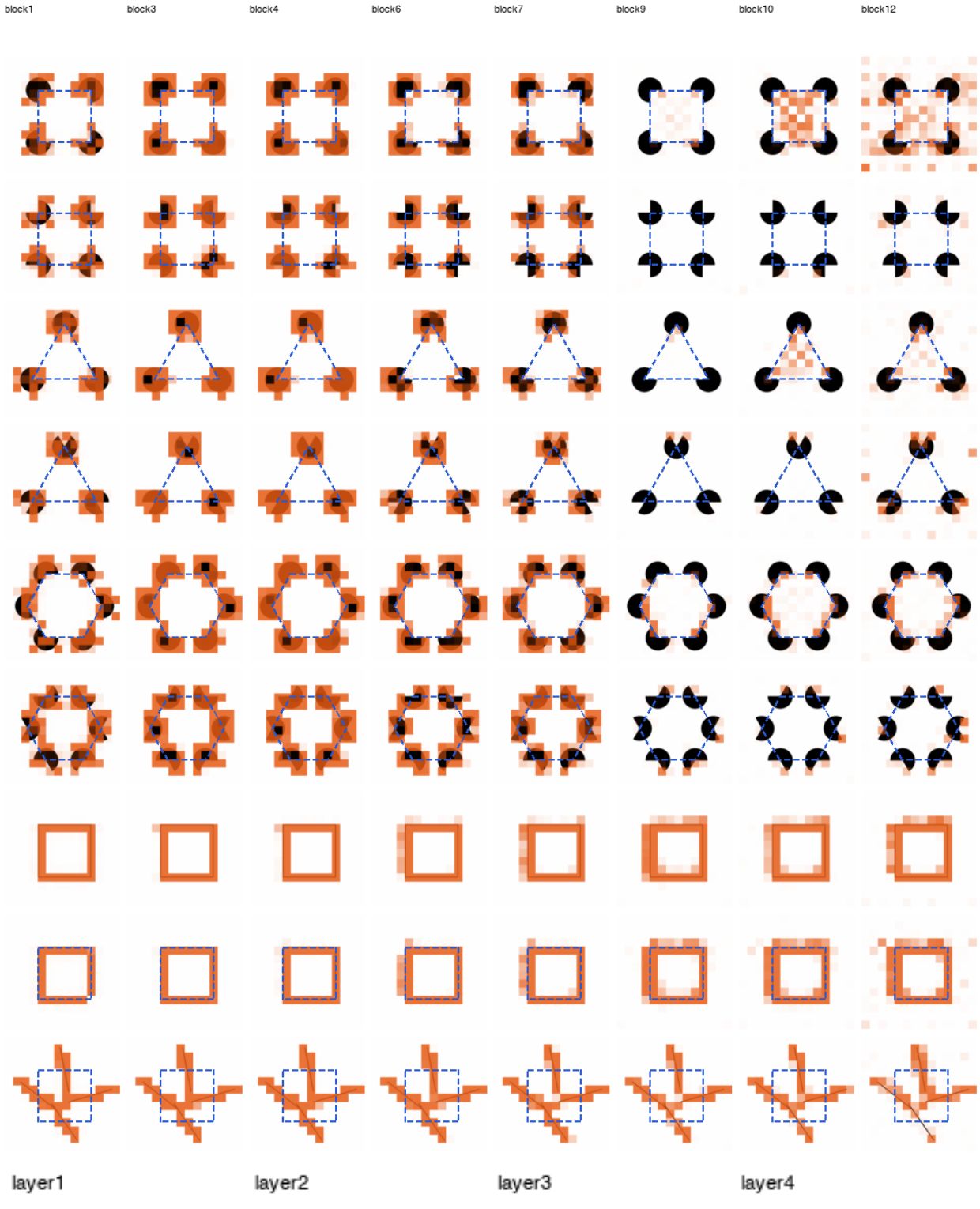
6.1.2 Fine-tuned transformers integrate fragments, and the arrangement matters

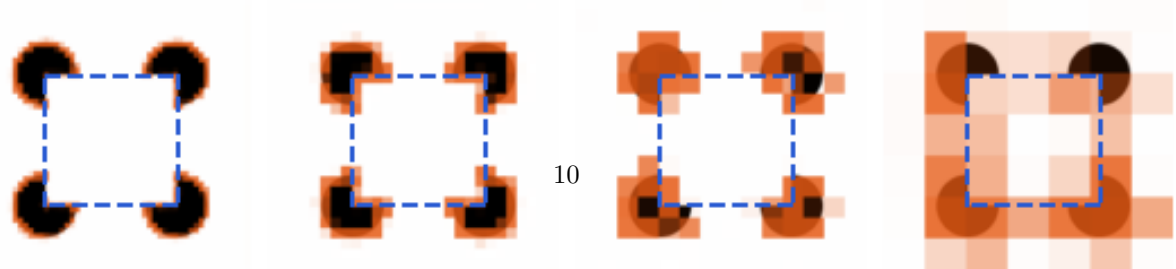
With corners removed, fine-tuned ViT-B and DINOv2 (plain and looped) classify fragmented outlines at 0.84–0.96 in the orientation-control battery (a separate test set of corner-removed fragments in place, scrambled and cue conflict, gaps of 0.5–2P), and their closure index over all gap sizes of the main battery is +0.60 to +0.80. Frozen backbones reach half to 70% of that, and fine-tuning raises the index 1.4–2× within each family. Scrambling the pieces' positions while keeping their orientations drops the fine-tuned transformers to 0.27–0.39 (Fig. 5B): the arrangement adds +0.46 to +0.65 beyond orientation statistics. When positions and orientations conflict, these networks pick the class implied by positions only 0.03–0.09 of the time and the class implied by orientations 0.18–0.42 (Fig. 5C). Pieces must therefore be both placed and oriented along the contour, the signature of contour integration by good continuation (Field et al., 1993) rather than of position or orientation alone.

The manipulation used by Zhang et al. (2024), which keeps the corners, behaves differently: scrambling corner pieces costs fine-tuned transformers only +0.12 to +0.17, so that test is largely solved from corner angles and cannot separate closure from corner reading.

6.1.3 Where closure emerges, and what the maps show

Applying the linear probe at every layer (Fig. 6), fragment closure in fine-tuned transformers rises through the middle blocks and peaks late (ViT-B +0.68, DINOv2-S +0.79). In the looped ViT it reaches +0.73 by the fourth pass through the shared core and gains at most about 0.02 with further iterations, so additional loops add depth without adding closure. Frozen ViT-B peaks at +0.60 and fine-tuned ResNet50 at +0.52 in its last stage. Relevance maps make the dissociation between fragments and illusory contours visible (Fig. 7): for fragmented outlines, relevance follows the whole contour including its gaps, whereas for Kanizsa figures it stays on the inducers, and Kanizsa triangles and squares are labelled "hexagon" whether their inducers face inward or are rotated.


block1
block3
block4
block6
block7
block9
block10
block12
layer1
layer2
layer3
layer4



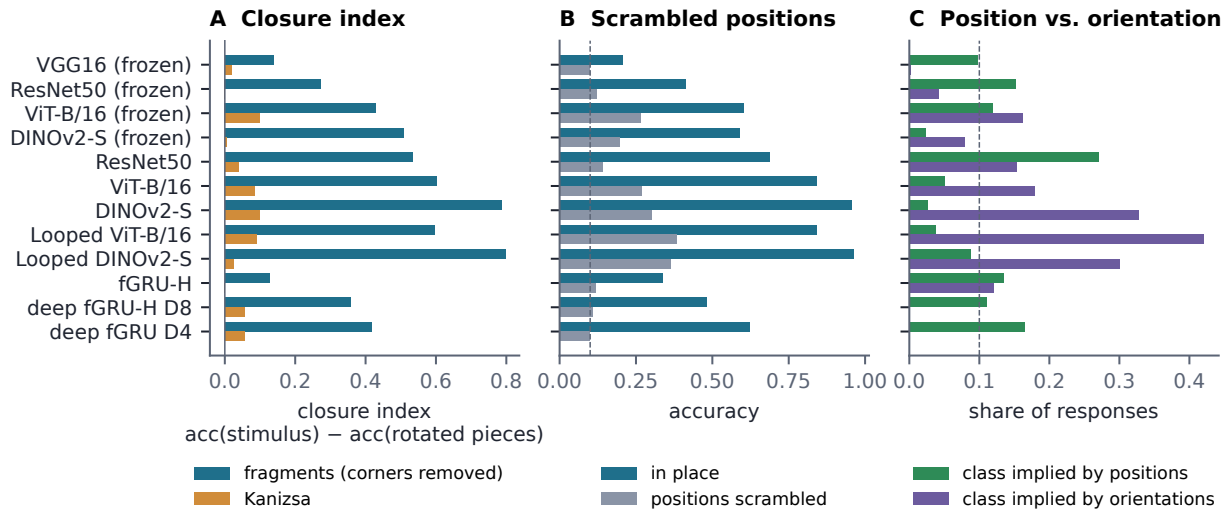


Figure 5: (A) Closure index for corner-removed fragments (all gap sizes) and Kanizsa figures. (B) Corner-removed fragments in place versus the same pieces with scrambled positions and unchanged orientations; dashed line, chance. (C) Cue-conflict stimuli: how often the network picks the class implied by the pieces' positions versus their orientations.

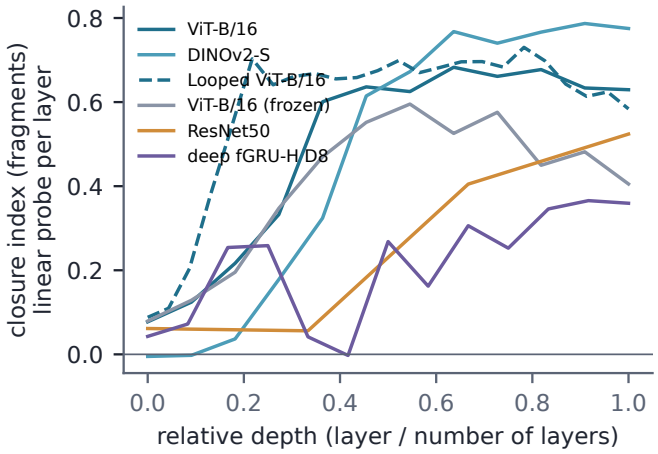


Figure 6: Fragment closure index from a linear probe trained on complete outlines at every layer, against relative depth. For the looped ViT, depth counts executed blocks (loops unrolled).

6.1.4 Masked autoencoders draw the illusory edge only when the inducers are hidden too

Li et al. (2026) report that a masked autoencoder (MAE) continued on natural images reconstructs boundary-like edges between Kanizsa inducers when the gap is masked (boundary F1 0.52, edge strength 0.15), and much weaker ones after continued training on texforms; they compare the two models but test no display without an illusory figure. We rebuilt their protocol from the paper (their code is not released): virtual squares with straight, concave or convex sides, inducer radius 24–42 px and a 48 px gap, their patch mask, their Canny settings, evaluation region, boundary F1 and edge strength (stimulus details the paper leaves open are listed in the code), and applied it to our four MAEs. On the Kanizsa displays it gives boundary F1 0.47–0.55 and edge strength 0.13–0.17, close to their values; with the inducers rotated in place it gives 0.14–0.19 and 0.02–0.04, with full discs 0.08–0.13 and 0.00–0.01, and with no inducers 0 (Fig. 8). Their result therefore holds against these controls: the reconstruction fills the hidden region just outside each side with grey and the region inside with white, and the edge between them lies on the virtual square. Their mask,

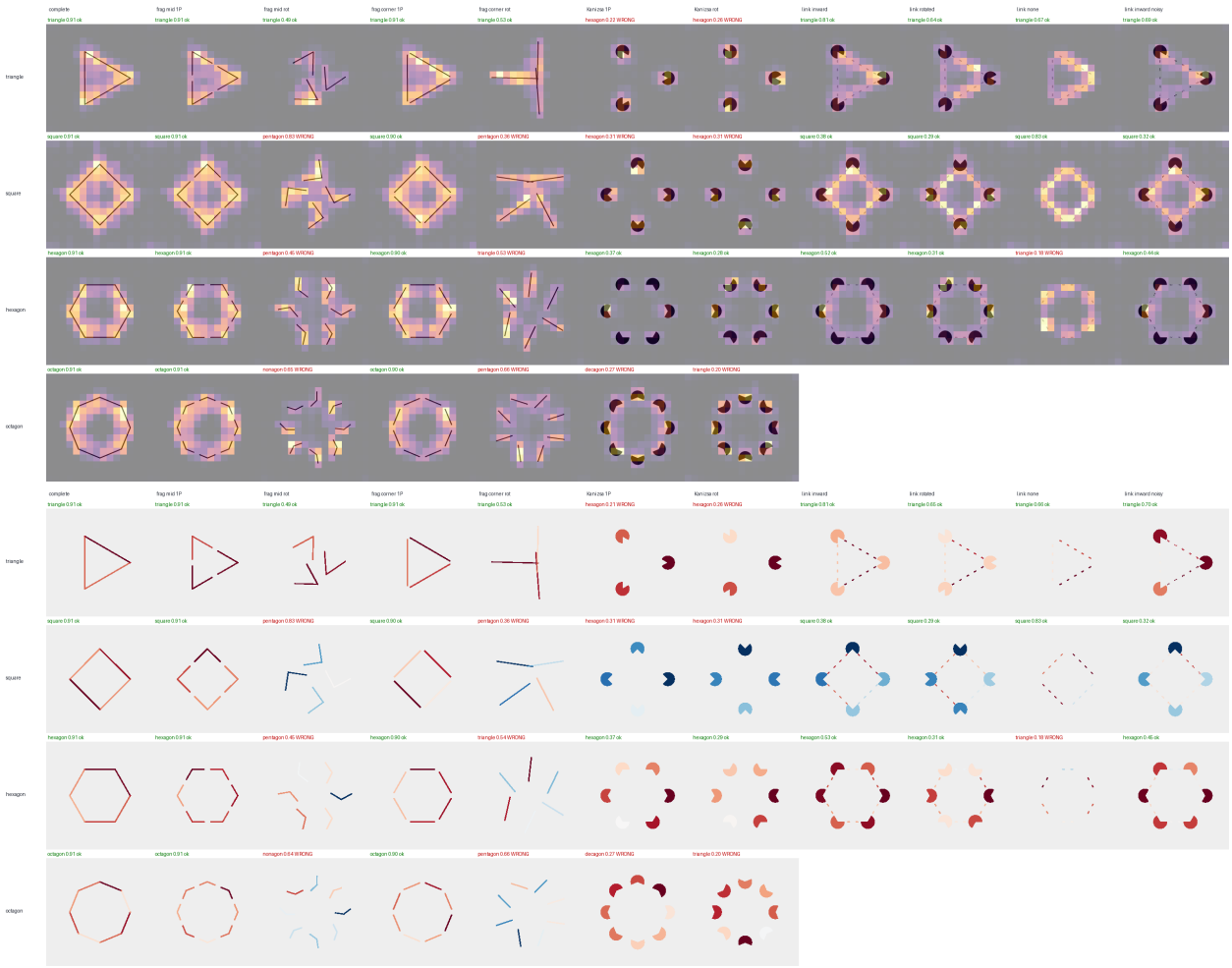


Figure 7: Fine-tuned DINOv2-S. Top: class-specific relevance maps (Chefer et al., 2021a) for the top-1 class per condition (columns) and true class (rows); cell titles give the prediction and its probability. Bottom: Shapley values (Shapley, 1953) of the stimulus's pieces for the true label (red supports, blue opposes). Illustrative only: these maps are weakly faithful (Section 9); the behavioural evidence for integration is in Fig. 5.

however, extends into the inducers: 29 of its 73 patches contain inducer pixels, including the notches' own straight edges. With the same patches minus those holding inducer pixels (44 patches), so that only the gap is hidden and the notch edges stay visible, the reconstructions contain no edge in the gap (F1 0.01–0.05, edge strength 0.003), while a line drawn along the sides is continued across the same hidden patches (F1 0.96–0.99, edge strength 0.29–0.30). The edge is not confined to the hidden inducer parts: under their mask, F1 in the middle of the gap alone is 0.49–0.56. Our earlier test with our own displays and a mask of the gap alone (Appendix I) agrees: no line along the illusory edge (at most 0.8 grey levels, against about 150 for a real line). The MAEs thus produce the illusory edge when they must reconstruct the inducers' notches together with the gap, and not when the notch edges are visible and only the space between them is missing, the case in which a drawn line is continued. Fine-tuned for classification, the same four MAEs behave like the supervised ViT-B/16: closure index 0.58–0.67 for corner-removed fragments (ViT-B/16, 0.60–0.65) and 0.01–0.04 for Kanizsa figures (ViT-B/16, 0.08–0.14).

6.1.5 The transformers' Kanizsa evidence does not pass through the gaps

The probes of Section 7.2 ask whether the empty cells represent an edge; activation patching (Meng et al., 2022; Heimersheim & Nanda, 2024) asks whether the evidence the classifier uses passes through them. In a

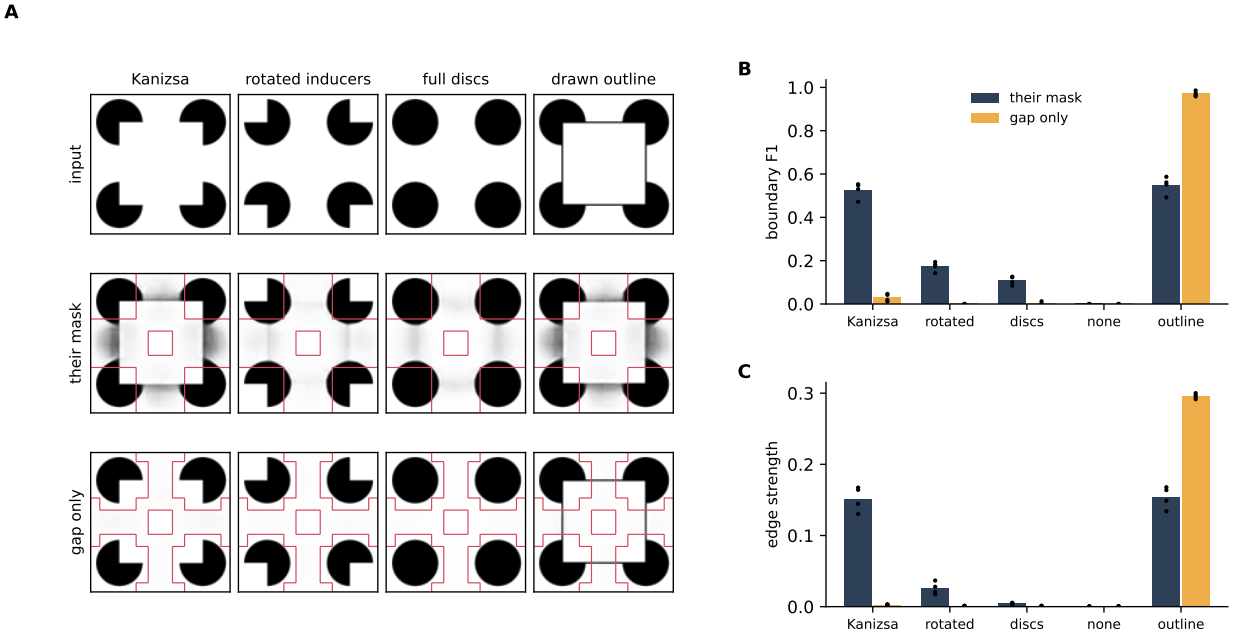


Figure 8: Reconstruction of masked Kanizsa displays by a masked autoencoder, scored with the protocol of Li et al. (2026). (A) Inputs (inducer radius 30 px, straight sides) and the official MAE ViT-B/16's reconstructions; red outlines, the masked patches. Their mask hides every patch near the middle of each side, inducer patches included; the gap-only mask hides the same patches minus every patch with an inducer pixel. Rotated inducers, full discs and a drawn outline get the same mask as the Kanizsa display. (B) Boundary F1 of Canny edges against the virtual square's sides, (C) edge strength across them, in their evaluation region (their mask) or the middle of each gap (gap-only mask); bars, mean of the four MAEs; dots, each MAE; 20 configurations, both polarities.

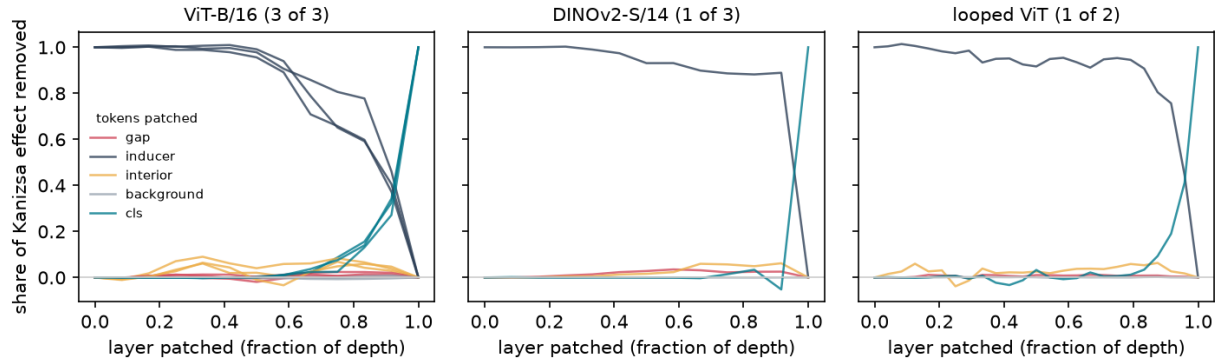


Figure 9: Activation patching in the transformers with a Kanizsa effect (log-odds of the true class at least 0.5 higher than with rotated inducers; inducer radius at least 1.5 patches). Share of that difference removed when one set of tokens at one block is replaced by the rotated-inducer display's (gap, inducer, interior and background tokens; class token), against the block's depth; one line per network.

transformer every token reads every other at every block, so a Kanizsa figure could be recognized through the gap tokens even if no edge is represented there. For each Kanizsa display of the battery and its rotated-inducer control, which differ only in the inducer pixels, we ran the fine-tuned network on one display, replaced one set of tokens with the other display's at one block, or at that block and every later one, and measured the change in the log-odds of the true class. The sets are the gap tokens (cells crossed by the edge between neighbouring inducers that contain no inducer pixels in either display), the inducer tokens, the interior tokens, and as many background tokens outside the shape as there are gap tokens; we report the share of the Kanizsa-minus-rotated difference that the replacement removes.

A Kanizsa effect large enough to localize (at least 0.5 in log-odds) appears only with inducers of radius 1.5 patches or more, and there in 5 of the 12 transformers: the three ViT-B/16 seeds, one of three DINOv2-S seeds and the looped ViT-B/16 (accuracy 0.53–0.78 on these Kanizsa figures, 0.18–0.45 with rotated inducers); the others, all four MAE networks among them, reach at most 0.33. In these five networks, giving the gap tokens the rotated display's content at every block removes at most 0.05 of the effect, no more than the background tokens (at most 0.01), and the interior tokens remove 0.05–0.15; the inducer tokens carry all of it (1.00) until the last blocks, where it passes to the class token (Fig. 9). When the same gaps are closed by drawn lines (inducer radius up to 1 patch), the gap tokens carry 0.38–0.88 of the effect in all twelve networks, so the test finds evidence in the gaps where there is some. Where a Kanizsa effect exists the gaps are short, about six gap tokens per display against about eighty inducer tokens; within that limit, the transformers read their preference for aligned inducers from the inducers themselves, not from the space between them.

6.2 Looped transformers and recirculation

Two ways of adding recurrence to a pretrained transformer were tested: looping a weight-tied core (Section 5.1; the contour probe over its executed blocks is in Fig. 10, its per-layer closure probe in Fig. 6) and training-free recirculation (Appendix F). Looping a fine-tuned transformer changes the fragment closure index by -0.01 to +0.01. **Recirculation.** Recirculation (Mozer et al., 2026) turns a frozen transformer into a dynamical system by leaking a renormalized deep-layer activation back into a shallow layer at every step. On a single static image there is no step dimension, so what we test is training-free recurrence in depth (looping, in the terms of Mozer et al., 2026) with recirculation's mixing rule, in which the source activation is rescaled to the destination's norm before being added: fine-tuned and frozen ViT-B/16 and DINOv2-S re-process the same image with the previous pass's deep state mixed into the shallow layer, over 60 settings per network of source layer, destination layer, mixture coefficient and number of passes. Among settings that lowered accuracy on complete outlines by at most five percentage points, none raised the Kanizsa closure index by more than +0.03 or the fragment index by more than +0.03 (largest Kanizsa index of a fine-tuned network over all settings +0.14), small differences, selected post hoc, whose reliability we did not estimate.

6.3 Horizontal and top-down recurrent circuits

The six recurrent circuits with single-iteration controls are compared with those controls in Fig. 12A, their gap similarity over iterations in Fig. 12B, and the contour probe on their recurrent state at every iteration in Fig. 11. For fGRU-H and fGRU circuits, recurrence increases the gap similarity: it rises from 0.28 to 0.35 (fGRU-H), 0.17 to 0.42 (γ-net) and 0.35 to 0.37 (eight-layer deep fGRU-H) over iterations (Fig. 12B). Recurrence does not consistently improve the fragment closure index: in 5 of 6 circuits the control trained with a single iteration is within 0.02 of, or above, its eight-iteration counterpart (for example the fGRU-H, +0.13 versus +0.18; Fig. 12A), and the one gain, +0.08 in the deep fGRU D4, is from a single seed. **Divisive normalization with a CNN stem only.** When the CNN is reduced to a stem and every further layer is a recurrent fGRU preceded by divisive normalization, the corner-removed closure index is +0.29 for the four-layer and +0.26 for the eight-layer stack, well below the fine-tuned transformers; the single-iteration controls score +0.30 and +0.40, and the Kanizsa index of both stacks and their controls stays at or below +0.01. Across recurrent circuits the Kanizsa index is at most +0.06 (single-iteration controls +0.04), so recurrence adds no illusory-contour signal either.

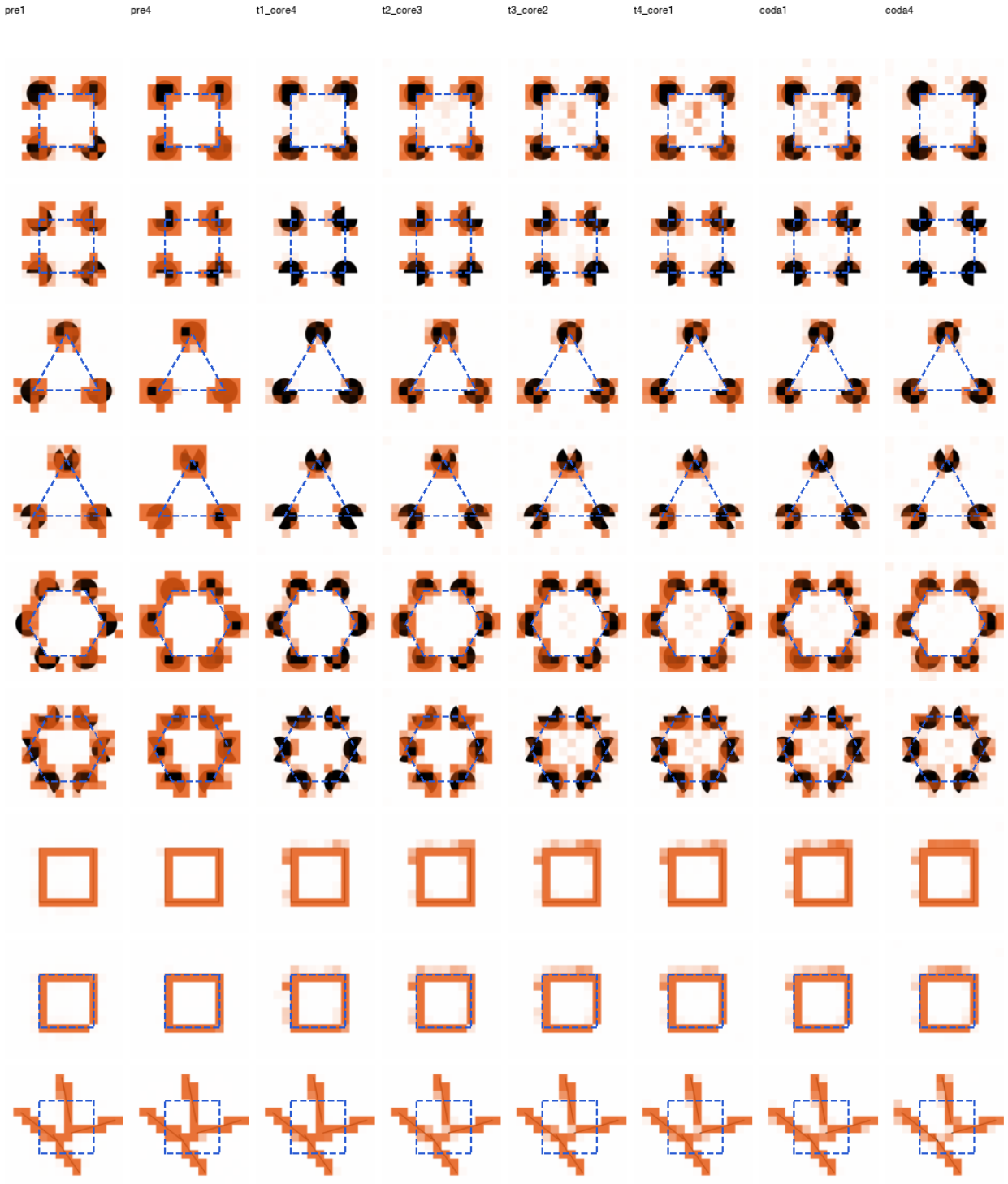
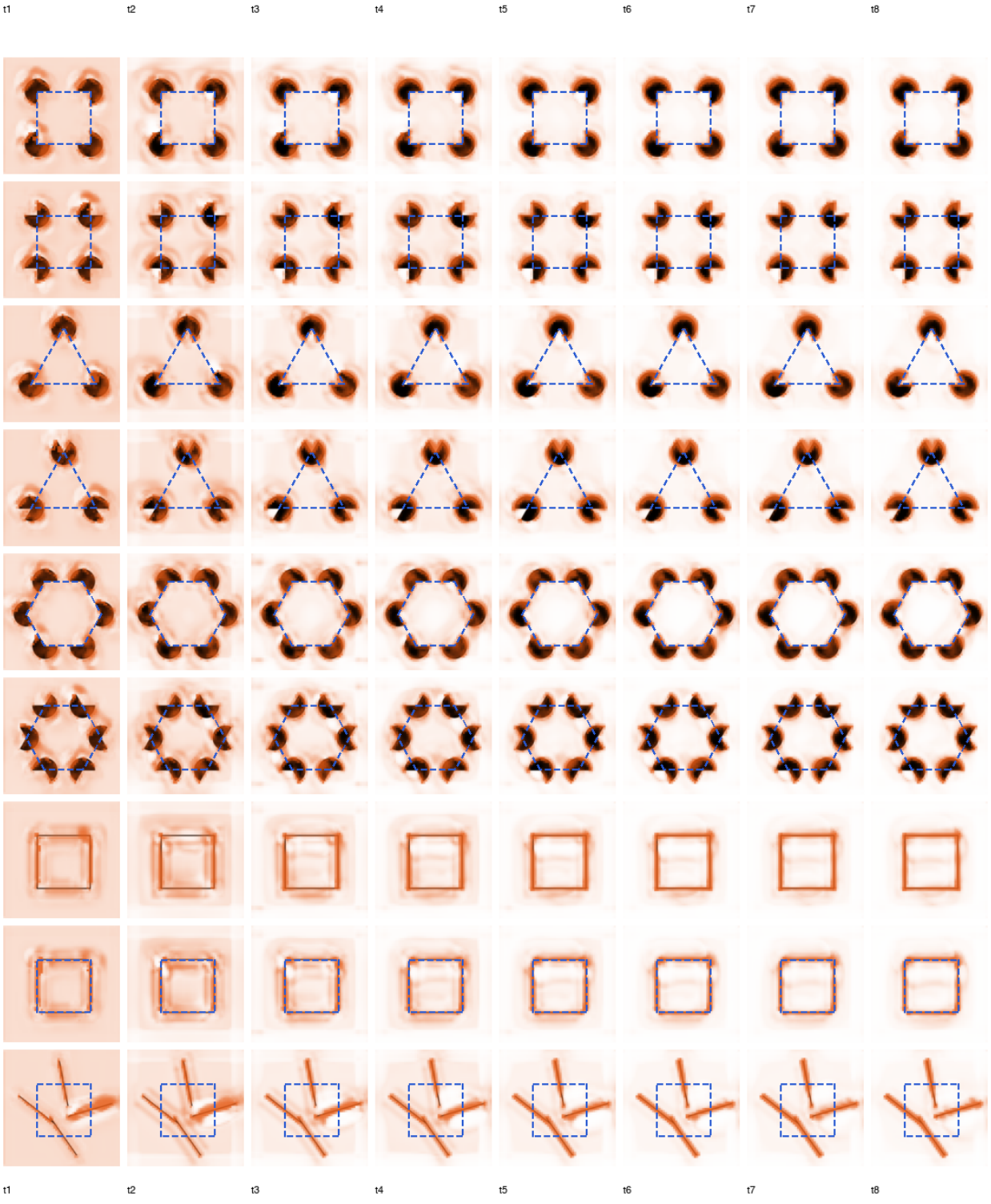


Figure 10: The contour probe of Fig. 4 for the looped ViT-B/16 (prelude blocks 0–3, the weight-tied core of blocks 4–7 applied four times, coda blocks 8–11; 24 executed blocks, every third shown). Rows as in Fig. 4. The passes through the core repeat the same map; nothing accumulates on the illusory edges.


t1
t2
t3
t4
t5
t6
t7
t8
t1
t2
t3
t4
t5
t6
t7
t8

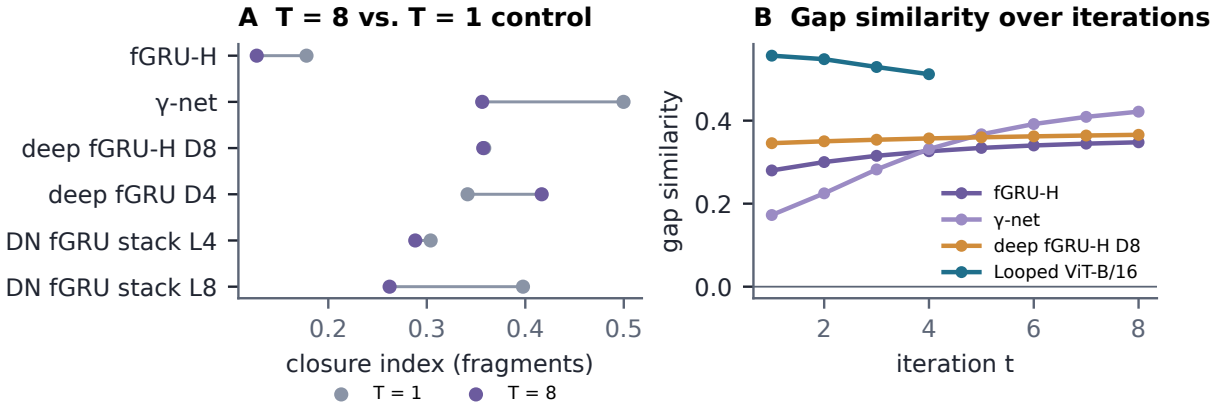


Figure 12: (A) Fragment closure index for the six recurrent circuits that have single-iteration controls (T = 8) and those controls (T = 1). (B) Gap similarity in the gap region of the recurrent state (first recurrent layer, one-patch corner-removed fragments) across the training iterations; for the looped ViT-B/16, across passes through the core.

6.4 Contour tracers

Our recurrent circuits were trained on complete outlines, a task that never demands long-range integration. We therefore probed networks trained for exactly that: three hGRUs (one as published, and two whose per-iteration batch normalization is replaced by group normalization (Wu & He, 2018) shared across iterations, trained on contours of 6 and 14 dashes), an fGRU/γ-net and weight-tied looped local-attention transformers trained to segment the dashed contour on which a marker sits, a Pathfinder-style curve-tracing task (Jolicoeur et al., 1986; Roelfsema et al., 1998) (validation intersection-over-union, IoU, 0.92–0.996 on 128 px images, 0.80 for the untied control below; taken from a companion project without retraining; the training target is the full dashed contour, so a network that completes gaps has no reason to label them). Each saw a marker on a short dashed segment followed, along the same line or arc, by a second segment after an extra gap, or by the same segment rotated in place (segment-pair probes; Fig. 13). *Traced* is the share of the target segment's pixels labelled as part of the marked contour, and *gap coverage* the share of the missing contour labelled as contour (Appendix E). At the dash spacing of their training data, the tied transformer and the three hGRUs extended the trace onto the aligned segment (mean 0.55–0.75 of its pixels) and almost never onto the rotated one (mean at most 0.06): they link by good continuation. For these four tracers, pooling aligned straight and curved segment pairs, linking halved after only 5–6 px of extra gap, averaged 0.02 at 12 px, and mean coverage of the missing contour stayed at about 1% of gap pixels or less (at most 0.01; the fGRU reaches 1.5%). Running the looped transformer for twice its trained iterations reduced tracing of the aligned target at native spacing (0.55 to 0.41) and did not extend linking across larger gaps. An untied 24-block transformer of the same depth traced only 0.08 of the aligned target at native spacing. The fGRU/γ-net labelled the rotated segment nearly as often as the aligned one (0.75 versus 0.82), so its spread is not contour-specific. In their segmentation output, networks that solve contour tracing therefore show the same profile as our classifiers: linking of present, aligned segments over short distances, and almost no labelling of absent contour; their training target did not require such labelling, so this does not rule out completion in their internal states.

6.5 Cued segmentation circuits

In incremental grouping, a label spreads from a cued point across the object it belongs to and stops at the object's boundary (Roelfsema, 2006). Mollard et al. (2026) trained a recurrent γ-net (Linsley et al., 2020b) to segment the object at a cued location in natural images and found that the time its grouping signal takes to reach a point predicts human reaction times. No code or weights have been released, so we reimplemented and trained it from the paper's Methods: frozen Panoptic FPN ResNet-101 features (Kirillov et al., 2019), four recurrent levels of hGRU units with top-down connections, a 5×5 horizontal kernel at the finest level, 40 timesteps, and contractor recurrent back-propagation (C-RBP; Linsley et al., 2020a) on COCO (Lin et al.,

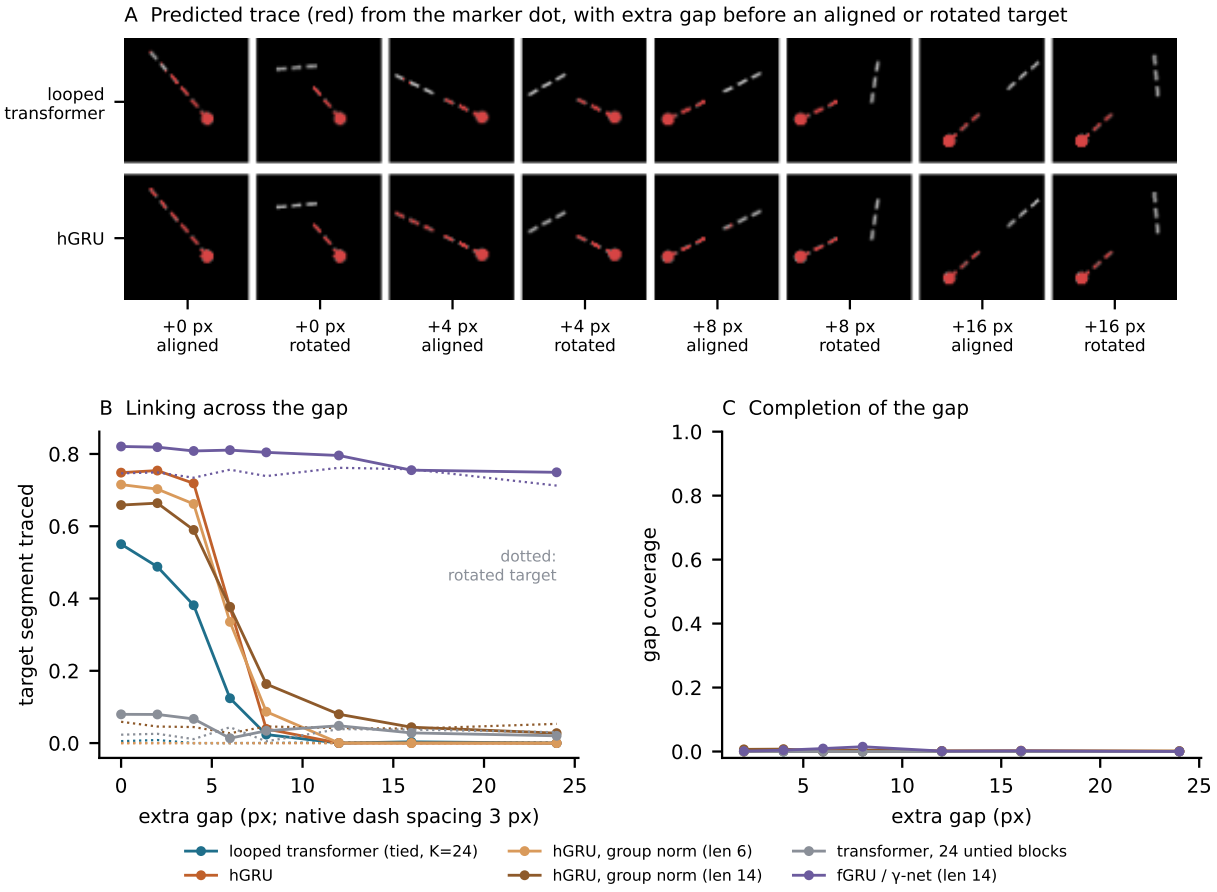


Figure 13: Contour-tracing networks (trained to segment the dashed contour a marker sits on) probed with a marked dashed segment, an extra gap, and a target segment that continues the path or is rotated in place. (A) Predicted traces. (B) Share of the target traced; solid, aligned; dotted, rotated. (C) Share of the missing contour in the gap predicted as contour.

2014) (Appendix J). To find what the circuit's behaviour depends on, we also trained it with the reference code's penalty weight, with 3×3, 9×9 and 15×15 finest-level kernels, and with the fGRU cell of our other circuits in place of the hGRU (also on ViTDet ViT-B features, trained on COCO masks; Li et al., 2022); we replaced the recurrent hierarchy with one weight-tied transformer block looped for 40 steps, in which every position attends to every other at every step; and we trained single-step controls of the hGRU and of the looped transformer, which read out after one step. On cued COCO validation objects the 12 networks reach an IoU of 0.43–0.55 (Table 1; one 9×9 run, whose Lipschitz penalty rose late in training, 0.28), against 0.74 reported on the images used by Mollard et al. (2026).

We placed the cue at the centre of each polygon of the battery ($n = 3$–6, inducer radius 1–2 patches) and compared the region labelled as the cued object with the polygon. The cue is ours, and it sits at the same place in every display: the circuit is told where to look and asked what object is there, so it never has to find the Kanizsa figure; the test is whether its answer covers the figure between aligned inducers and not the same region between rotated inducers or full discs. Run with the cue removed (a zero cue map, which no training image had), the circuits divide: the looped transformer, its single-step control and the 15×15 hGRU label nothing on any display, whereas the 3×3 hGRU, the single-step hGRU and the fGRU trained with the weighted loss label the Kanizsa hexagon as much as with the cue or more (uncued 0.68, 0.96 and 0.72 at 4 px gaps, against 0.61, 0.97 and 0.63 cued), with the same gap limit, and label less between rotated inducers and full discs than when cued (Appendix J). In the hGRU and fGRU circuits the figure label is thus a read-out of

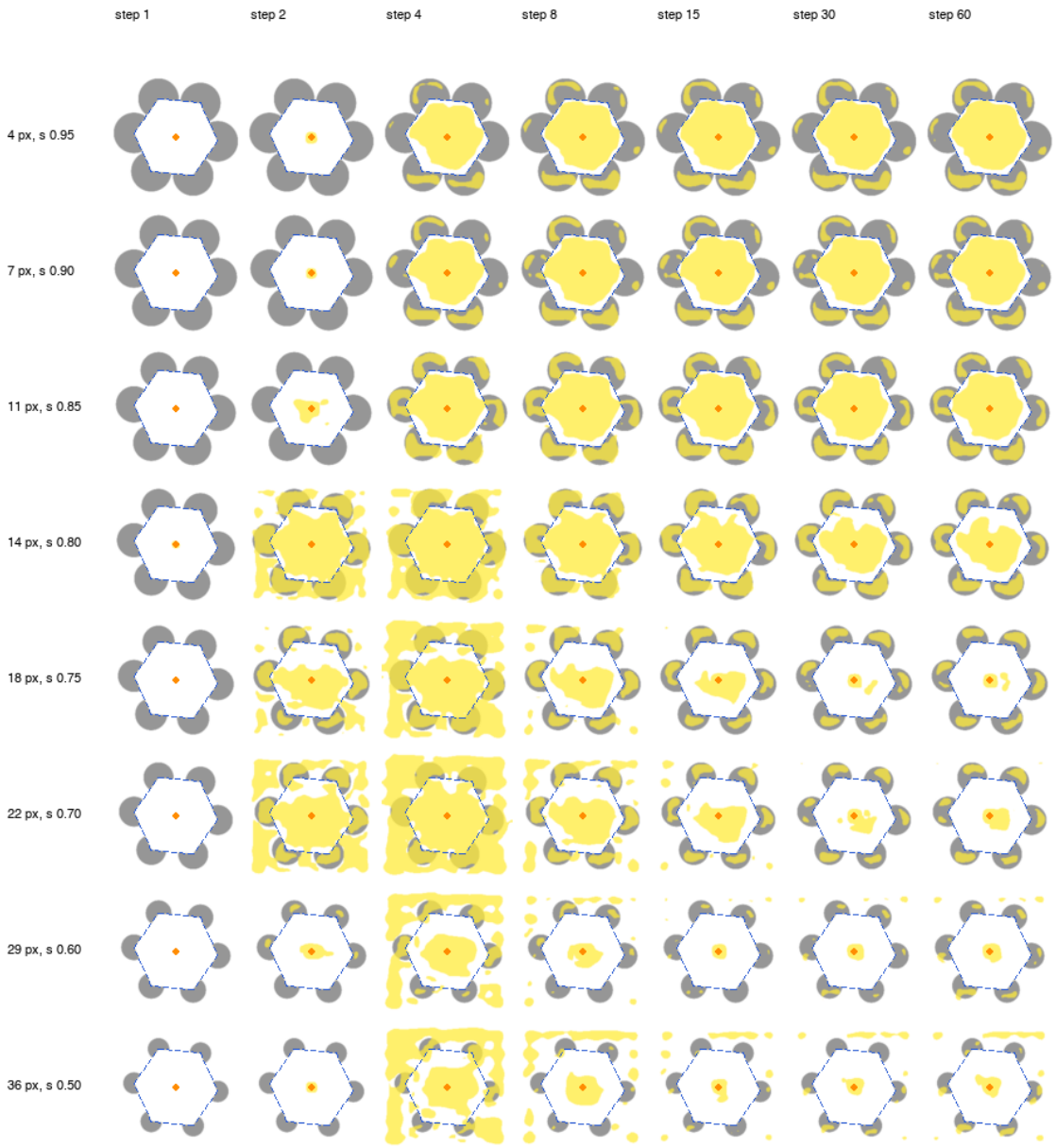


Figure 14: The looped transformer's cued label (yellow) growing from the cue (orange dot, placed by us at the hexagon's centre in every display, so that the network is answering what object lies there rather than finding the hexagon) on the Kanizsa hexagon, whose sides are drawn as a dashed blue outline, at gaps of 4 to 36 px between the inducers (support ratio 0.95 to 0.50; rows), after 1, 2, 4, 8, 15, 30 and 60 steps (columns). The label floods in steps 2–4, between the inducers and well outside the figure, and is then pruned: at gaps up to 11 px it settles on the hexagon, at 14 px it keeps part of it, and from 18 px it recedes to a patch around the cue. The animated version, with the closed-gap, dashed-line, full-disc and rotated-inducer displays and the other circuits, is in the supplementary material.

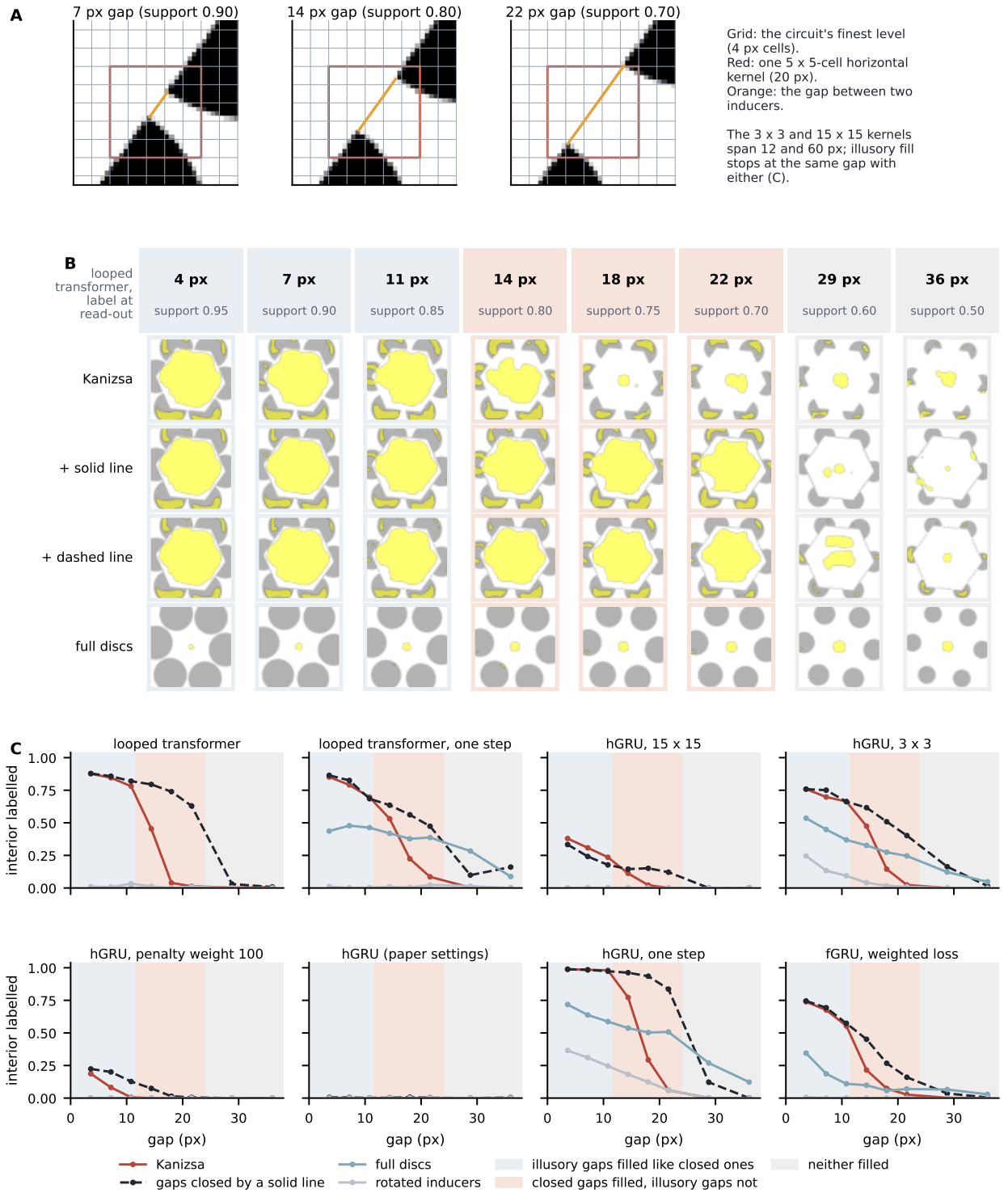


Figure 15: Cued segmentation circuits (after Mollard et al., 2026) on a large hexagon (circumradius 4.5 patches) at eight gap sizes, cue at the centre. (A) One gap enlarged on the 4 px grid of the circuits' finest recurrent level, with the 20 px reach of the paper's 5×5 horizontal kernel (red) and the gap (orange). (B) The looped transformer's label (yellow) at its read-out step for the Kanizsa display, the same inducers with the gaps closed by a solid or dashed line, and full discs at the same positions; header colours, the three gap ranges of (C). (C) Share of the hexagon's interior labelled against gap size (mean over four orientations) for eight circuits: Kanizsa, gaps closed by a solid line, full discs and rotated inducers. Shaded ranges follow the looped transformer: illusory gaps filled like closed ones; closed gaps filled and illusory gaps not; neither.

the backbone's features to which the cue adds; in the looped transformer nothing is labelled without the cue. On the CNN backbone no network segments the Kanizsa figure (IoU at most 0.25), a thin real outline is segmented no better (at most 0.44), and most networks segment a mid-grey filled polygon (at least 0.48; Appendix J). At these proportions, then, the method cannot tell an illusory boundary from no boundary, because it does not register a drawn one either. We therefore used the large hexagon and varied the support ratio s, the share of each side bounded by contour pixels (Shipley & Kellman, 1992), from 0.50 to 0.95, which narrows the gap between neighbouring inducers from 36 to 4 px, and compared the Kanizsa display with rotated inducers, full discs at the same positions (which specify no polygon), thick corner segments, the same aligned inducers with each gap closed by a real solid or dashed line of the battery's outline width, and a blank page (Fig. 15). We report *fill*, the share of the hexagon's white pixels labelled as the cued object, beyond one patch from the cue, at the network's read-out step.

The looped transformer fills 0.85 of the hexagon with aligned inducers and 7 px gaps, as much as when the gaps are closed by a line (0.86), against 0.01 with rotated inducers, 0.00 with full discs and 0.02 with corner segments, and the label stays inside: just outside the gaps 0.08 is labelled, against 0.36 on a blank page, where the label spreads (0.63 of the hexagon's area; Figs. 15B, 16 and 14). This specificity holds only for small gaps. At 18 px a closing line is still filled (0.74) but the illusory gap is not (0.04), and at 29 px even the closing line is not (0.03). The circuits thus show three ranges (Fig. 15C): up to about 11 px, illusory gaps are filled like closed ones; from 14 to 22 px, closed gaps are filled and illusory ones are not; from 29 px, neither. All but two of the battery's twelve combinations of n and inducer radius have gaps wider than 22 px.

Four controls locate what sets this limit. *Recurrence is not needed.* The single-step controls fill the Kanizsa hexagon as much as the looped transformer (looped transformer 0.79, hGRU 0.98 at 7 px) and stop at the same gaps (18 px: 0.22 and 0.29, against 0.56 and 0.94 with closing lines). What they lack is selectivity: they also label the hexagon between full discs (0.48 and 0.64), the hGRU between rotated inducers (0.31), and the looped transformer the whole blank page (1.00). Over 40 steps the looped transformer removes the fill between full discs (0.00) and reduces the spread on a blank page to 0.63 while keeping the aligned fill; in the fGRU trained with the weighted loss, by contrast, a label that fills the Kanizsa hexagon at step 40 (0.68; discs 0.19) floods the image by step 60 (Fig. 16). *The kernel's reach does not set the limit.* hGRUs with 3×3 and 15×15 finest-level kernels (12 and 60 px) both stop between 14 and 18 px (18 px: 0.14 and 0.02, against 0.51 and 0.15 with closing lines), as the single-step controls do, consistent with the bridgeable gap's independence from the receptive field in the battery (Section 7.1, Fig. 22). *The backbone does.* On ViTDet features the fGRU's fill does not depend on the gaps: at 29 px it labels 0.78 of the Kanizsa hexagon, where a closing line gets 0.51, and it fills between full discs (0.83 at 7 px), so it labels the region the inducers surround whether or not the gaps are closed. *Contrast polarity changes the outcome.* With white inducers on black, the looped transformer labels most of the hexagon whatever the inducers (Kanizsa 0.88, rotated 0.72, discs 0.59, blank 1.00); only containment remains: at 14 px, 0.09 of the strip just outside the sides is labelled with aligned inducers and 0.56 with rotated ones.

The circuit of Mollard et al. (2026) varies most between runs. With the paper's settings it segments COCO objects (0.48) but labels nothing on any display, not even the filled hexagon (0.00), so it cannot be tested; with reversed contrast it labels the filled hexagon (0.65) but not the Kanizsa one (0.01). With the reference code's penalty weight it labels the filled hexagon (0.68) and the Kanizsa one only at the smallest gaps (0.08 at 7 px, closing line 0.20). The 3×3 and 15×15 runs, trained identically apart from the kernel, fill 0.70 and 0.31 at 7 px, the label growing over steps (gaps up to 11 px, 3×3: 0.15, 0.71 and 0.84 after 20, 40 and 60 steps); the 15×15 run leaves full discs and rotated inducers empty (0.00 and 0.00), the 3×3 run does not (discs 0.45). The 9×9 run labels nothing, also at iteration 12,000, before its penalty rose (filled hexagon 0.02). With one run per setting, these differences cannot be attributed to the settings. The looped transformer's selectivity is not reproducible either: a second run, identical apart from the random seed, labels the Kanizsa hexagon at 7 px gaps (0.96) but also the hexagon between rotated inducers (0.98) and full discs (0.98), and still 0.78 of the Kanizsa hexagon at 29 px; like the fGRU on ViTDet features, it labels the region the inducers surround (Table 1).

Trained on COCO alone, most of these circuits do not segment a thin drawn outline, so we also trained the looped transformer and the 15×15 hGRU, each with 40 steps and with one, on batches half of which were drawn polygons (filled, outlined, or outlined with inducers at the vertices, aligned or rotated; never a Kanizsa



Figure 16: The cued label (yellow) step by step on the large hexagon with 7 px gaps (support ratio 0.90), cue (orange) placed by us at the centre, the point the circuit is asked about. Left, Kanizsa display after t steps; right, rotated inducers, full discs and the gaps closed by a solid line at the network's read-out step (40, or 1 for the single-step controls, which are trained and read out after one step).

figure without its outline). We counted a polygon as seen when at least half of its interior and at most a tenth of a band just outside its sides are labelled, over $n = 3$–8 sides, four sizes, support ratios 0.50–0.90 and both polarities. These circuits see nearly every drawn outline (looped transformer 99%, hGRU 98%) and most Kanizsa figures (87% and 70%), but also many figures whose inducers are all rotated by 15–180° (27–57% and 9–37%; full discs 10% and 8%, blank page 0%); after one step they see rotated inducers more often (41–65% and 49–77%). Recurrence reduces this response to arrangements that specify no polygon without removing it.

Within the limits of single training runs of a reimplementation that differs from the original in how the cue enters, in its training images and in its accuracy, circuits trained to spread a cued label across objects label the surface of a Kanizsa figure, and in one of two runs not its controls, only where the gaps are a few pixels wide, under one patch of the battery; a single step does as much, any horizontal kernel from 3×3 to 15×15 stops at the same gap, and a ViT backbone fills regardless of the gaps. What bridges these gaps are the feed-forward features; recurrence reduces, without removing, the label in regions that specify no shape. At the proportions of standard Kanizsa figures, these circuits treat the illusory surface as empty space.

6.6 Feedback networks on frozen fronts

Table 9 lists every selected network of the family, and Figs. 17 and 18 show the time course of the main ones. In all of them the cue sits at the figure's centre, placed by us; what is measured is the network's answer to "what object is here?", so a filled square means that the network, told where to look, labels the region bounded by the inducers. The pointing is not what produces the label, however: with the cue removed altogether (a zero cue map, a condition never seen in training), the networks label the line square as completely as before

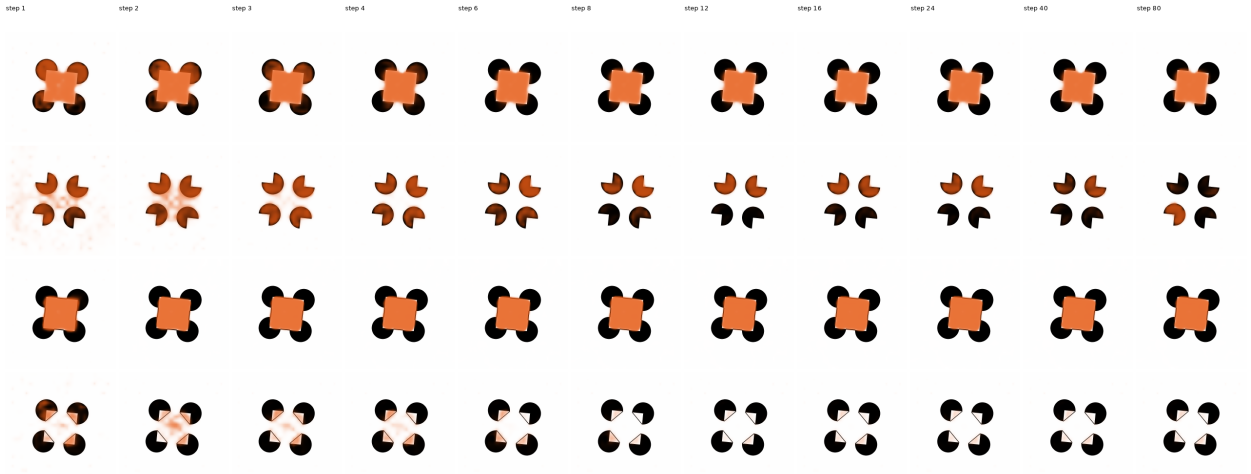


Figure 17: SAM 3 with its last six blocks looped (seed 0), step by step. Rows: the Kanizsa square (support ratio 0.7), the inducers turned by 180°, the gaps drawn as real lines, and the inducers closed into discs by hairlines; columns: recurrent steps 1 to 80; orange, the network's probability that a pixel belongs to the cued object; dashed blue, the virtual square's sides, where an illusory contour would be; the cue, placed by us at the centre of every display, is the point the network is asked about, so the network is not finding the square, it is answering what lies at the cued point. The single pass through the looped blocks (step 1) labels the Kanizsa square and spreads onto the rotated inducers and the closed discs; the loop removes the label from the controls within four to eight steps and keeps it on the Kanizsa square. Animated versions for every network are in the supplementary material.

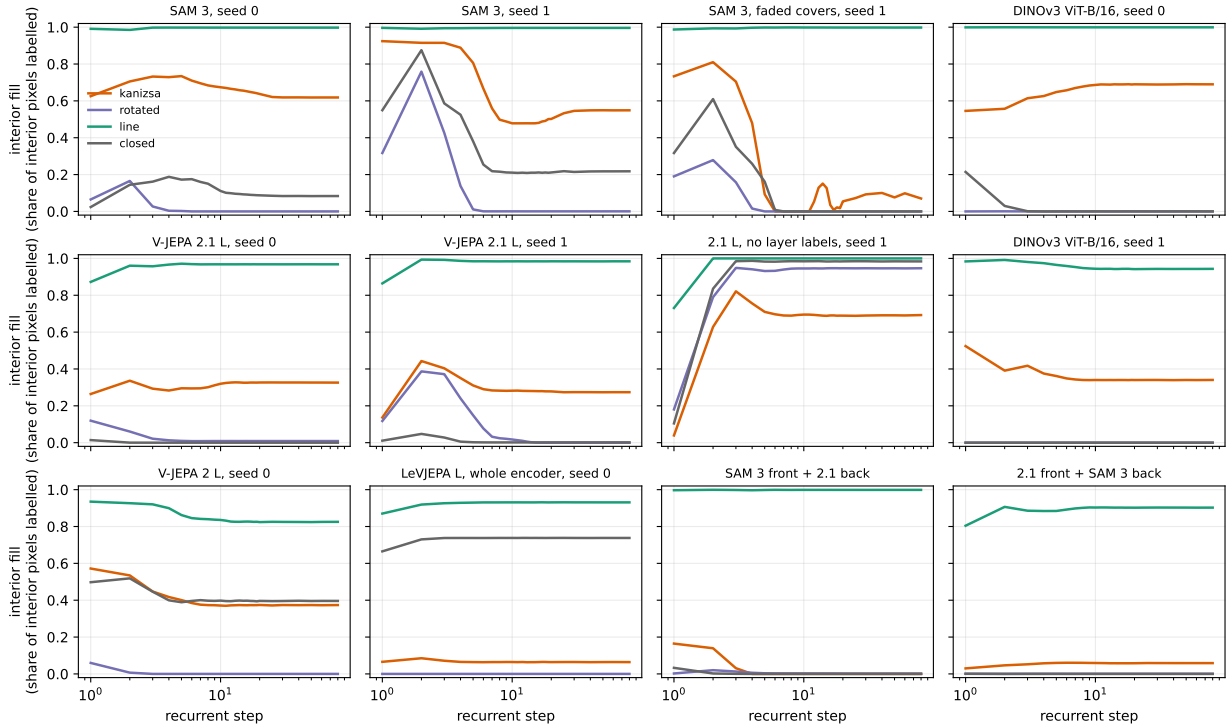


Figure 18: Interior fill (share of the figure's interior pixels labelled as the cued object, 0–1; the cue is at the figure's centre in every display) against recurrent step (log axis) for the single displays of the battery, mean over support ratios and orientations, for nine networks: the two SAM 3 seeds and SAM 3 trained with faded covers (top), V-JEPA 2.1 L and its no-layer-label control (middle), V-JEPA 2 L, the LeVJEPA whole encoder and the SAM 3-front hybrid (bottom).

Table 1: All trained cued segmentation circuits: IoU on cued COCO validation objects (all, and objects under 5% of the image), and the share of the large hexagon's interior labelled at the read-out step with 7 px gaps (filled hexagon, Kanizsa, rotated inducers, full discs, thick corner segments, gaps closed by a solid line, blank page) and with 18 px gaps (Kanizsa, solid line); mean over four orientations. Kernel sizes are those of the finest recurrent level. †Lipschitz penalty rose after iteration 12,000; two rollbacks. ‡Trained before the C-RBP corrections of Appendix J.

	COCO IoU		fill, 7 px gaps							fill, 18 px gaps	
network	all	small	filled	Kanizsa	rotated	discs	corners	line	blank	Kanizsa	line
hGRU, 5×5 (paper settings)	0.48	0.41	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.01
hGRU, 5×5, penalty weight 100	0.50	0.42	0.68	0.08	0.00	0.00	0.00	0.20	0.00	0.00	0.01
hGRU, 3×3	0.55	0.44	0.64	0.70	0.13	0.45	0.37	0.75	0.01	0.14	0.51
hGRU, 9×9 †	0.28	0.18	0.06	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.01
hGRU, 15×15	0.54	0.44	0.54	0.31	0.00	0.00	0.19	0.24	0.00	0.02	0.15
hGRU, 5×5, one step	0.55	0.45	0.74	0.98	0.31	0.64	0.10	0.98	0.00	0.29	0.94
fGRU (paper settings) ‡	0.43	0.35	0.28	0.28	0.01	0.10	0.00	0.34	0.00	0.01	0.27
fGRU, weighted loss ‡	0.50	0.35	0.40	0.68	0.00	0.19	0.03	0.69	0.00	0.07	0.27
fGRU on ViTDet ViT-B	0.50	0.40	0.86	0.55	0.06	0.83	0.43	0.64	0.00	0.92	0.54
looped transformer	0.46	0.32	0.99	0.85	0.01	0.00	0.02	0.86	0.63	0.04	0.74
looped transformer, second seed	0.45	0.30	0.76	0.96	0.98	0.98	0.87	0.97	0.00	0.83	0.93
looped transformer, one step	0.51	0.38	0.99	0.79	0.00	0.48	0.66	0.83	1.00	0.22	0.56

(0.94–1.00), nothing between rotated inducers, and the Kanizsa square at 0.61 (SAM 3 seed 1; cued 0.55), 0.18 (seed 0; cued 0.62), 0.32 and 0.15 (2.1 L; cued 0.33 and 0.28) and 0.33 (V-JEPA 2 L; cued 0.41), with the label confined to the figure (0–8% of the image) and the LeVJEPA whole encoder unchanged (closed discs 0.73, Kanizsa 0.07, cued or not). The cue thus matters for two of the six networks and never for the selectivity between displays.

Untouched encoders already differ. Before any training of ours, one linear read-out per token, fitted on clip frames to mark the object and applied to the line drawings, separates the inside of the line square from its outside in every encoder (inside minus outside: SAM 3 0.60, V-JEPA 2.1 L 0.46, V-JEPA 2 L 0.24, LeVJEPA 0.22), and for the Kanizsa square the same read-out gives 0.14 (SAM 3), 0.31 (2.1 L), 0.03 (2 L) and 0.07 (LeVJEPA); only SAM 3's front separates the Kanizsa square from the closed discs (-0.04 for the discs, 0.18 apart), and in 2.1 L the four discs and the square between them are marked as one blob (0.25 for the discs). Measured block by block (Section 7.5), the separation appears in the middle of the 2.1 L and SAM 3 encoders and declines in their last blocks, and stays low throughout V-JEPA 2 L and LeVJEPA.

What stage 2 makes of it. Table 9 gives every selected network. Trained on plain unity clips, V-JEPA 2 L labels the Kanizsa square (0.40–0.77 across three seeds) and nothing between rotated inducers, but also the closed discs (0.26–0.70 at the selected checkpoints), and the no-loop control, the same network trained without feedback, matches or exceeds it on every display (Kanizsa 0.77 and 0.73, closed 0.54 and 0.42): training with recurrence was not needed for any test in the battery. V-JEPA 2.1 L and B fit the clips better and faster and transfer worse: after 4,000 iterations the 2.1 L network labels neither the line square (0.40) nor the Kanizsa one (0.00), and its checkpoints lose the transfer as the clip loss falls. The outline clips repair this, and with them 2.1 L passes the three transfer criteria on both seeds (line 0.97 and 0.98, Kanizsa 0.33 and 0.27, rotated 0.01 and 0.00, closed 0.00 and 0.00), though with a small Kanizsa fill and a new failure: it fills the square between drawn corners turned outward (0.25 and 0.81). The texture clips and the combined clips give similar Kanizsa fill with more fill between turned corners. LeVJEPA's frozen front cannot be trained for the task at 224 px (validation IoU below 0.1 after both stages); its whole encoder learns real contours (line 0.93 and 0.97) and then fills the closed discs (0.74 and 0.64) and not the Kanizsa square (0.06 and 0.01; Fig. 35): the reverse of an illusory contour. DINOv3 on plain unity clips fills nothing after one stage, everything with the layered objective, and after two stages labels the Kanizsa square in one of three seeds (0.35; the others 0.00 and 0.20) but no plain drawn outline.

SAM 3 and DINOv3 pass the battery; SAM 3 passes the selection gate on both seeds, DINOv3 on one. With the outline clips, both SAM 3 seeds pass every transfer criterion (line 1.00 and 1.00, Kanizsa 0.62 and 0.55, rotated 0.00 and 0.00, closed 0.08 and 0.22, Kanizsa minus closed 0.53 and 0.33; nothing

between turned corners), the only network of the family, and of the paper, to do so on two of two seeds. The two training seeds agree in their stage-1 fit (validation IoU 0.70 and 0.72 at iteration 3,000; 2.1 L 0.41). Faded covers do not improve it: seed 0 is unchanged (0.54) and seed 1 loses the fill (0.07), so that SAM 3 passes on three of its four stage-2 runs; faded covers changed nothing on 2.1 L either (0.30 and 0.33). DINOv3, run under this same recipe after the SAM 3 result (its earlier runs used the plain clips, where it reached 0.35, 0.00 and 0.20 across seeds), passes the battery on both seeds: Kanizsa 0.69 and 0.34, rotated 0.00 and 0.00, closed discs 0.00 and 0.00, full discs 0, nothing between turned corners, nothing outside, and the lowest label on the inducers' bodies of any network that fills (0.07 and 0.03 on the Kanizsa display, 0.06 and 0.01 on the rotated one); without any cue they label the Kanizsa square at 0.52 and 0.32 (cued 0.69 and 0.34), the rotated inducers and closed discs at 0. Split by support ratio, seed 0 fills 0.46 at 0.5 and 0.92 at 0.7, the steep dependence of SAM 3, whereas seed 1 fills 0.32 and 0.36, a third of the square at either support ratio. The two seeds differ in what else they do. Seed 0 labels 0.34–0.80 of the interior of the validation set's scrambled outlines, drawn outlines whose sides are cut in half and each half turned and shifted, at every one of its sixteen checkpoints, against the bound of 0.25 that every other selected network met, so under the paper's rule it counts as a failed network; seed 1 meets the bound from iteration 1,000 on (0.14 at the chosen checkpoint) but fills a plain drawn outline only at 0.57. Their time courses differ too: on seed 0 the Kanizsa fill rises from 0.55 at step 1 to 0.68 at step 8 while the closed discs are pruned from 0.21 to 0, and cutting the feedback lowers it to 0.51, the one case in which the loop adds to the figure; on seed 1 the fill falls from 0.52 to 0.34 and cutting the feedback raises it to 0.53, as in SAM 3. Under the identical recipe, then, a self-distilled image encoder gives Kanizsa-minus-closed margins of 0.69 and 0.34 against SAM 3's 0.53 and 0.33; what separates SAM 3 is that both its seeds pass the scrambled-outline gate and fill a plain outline completely.

The loop is a veto, not a generator. Figure 18 and the contact sheet of Fig. 17 show what the recurrence does. In every SAM 3 run the single pass through the trained looped blocks (steps 1–2) already labels the Kanizsa square (0.63–0.71 on seed 0, 0.92–0.92 on seed 1) and also the rotated inducers (seed 1: 0.76 at step 2) and the closed discs (seed 1: 0.88); over steps 3 to 8 the loop removes the label from the controls (rotated 0.00 and 0.00, closed 0.15 and 0.22 at step 8) and keeps about two thirds of it on the Kanizsa square (0.68 and 0.50, flat to step 80). Cutting the feedback from the first step therefore gives a *higher* Kanizsa fill (0.65 and 0.93), with the controls filled as well. V-JEPA 2.1 L shows the same shape at a third of the amplitude (step-1 Kanizsa 0.26 and 0.14, rotated 0.12 and 0.12, both gone by step 8), and V-JEPA 2 L on plain clips prunes a step-1 label that covers Kanizsa and closed displays alike. Across the four SAM 3 runs the step-1 proposal is consistent (0.63–0.92) and what varies is how much of it survives the veto (0.62, 0.55, 0.54 and 0.07 at step 80); in the run that fails, the loop removes the Kanizsa label together with the controls. The recurrence thus performs the figure/non-figure decision; it does not create the fill, and it is not needed for it.

The layer labels are required. Without the layer-label loss, no checkpoint of either 2.1 L seed keeps its false fill under 0.25 (0.34–0.96), the selected checkpoints fill every display (seed 1: rotated 0.95, closed 0.98, Kanizsa 0.69), and the time course (Fig. 18) shows the loop building an indiscriminate fill rather than vetoing one. The Kanizsa selectivity of these networks is conditional on being trained to label hidden and occluding layers, not on the clips alone.

Neither half of SAM 3 transplants. 2.1 L's front with SAM 3's looped blocks gives no Kanizsa fill (0.06; controls 0), less than plain 2.1 L; SAM 3's front with 2.1 L's looped blocks gives none either (0.00), and that network over-fills the scrambled validation contours at 13 of its 16 checkpoints, so that selection is forced to a barely trained one. Either half lifts 2.1 L's stage-1 fit (0.66 with SAM 3's front, 0.58 with its back, against 0.41), but the fill appears only with the matched encoder, front and looped blocks pretrained together. With one run per hybrid and an adapter trained for 4,000 iterations, this is a failure to transplant, not proof that it cannot be done.

Support ratio, and the inducers' bodies. Split by support ratio, the SAM 3 fill is near complete at 0.7 (0.95 and 0.96) and weak at 0.5 (0.28 and 0.14); the probes of Appendix L show it absent at 0.41 for every figure size, weak at 0.5 and saturated by 0.6, and show that the single pass fills at every support ratio and the loop removes the low-support fill. Human illusory contours grow in clarity with support ratio (Shipley & Kellman, 1992); the network switches. At support ratios of 0.9 and above, where the inducers nearly close the figure, a surface probe on the last looped block also reads a filled region between rotated inducers (0.51 and 0.68 at 0.90 and 0.95) and between full discs. Finally, the networks put some of their label on the

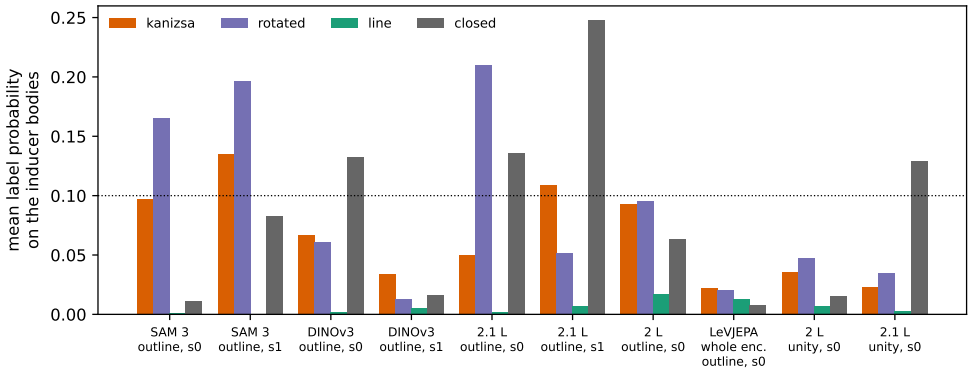


Figure 19: Label on the inducers' own bodies: the read-out's mean probability that a pixel belongs to the cued object over the inducer pixels more than 6 px from the figure's sides, 0–1, at step 80, per network and display; dotted line, 0.10.

inducers themselves (Fig. 19): 10–14% of the inducer bodies in the Kanizsa display and 17–20% in the rotated display, where nothing is filled between them, so on that display the label marks the inducers as objects; 2.1 L does the same on the rotated display in one seed (21%) and on the closed discs in the other (25%), and the LeVJEPA whole encoder and the V-JEPA 2 network trained on plain clips stay at or under 5% everywhere.

What this family adds. Across five encoders, four clip sets, three controls, two hybrids and one to three seeds per cell, two encoders label a Kanizsa figure selectively under the recipe with outline clips and layer labels, SAM 3 (trained with object masks) and DINOv3 (self-distilled; one seed over the bound on scrambled outlines), and in both the single pass already carries the label; the recurrence removes it from the non-figures, and in one DINOv3 seed adds to it on the figure. What that implies is taken up in the Discussion.

7 Results by probe

Section 6 read each family against the probes it was built for. This section reads each probe across the families: the gap and the receptive field for every classifier and circuit, the empty cells between the inducers for every network with spatial feature maps, the surface fill under a cue for the cued circuits and the feedback networks together, the step dynamics of every recurrent family on one axis, and where in depth the relevant signals appear. Table 2 gives the answer of each family to each question in one place; Table 3 and Table 9 hold the numbers behind every network mentioned.

7.1 Patch size, receptive field and the empty gap

A gap in a contour is defined in pixels, but each network sees it through its own sampling grid. A vision transformer cuts the image into $P \times P$ patches ($P = 16$ for ViT-B/16, 14 for DINOv2-S/14) and starts from one token per patch; empty patches share the same content embedding and differ only in their positional embedding, so information about the missing contour can reach them only through attention. A convolutional network has no such grid at its input, but each unit integrates a bounded region that grows with depth. We summarize it by ERF_{50}, a summary derived from the effective receptive field of Luo et al. (2016): the gradient of the central unit's activation (summed over channels; for the CNNs the last convolutional map, for recurrent circuits the recurrent state at the centre, for the deep hierarchies and DN stacks the first recurrent layer after the top-down pass) with respect to 16 white-noise images, and the radius around the gradient centroid that contains half of its absolute mass. It is a summary of how far input influences one unit, typically much smaller than the theoretical receptive field, and not a guarantee that every unit integrates that region on our stimuli. Horizontal recurrence enlarges the receptive field over iterations: in the stride-4 recurrent layers, each fGRU iteration applies two 9×9 horizontal convolutions in sequence (32 px of additional radius from these kernels alone; deeper layers of the hierarchies use smaller kernels at coarser strides; the 3×3 suppression gate

Table 2: What each family showed, by question. Sections and tables in brackets hold the numbers; "–" means the probe does not apply to that family.

family	Kanizsa over rotated inducers	anything on the illusory edge	does recurrence or depth add it	where the signal is
feed-forward classifiers (CNNs, ViT, DINOv2, MAEs) [6.1]	small, seed-consistent only in ViT-B/16 (index $\leq$ 0.1); accuracy $\leq$ 0.4	no: contour probe $\leq$ a few % of a real line; MAEs draw the edge only when inducers are hidden too	no: per-layer probe peaks late and is flat afterwards	inducer tokens (activation patching); an interior patch at blocks 9–12 of ViT-B/16
looped transformers, recirculation [6.2]	as the frozen / fine-tuned ViT	no	no: loops add depth, not closure; recirculation $\leq$ 0.02	as above
fGRU-H, γ-net, deep and DN stacks [6.3]	$\leq$ 0.07	gap similarity rises over iterations in three circuits	no: single-iteration controls match in 5 of 6	within the receptive field; gaps bridged are smaller than ERF_{50}
contour tracers [6.4]	link aligned pieces (0.55–0.75), rotated $\leq$ 0.06	$\leq$ 1% of the missing contour labelled	no: twice the iterations trace less	a few px beyond training statistics
cued circuits (Mollard reimplementation) [6.5]	fill like a closed figure only at gaps $\leq$ 11 px; one of two runs selective	– (surface label, not edge)	no: single-step controls fill as much; 40 steps only prune non-figures	the backbone's features; cue gates the looped transformer, not the hGRU/fGRU
feedback networks on frozen fronts [6.6]	SAM 3 yes on 3 of 4 runs (0.55–0.62 vs 0); DINOv3 0.69 and 0.34 vs 0, one seed over the scrambled-outline bound; V-JEPA 2.1 0.27–0.33; others no or the reverse	– (surface label); fill absent at support 0.41, saturated by 0.6	no: the first pass proposes, the loop vetoes (and adds in DINOv3); no-loop training passes the same battery	the encoder's middle blocks; outline clips and the layer-label loss; SAM 3 and DINOv3 features

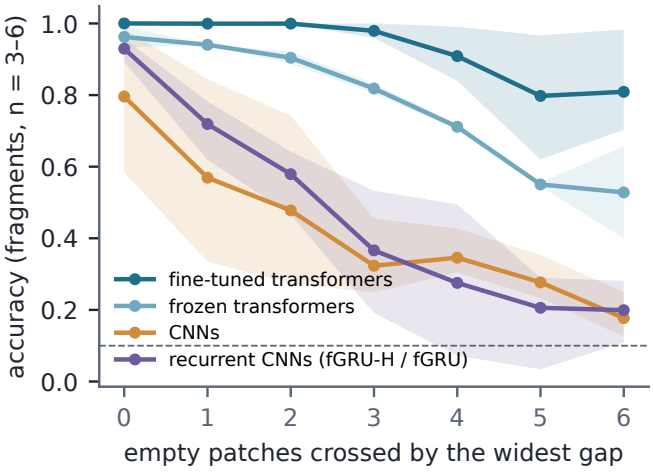


Figure 20: Accuracy on fragments ($n = 3$–6, all gap sizes valid) against the largest number of empty patches crossed by any gap. Lines: group means; bands: range across networks.

and instance normalization add further, partly global, dependencies). We ask three questions: does failure track the patch grid, does it track receptive-field size, and what happens inside the empty part of the gap?

7.1.1 Accuracy falls with the number of empty patches

Plotted against the number of empty patches a gap crosses, accuracy separates the network families (Fig. 20); we did not test whether this count predicts accuracy better than gap size. Averaging, for 3–6-sided polygons,

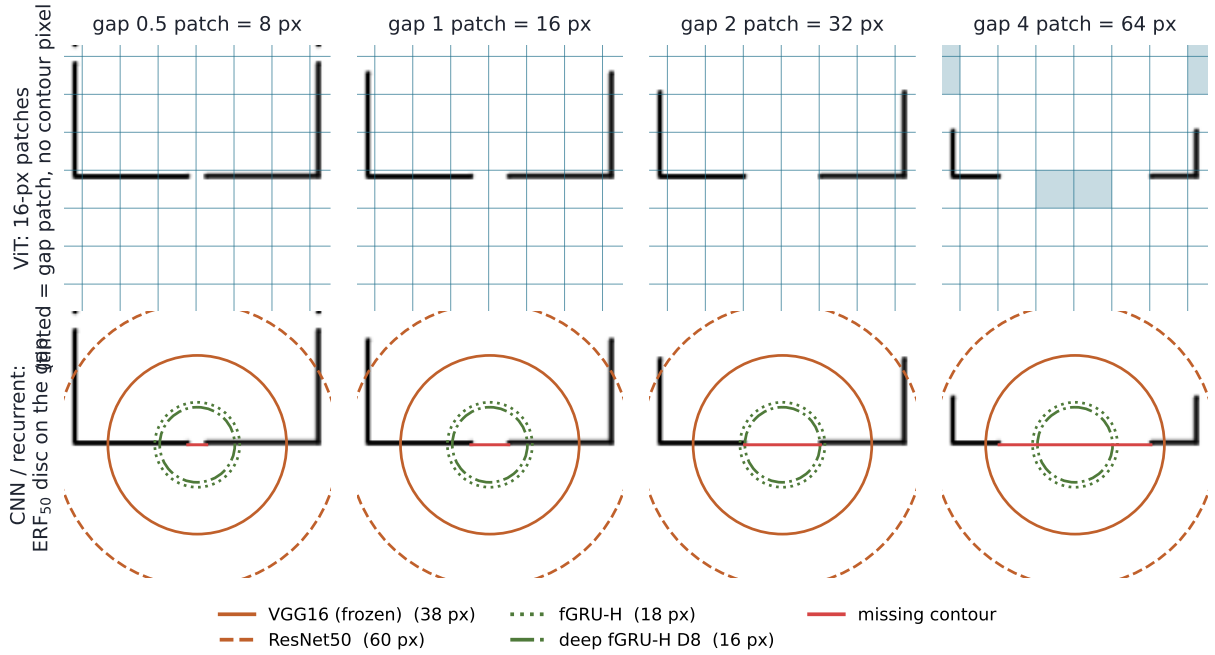


Figure 21: The same mid-gap fragment at four gap sizes, zoomed on one gap. Top: the 16 px patch grid a ViT-B/16 tokenizes; tinted patches are crossed by the missing contour and contain no contour pixels. Bottom: the measured 50% effective receptive field (ERF_{50}) of four networks, centred on the gap (red: missing contour). Even the 64 px gap lies inside the ERF_{50} of two of the four networks shown.

over gaps that cross three or more empty patches (bins of 3, 4, 5 and 6 or more empty patches weighted equally), fine-tuned transformers score 0.78–0.99, while the CNNs and the representative fGRU-H/fGRU circuits score 0.10–0.39.

7.1.2 The bridgeable gap is far smaller than the receptive field

For each network we define a critical gap g^*, the gap at which the mid-gap closure index falls to half its value at the smallest gap (linear interpolation in log gap; a network whose index at the smallest gap is below 0.05 would be excluded, which none is), and compare it with the measured ERF_{50} for CNNs and recurrent circuits and with the patch size for transformers (Figs. 21, 22). A transformer's attention can span the whole image, and the analogue of receptive-field size proposed by Dosovitskiy et al. (2021), the mean attention distance, varies across heads and layers; we therefore use the patch size, the resolution at which the gap is tokenized, as the reference. We compute the index on the fixed set of 3- to 6-sided polygons, for which every gap size is valid; pooling all classes would change the mix of classes with gap size, because larger gaps exceed the edges of many-sided polygons. For CNN and recurrent networks other than the single-iteration controls, g^* lies between 6 and 50 px; it marks where the index halves, and a smaller residual effect can persist beyond it. For 9 of 10 CNN and recurrent networks it is smaller than ERF_{50} (median ratio 0.44; 14 of 16 including the single-iteration controls), and its rank correlation with ERF_{50} is only +0.33; the exception, excluding single-iteration controls, is the fGRU-H with a 1×1 readout, whose readout adds no spatial convolution after the recurrence and which has the largest fragment closure index of any recurrent circuit (+0.63). In 9 of 10 CNN and recurrent networks (excluding single-iteration controls) the index therefore halves at gaps whose full length fits inside the measured ERF_{50}, a radius; a smaller effect can remain at larger gaps. ERF_{50} is measured on noise images, and for the deep hierarchies at the first recurrent layer, so this comparison is descriptive: it shows that the gap lies within the region that influences a unit, not that the network uses that region for the task (Section 7.1.3). Transformers behave differently: 3 of 6 keep more than half their mid-gap index up to the largest gap tested (four patches), and the others halve at 3.4–3.6 patches. Mid-gap fragments

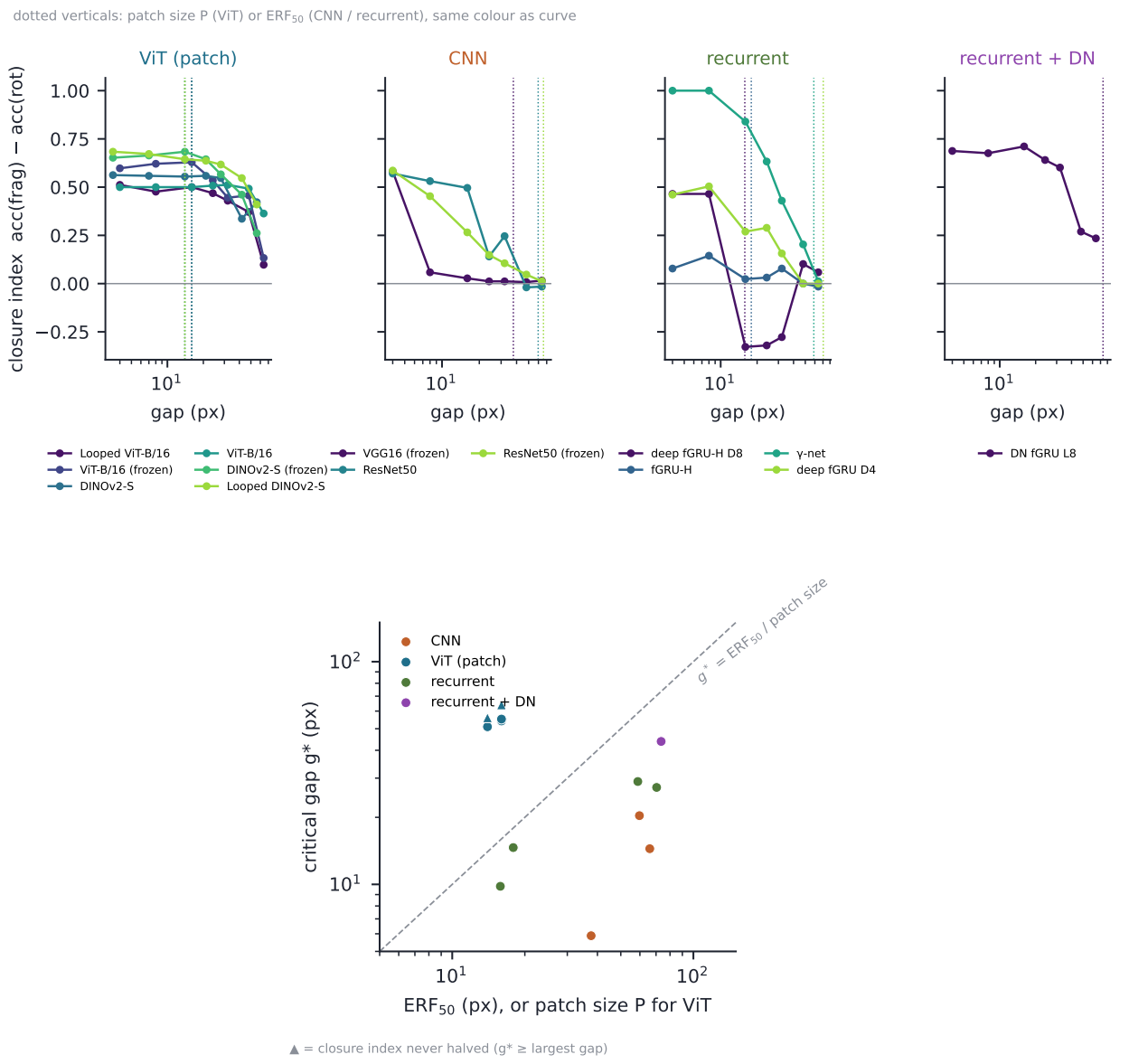


Figure 22: Top: mid-gap closure index (3- to 6-sided polygons) against gap size, by family, for a representative subset of networks (all networks in Table 4); dotted verticals mark each network's ERF_{50} (patch size for transformers). Bottom: critical gap g^* (closure index halved) against ERF_{50} or patch size for the same networks; dashed line, equality; triangles, index never halved within the tested range. Over all networks (Table 4), 9 of 10 CNN and recurrent networks lie below the diagonal excluding single-iteration controls (the exception is the fGRU-H with a 1×1 readout) and 14 of 16 including them.

keep the corners, which these networks read directly (Section 6.1.2), so this index overstates how far they bridge gaps.

7.1.3 What happens inside the gap

A network that completes a contour should represent something at the location of the missing piece. We measured this directly. For mid-gap fragments of 3- to 6-sided polygons, their rotated controls and the matching complete outlines (up to 48 image triplets per gap size), we took the spatial feature maps of each network (patch tokens after each transformer block, before the final normalization; the four ResNet50 stages; for recurrent circuits, the state of the layer the classifier reads, at every iteration) and selected the cells that

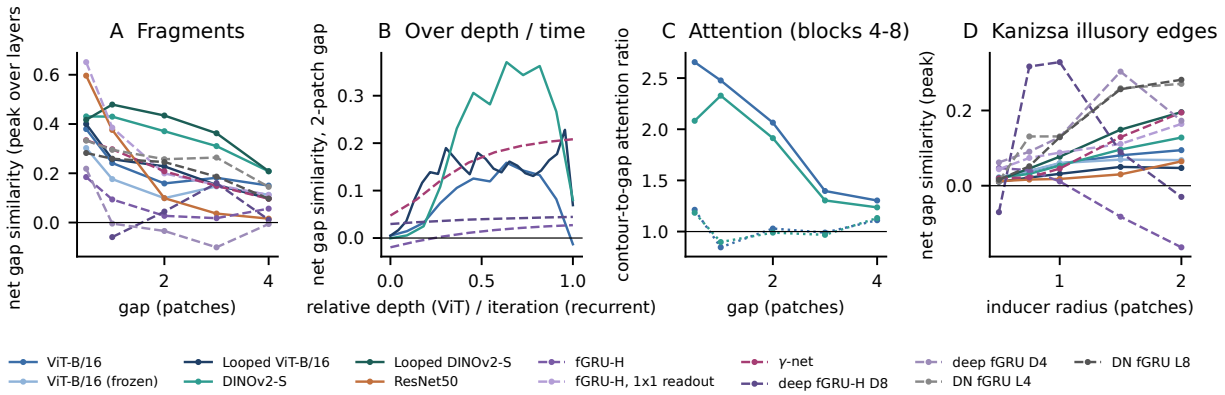


Figure 23: Inside the gap. Net gap similarity: how much more the empty feature-map cells on the missing contour resemble the same cells in the complete outline for the stimulus than for its rotated control (cosine similarity of the concatenated cell features), minus the same difference on background cells at least 32 px from any contour pixel. (A) Peak over layers (iterations for recurrent circuits) against gap size. (B) Across relative depth (transformers, solid) or iterations (recurrent circuits, dashed) at a 2-patch gap. (C) Attention from contour tokens to the empty gap tokens relative to empty tokens at the same distance from the drawn contour, blocks 4–8; solid, stimulus; dotted, rotated control. (D) As A, for the illusory edges of Kanizsa figures against rotated inducers.

the missing contour passes through and that contain no contour pixels in either the stimulus or the control. The *gap similarity* is the closure measure $\bar{C}$ of Kim et al. (2021) restricted to these cells: the cosine similarity of their concatenated features to the same cells in the complete outline, minus the same similarity for the rotated control. Because late transformer tokens carry global class information, we subtract the same index computed on background cells at least 32 px from any contour pixel (*net gap similarity*; Fig. 23). This is a similarity measure: it shows whether the gap cells become more like the drawn contour, not that they encode its position and orientation, and it does not show whether the classifier uses them. Requiring cells to be empty in both images removes most half-patch gaps for transformers (8 and 18 of 48 triplets remain at half a patch and one patch for ViT-B/16; 47–48 from two patches on), so we concentrate on gaps of one patch and more. For transformers we also measured attention in both directions: from contour tokens (tokens containing contour pixels) to the empty gap tokens, relative to other empty tokens at the same distance from the nearest contour pixel, and from the gap tokens to the contour tokens adjacent to the gap (the contour ends in the stimulus, the rotated pieces in the control).

Similarity to the drawn contour is high only near it. In transformers and ResNet50, net gap similarity falls with gap size (Fig. 23A): transformers reach 0.18–0.48 at one patch and 0.10–0.21 at four, and ResNet50 falls from 0.38 at one patch to 0.10 at two and 0.02 at four, although its ERF_{50} spans far larger gaps. In transformers the effect peaks in the middle blocks (in the looped ViT-B at a two-patch gap, in the coda; see below) and is mostly gone in the last block (net gap similarity at most 0.11 for gaps of one patch or more). Recurrent circuits are more varied: most fall with gap size, several (like the transformers) with a small rebound at three patches (Fig. 23A).

Attention targets the gap; gap tokens prefer the aligned contour only modestly, and only when the gap is small. In blocks 4–8 (executed blocks counted from 1), contour tokens attend 2.5 (ViT-B/16) and 2.3 (DINOv2-S) times more to empty tokens on a one-patch gap than to equally distant empty tokens, against 0.8 and 0.9 for the rotated control (Fig. 23C); at four patches the ratios are 1.3 and 1.2 (controls 1.1 and 1.1). This direction, however, updates the contour tokens with what is in the gap. In the direction that would carry the contour into the gap, gap tokens put 0.16 (ViT-B/16) and 0.23 (DINOv2-S) of their attention on the adjacent contour ends at one patch, against 0.06 and 0.09 on the rotated pieces in the control (over triplets in which both images have contour pixels adjacent to the gap; the rotated pieces leave none in 17% and 15% of one-patch cases), a two- to three-fold difference in total attention mass. Part of it is token count: the aligned contour leaves 5.6 contour tokens adjacent to a one-patch gap against 3.1 for the rotated pieces (ViT-B/16), and per adjacent token the preference is 1.6× (ViT-B/16) and 1.4× (DINOv2-S).

At four patches the mass is 0.23 and 0.26 against 0.21 and 0.23 (1.0× and 1.1× per token): a much smaller preference. Gap tokens therefore attend somewhat more to aligned contour ends than to misaligned pieces when the gap is a single patch, and only slightly when it is four; this does not test whether the gap tokens encode the contour's position and orientation.

Iteration propagates little, and inconsistently. If completion were a dynamic process, repeated processing should increase net gap similarity, especially at larger gaps. In the looped ViT-B, net gap similarity at a two-patch gap reaches 0.19 at the end of the first pass through the shared core, stays between 0.13 and 0.18 over the remaining passes (0.14 at the end of the last), rises to 0.23 at block 23 in the coda and falls to 0.07 after the last block (Fig. 23B): further passes through the core do not raise it. Among recurrent circuits, iterations raise net gap similarity in the fGRU-H (0.02 to 0.09 at one patch, -0.01 to 0.02 at three), more in the γ-net (0.05 to 0.21 at two patches, 0.03 to 0.15 at three), and in the four-layer DN fGRU stack at gaps of one patch and more (0.26 to 0.28 at one patch, 0.14 to 0.26 at three); it falls in the eight-layer DN stack (0.19 to 0.04 at three patches), remains negative or near zero at every gap in the deep fGRU, with small changes in either direction (-0.01 to -0.02 at one patch), and changes little in the fGRU-H with a 1×1 readout and the deep fGRU-H. The circuits in which iterations increase net gap similarity gain no closure index from recurrence: the single-iteration controls of the fGRU-H, the γ-net and the DN stacks have equal or larger fragment closure indices (Section 6.3); the one circuit whose index rises with recurrence, the deep fGRU, shows small increases at some gaps, but its net gap similarity remains negative at gaps of one patch or more. Separately trained controls cannot show that the eight-iteration networks do not use the gap activity; that would require intervening on the trained network.

Illusory edges. On the illusory edges of Kanizsa figures (inward against rotated inducers), transformers show little: net gap similarity peaks at 0.20 (ViT-B/16: 0.09) and is at most 0.06 in the last block; ResNet50 peaks at 0.06 (Fig. 23D; at least 28 triplets per radius). Several recurrent circuits reach values like those on real gaps: 0.27 for the four-layer DN fGRU stack at the largest inducers, where the illusory edge is short, and 0.33 for the deep fGRU-H at an inducer radius of one patch, at its coarse 7×7 readout layer (from 42 usable triplets), where the inducers' straight notch edges lie within one or two feature cells of the illusory edge. No recurrent circuit has a Kanizsa closure index above +0.06 (Section 6.3), so this similarity has no demonstrated benefit for recognition.

7.1.4 Little evidence of boundary completion

By boundary completion (Grossberg & Mingolla, 1985) we mean here a process that extends a contour from its ends into the empty gap over successive layers or iterations until the two ends meet. These measurements fit a simple picture. Near the drawn contour, overlapping receptive fields and attention to neighbouring tokens make empty gap cells resemble the drawn contour; farther away the resemblance declines (though in transformers it remains above zero at four patches), and we find no evidence of a process that carries a contour end across the empty region to meet the other end. Attention gives transformers the connectivity for such a process, and at a one-patch gap the gap tokens do attend more to aligned contour ends than to misaligned pieces (about 1.5 times per adjacent token); at four patches this preference falls to about 1.0–1.1 times per token, and looping does not raise the similarity. Horizontal and top-down recurrence gives fGRU circuits the dynamics, and in the γ-net it does increase similarity in the gap over iterations, but this brings no gain in the fragment closure index over a single iteration. Models of human contour integration and completion differ in exactly this respect: association fields link elements by their relative position and orientation (Field et al., 1993), and boundary-completion models complete a contour only when aligned edge signals are present on both sides of the gap (Grossberg & Mingolla, 1985). Our measures do not test position or orientation coding in the gap, nor what the classifier uses, and cannot rule out that another read-out would show completion; Section 7.2 applies two that read the empty cells directly. They are, however, consistent with the behavioural results: in 9 of 10 CNN and recurrent networks the closure index halves at gaps inside the measured ERF_{50}, looping and recurrence add little to the similarity in the gap and nothing consistent to the closure index, and Kanizsa figures are mostly misclassified (accuracy at most 0.39).

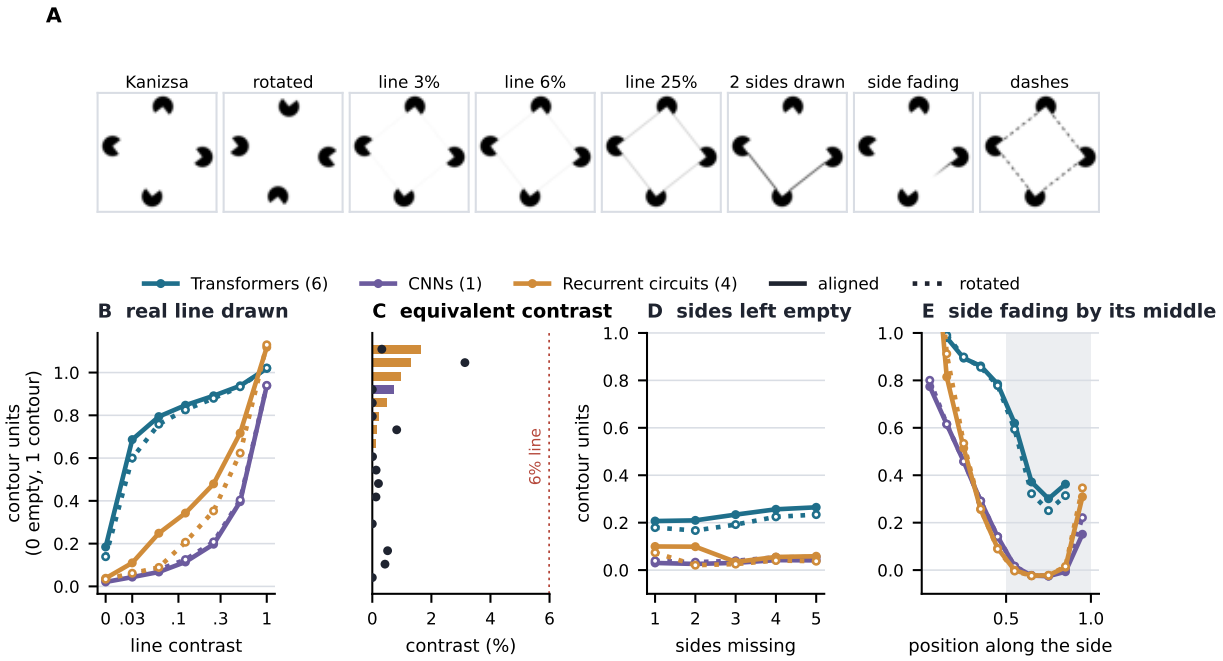


Figure 24: The illusory edge against physically drawn versions of the same shape. (A) Displays: the Kanizsa square and its rotated control, a real line of increasing contrast between the same inducers, some sides drawn and the others left empty, a side fading to nothing by its middle, dashes. (B) Contour units (the 5% most contour-selective units of a layer, scaled so that an empty cell is 0 and a drawn contour 1) in the empty cells on the illusory edges, against the contrast of a real line drawn there; contrast 0 is the plain Kanizsa figure (solid) or its rotated control (dotted). Bold: median of each family; networks whose read-out does not rise with line contrast are omitted. (C) Equivalent contrast of the illusory edge per network: bars, contour probe; dots, contour units. (D) Contour units on the sides left empty, against how many sides are missing (the others drawn). (E) Contour units along a side drawn at full contrast at one end and fading to nothing by its middle; grey, no contour pixels.

7.2 The empty cells between the inducers

The published studies probe representations; the most direct probe is the activity in the empty cells where an illusory edge would be. For 20 networks (all with spatial feature maps) we read every layer, block or iteration in two ways, both fitted only to complete outlines and never to Kanizsa figures: a linear *contour probe*, a logistic regression that labels feature-map cells as contour or empty (trained on two of the four grid shifts and tested on the others), and the raw activity of the 5% most *contour-selective units*, those that best separate contour cells from empty cells, scaled so that an empty cell reads 0 and a drawn contour 1. We compare the illusory edge with physically drawn versions of the same shape: a real line of contrast 3–100% drawn between the same inducers, dashes, a faint filled surface, some sides drawn and the others left empty, and a side that fades to nothing by its middle (Fig. 24A); every display also appears with the inducers rotated in place. A read-out is used only where it rises with the contrast of the real line (rank correlation ≥ 0.8), which holds for the probe in 18 networks and for the units in 14. The *equivalent contrast* of the illusory edge is the contrast of a real line that adds as much response as aligned inducers add over rotated ones.

A real line at 6% contrast raises the contour units by a median +0.44; aligned inducers raise them by +0.02 over rotated ones (at most +0.07; Fig. 24B). The illusory edge is therefore worth at most 1.7% of a real line's contrast to the contour probe and at most 3.1% to the contour units (Fig. 24C). Sides left empty stay empty however many of the others are drawn (aligned minus rotated, +0.02; Fig. 24D), and along a side that fades out the response ends where the contour pixels end: in the empty half it is 0.18 with aligned and 0.18 with rotated inducers (Fig. 24E). In real gaps the same probe responds: in the empty cells of one-patch gaps between fragments, aligned pieces add a median 0.43 of the drawn-contour response over

rotated pieces, against 0.02 between inducers at inducer radius 1 patch (largest, DINOv2-S, 0.34, from cells next to the inducers). One patch inside the virtual shape, cells resemble the complete shape slightly more with aligned than with rotated inducers (+0.09; one patch outside, +0.02), but these cells lie next to the inducers' inward-facing notches, so the difference does not separate a filled surface from the notches' own edges. Shapley interactions among the inducers, which unlike Shapley values of image regions can register completion, give the same answer: relative to a 180° control, fragment corners are superadditive but Kanizsa inducers are not, except weakly in transformers (Appendix H).

7.3 Surface fill under a cue in cued circuits and feedback networks

Two families label a surface from a cue: the cued circuits on a frozen convolutional backbone (Section 6.5) and the feedback networks inside a vision transformer (Section 6.6). Both were measured against the support ratio, and both show a threshold rather than the graded dependence of human illusory contours (Shipley & Kellman, 1992; Lesher & Mingolla, 1993). The cued circuits fill the Kanizsa hexagon like a closed one only while the gaps are under about 11 px (support ratio 0.85 and above for the large hexagon), and stop between 14 and 22 px whatever the size of their horizontal kernel; a single step does as much as forty. The SAM 3 feedback networks' fill is absent at a support ratio of 0.41, weak at 0.5 and saturated by 0.6, with a single pass that fills everywhere and a loop that removes the low-support fill; the V-JEPA networks fill less and with the same shape. In both families the step dynamics subtract: recurrence reduces the label on displays that specify no figure (full discs, rotated inducers, a blank page) without adding label where the figure is, and the backbone or encoder sets what is filled at all (a ViT backbone under the cued fGRU fills regardless of the gaps; the LeVJEPA whole encoder fills closed discs and not the Kanizsa square; SAM 3 fills the Kanizsa square and little else). Where the two families differ is what they treat as a figure at high support ratio: the cued circuits on the CNN backbone label the region between full discs after one step and unlabel it over forty; in the SAM 3 networks a surface probe reads a filled region between rotated inducers only once the inducers nearly touch (support ratio 0.90 and above), and the label covers a fifth of the inducers' bodies. Neither family labels a Kanizsa figure at the proportions of standard displays (support ratio 0.5 and below, gaps of several patches), which is where human observers see the clearest illusory contours of our battery's range.

7.4 Step dynamics across every recurrent family

Every recurrent family in this paper was read step by step, and the readings agree on one point: nothing grows towards completion. In the looped transformers net gap similarity is reached at the end of the first pass through the core and does not rise with further passes (Section 7.1.3); in the fGRU circuits the gap similarity rises over iterations without the single-iteration controls losing anything (Section 6.3); in the cued circuits and in the feedback networks the label floods in the first steps and recurrence then narrows it, so that a single step or a cut feedback gives as much Kanizsa label or more (Sections 6.5 and 6.6, Figs. 14 and 18). One network among those trained on the unity clips built its fill over steps, and its sibling seeds did not (Appendix L). The step-by-step maps of every network, in the supplementary material, show this directly: where a label appears, it appears in the first step or two, and recurrence afterwards narrows it.

7.5 Where in depth the separation appears

As in the classifiers, where the per-layer closure probe peaks late and net gap similarity peaks in the middle blocks and is mostly gone in the last (Sections 6.1 and 7.1.3), the relevant signal in the feedback networks' encoders sits in the middle of the stack. The untouched encoders of the feedback family show the same profile under the separation probe (one linear read-out per token, fitted on clip frames, read as the inside-minus-outside probability on the line drawings): in V-JEPA 2.1 L the separation rises from block 8, peaks at block 14 (line 0.60, Kanizsa 0.46, rotated 0.11, closed 0.25), holds through blocks 16–22 and drops at block 24, while the clip IoU keeps rising to block 18; in V-JEPA 2 L it stays at or below zero to block 14 and reaches 0.15–0.19 for the line and 0.04–0.07 for the Kanizsa square without separating it from the closed discs; in LeVJEPA it never exceeds 0.10. In SAM 3's trunk (Fig. 25) the separation is at or below 0.10 in the first ten blocks, appears at block 12 (line 0.58, Kanizsa 0.27), peaks at blocks 14–18 (line 0.67–0.70, Kanizsa 0.27–0.32, rotated 0.15–0.18, closed discs between −0.05 and 0.07), and declines through the last third: at block 26,

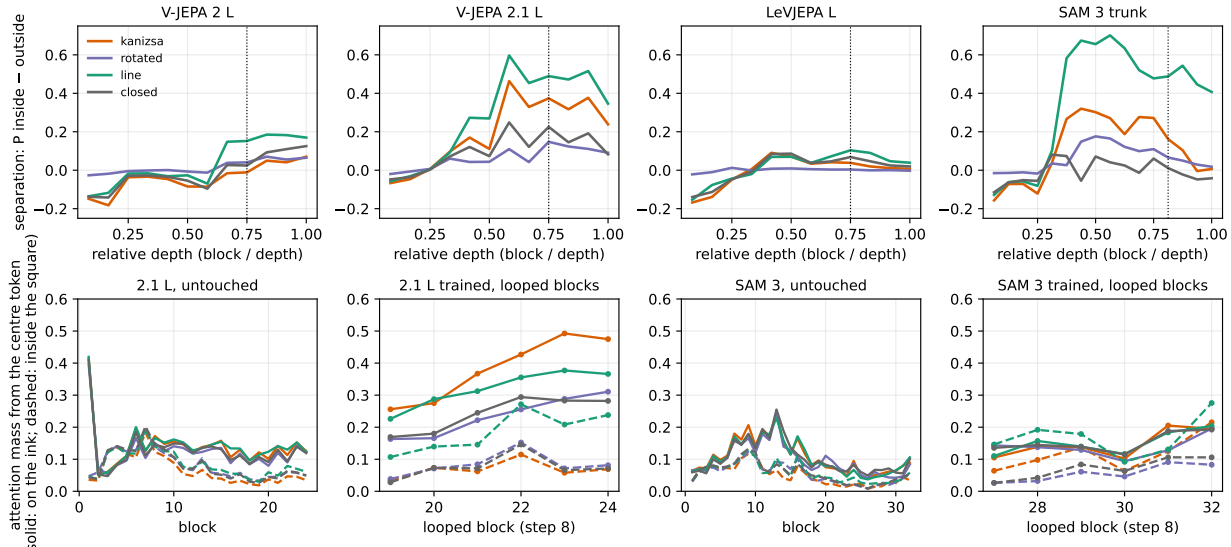


Figure 25: Top: separation (P inside minus outside under one linear read-out per token, fitted on clip frames) after every second block of the four untouched encoders, for the line square, the Kanizsa square, the rotated inducers and the closed discs, against relative depth; dotted, the block at which the looped networks are cut. Bottom: attention mass of the token at the square's centre on the ink (solid) and inside the square (dashed), head mean, per block of the untouched 2.1 L and SAM 3 encoders and per looped block of the trained networks at step 8.

where the loop is cut, the Kanizsa square is still separated (0.16, closed 0.01), and at blocks 30–32 it is not (0.00–0.01), while the line square keeps 0.41–0.54; the clip IoU of the same read-out rises to 0.46 at block 18 and stays there. The read-out that the loop trains thus starts from a front whose last frozen block still separates the Kanizsa square from the closed discs, after the encoder's own best blocks and before its last ones erase the difference. In the trained 2.1 L network the attention of the token at the square's centre, in the looped blocks, rises to about half of its mass on the inducers' ink for the Kanizsa display, 0.37 on the line square's ink (and 0.24 inside it), 0.30 on the rotated inducers and 0.28 on the closed discs, and the maps at steps 8 and 80 are the same: whatever sharpening the loop does is done within a few steps. A network trained on the plain unity clips shows a similar rise, spread over the whole discs and the image. SAM 3's attention is more diffuse: in the untouched trunk the centre token puts at most a quarter of its mass on the ink (block 13) and at most 0.13 inside the square at any block; in the trained looped blocks the mass on the ink stays at 0.09–0.22, the mass inside the square separates the line display from the others from the first looped block on, and separates the Kanizsa square from the rotated and closed displays (0.19–0.28 against 0.06–0.12) only in the last, global-attention block, again identical at steps 8 and 80. The signal that the trained read-out turns into fill is thus present in the middle of the frozen front, declines in the encoder's own last blocks, and is read back out by the looped blocks; nothing in depth, in iterations or in attention extends it across a gap that the front did not already span.

8 Audit of published closure measures

We recreated each measure's stimuli from its published specification and computed it on trained networks and on null observers that cannot close contours: untrained networks (three seeds for the closure-measure audit, one for the extended configural-effect controls; the brightness audit uses no networks), a Gabor energy bank, raw pixels, and a *pixel-growth* observer that encodes only the radial distribution of contour pixels about their centroid (our term; it differs from the pixel overlap that Kim et al. (2021) control, which compares images pixel by pixel; Fig. 26).

Null observers. The *pixel* observer is the image itself, average-pooled to 75×75. The *Gabor* observer is a fixed bank of quadrature-pair Gabor filters (wavelengths 4, 8 and 12 px, eight orientations) whose energy

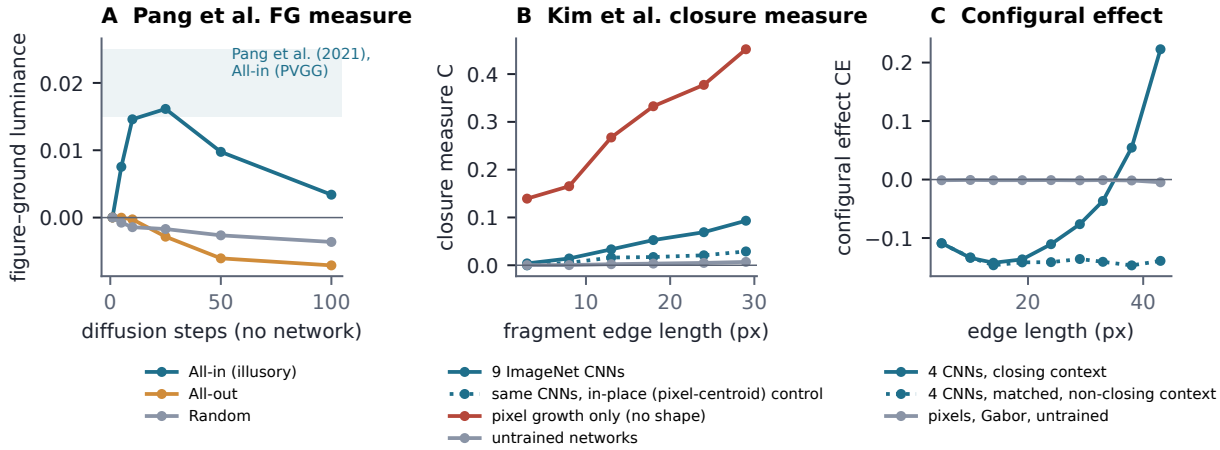


Figure 26: Published measures recomputed on null observers. (A) The figure–ground luminance difference of Pang et al. (2021) computed on input images smoothed by diffusion; shaded band, magnitude reported for the All-in condition. (B) The closure measure $\bar{C}$ of Kim et al. (2021) and Zhang et al. (2025) for nine ImageNet CNNs, for the same CNNs with a disordered control that keeps contour pixels in place, for an observer that only encodes the radial distribution of contour pixels, and for untrained networks. (C) The configural effect (CE) of Zhang et al. (2025) with a closing context and with the same components rotated so that no closed shape can form, against pixel, Gabor and untrained observers.

is average-pooled to a 14×14 grid, a model of oriented edge energy with no grouping. The *pixel-growth* observer encodes only the radial distribution of contour pixels: a 20-bin histogram of the distance of every contour pixel from the pixel centroid, normalized to sum to one, over distances of 0–120 px; it discards all angular information, although shape and arrangement still affect it when they change this distribution. *Untrained networks* are the same architectures as the trained ones with random initializations. For the illusory-brightness signature we used two observers without feedback or learned grouping: iterated diffusion of the input (the discrete heat equation, step 0.2) and a reconstruction from principal-component analysis (PCA) of greyscale CIFAR-10 images, with 32 components (16–256 tested). Trained networks are nine ImageNet CNNs (VGG16, AlexNet, ResNet50, DenseNet-121, Inception-v3, EfficientNet-B0, SqueezeNet 1.1, ShuffleNet v2 and MobileNet v3), read out at the layers used by Zhang et al. (2025) (the final convolutional stage for most, Mixed_7c for Inception-v3 and the final average pooling for ResNet50), except that for AlexNet we used the penultimate rather than the last fully connected layer, and ViT-B/16 (class token). The closure-measure and configural-effect stimuli are drawn at 300 px, as in Zhang et al. (2025) (Kim et al.'s originals are 150 px), and resized to each network's input (224 px; 299 for Inception-v3), so lengths quoted in stimulus pixels are 0.75 of a network-input pixel (1.0 for Inception-v3); the pixel-growth histogram is computed on the 300-px images. The brightness stimuli are 32 px as published.

Illusory brightness (Pang et al., 2021). The reported evidence is a figure-ground luminance difference (FG) of about 0.02 in reconstructions of Kanizsa squares whose inducers all face inward (All-in), which stays above zero after the first time step. Diffusing the input image produces an FG of +0.016 for All-in and -0.003 and -0.002 for the All-out and Random conditions after 25 steps, rising and then decaying as reported for their predictive-coding VGG (PVGG). A PCA reconstruction learned from natural images, with no feedback, gives +0.055 versus -0.032. The FG signature therefore does not require contour completion; it follows from smoothing inducers whose mass lies outside the illusory square.

Closure measure (Kim et al., 2021; Zhang et al., 2025). The closure measure is the cosine similarity of a network's representations of aligned fragments and a complete triangle minus that of disordered fragments (corners rotated about their vertices, as in Zhang et al., 2025) and the complete triangle:

$$\bar{C} = \mathbb{E}\big[\cos\big(f(x_{\text{aligned}}), f(x_{\text{complete}})\big) - \cos\big(f(x_{\text{disordered}}), f(x_{\text{complete}})\big)\big], \tag{1}$$

where f is the representation at the layer read out, and the expectation runs over image triples in which the complete triangle differs from the fragments in orientation and position and shares their background (Kim et al., 2021). Our replication reproduces the published pattern: trained CNNs score +0.019 to +0.087, rising with edge length, and untrained networks score +0.003 on average (individual seeds up to +0.018). But the pixel-growth observer, which discards all angular information, scores +0.289 (+0.452 at the longest edges), several times any network. The disordered control rotates each corner about its vertex and moves the contour pixels outward; when we instead rotate each corner in place about the centroid of its own pixels, VGG16 falls from +0.029 to +0.005 and DenseNet-121 from +0.025 to +0.005, the two networks reported to show closure most consistently (Zhang et al., 2025). A residual of +0.013 to +0.028 remains in 7 of the 9 networks. The pairing rule also matters: Kim et al. (2021) pair complete and fragment images that differ in global orientation and position, to ensure that there is no more pixel overlap between complete and aligned images than between complete and disordered images; when the paired images share orientation and position, raw pixels alone score +0.200 (versus +0.011 with the published pairing), and the published version of Zhang et al. (2025) does not state which pairing was used. Averaged over edge lengths, every trained network scores at most +0.014 on Kanizsa triangles, and ViT-B/16 scores +0.066 on line fragments.

Configural effect (Biscione & Bowers, 2023; Zhang et al., 2025). The network configural effect compares the distance between the representations of a base pair of images with that of a composite pair formed by adding the same components, the context, to both: $\mathrm{CE} = (D_{\mathrm{composite}} - D_{\mathrm{base}})/(D_{\mathrm{composite}} + D_{\mathrm{base}})$ with D the Euclidean distance between representations, positive for a configural superiority effect and negative for a configural inferiority effect. In the line-segment condition of Zhang et al. (2025) the context closes a square; we call it the closing context. This measure survives the null observers. Pixels, Gabor filters and untrained networks score near zero on average, and with components long enough that the gaps are under half a patch, ResNet50 and DenseNet-121 reach +0.30 and +0.29. Replacing the context with the same components rotated outward, so that no closed shape can form, reverses the effect (-0.17 and -0.18). Three further controls show what it measures (stimuli in Fig. 27; results in Fig. 28A–C). *Arrangement.* A context that keeps every component's position and reduces the radial differences between arrangements (components rotated in place about the centroids of their own pixels) is exceeded by the closing context by only +0.01 at an edge length of 5 px, and by +0.24 at 43 px; a fully closed square gives +0.23, on average no more than a nearly closed one (the closing-context CE itself still rises from 43 to 47.5 px in eight of the ten networks, on average from 0.12 to 0.13; it is the closing-minus-in-place contrast that stops growing), so CE grows as the outline approaches completion and the closing contrast gains nothing further, on average, from closing it. The pixel-growth observer shows no such advantage of the closing over the in-place context at 43 px (+0.01; +0.01 at 5 px, and negative at 10–33 px), so this observer does not account for the large contrast at 43 px, although centroid matching does not equalize the radial histograms exactly. The effect thus tracks how much of the outline is physically present, not gaps to be bridged. Averaged over all conditions, the ten trained networks (nine CNNs and ViT-B/16) score -0.16, mostly a configural inferiority effect, and in the line-segment condition CE is positive by a one-sample t-test across stimulus sets (nominal $p < 0.05$, uncorrected) in only 22% of network–edge cells with a closing context (0% in place). The pixel-growth observer scores -0.36 on CE itself, so radial pixel statistics alone can produce a configural inferiority effect of this size; the arrangement contrast above is what separates the networks from it. *Inducer shape.* Five notched inducers (notched disc, notched square, hexagon, cross and ring) have identical notch edges and hence the same ratio of luminance-specified to total edge length, which predicts the clarity of human illusory boundaries (Shipley & Kellman, 1992), but at an edge length of 43 px a network's closing-context CE varies across them by a median of 0.16 and changes sign (beyond ±0.02) in 5 of 10 networks (DenseNet-121: −0.07 with notched discs, +0.12 with notched squares; ResNet50 stays positive, +0.03 to +0.31). *Inducer size.* Enlarging the inducer disc while keeping the notch edges fixed, which leaves that ratio unchanged, moves CE from -0.18 to -0.11 (closing context, 19 px notch), while the closing context's advantage over the in-place control stays positive (at least +0.02). CE is therefore specific to closing arrangements of nearly complete outlines, but it is not invariant to manipulations that leave that ratio unchanged.

Accuracy under edge removal (Zhang et al., 2024). This protocol already concludes that CNNs lack closure. Because corners are kept, it can be passed by reading corner angles (Section 6.1.2); because gaps scale with edge length, gap size in patches is confounded with the number of sides; and because, in their

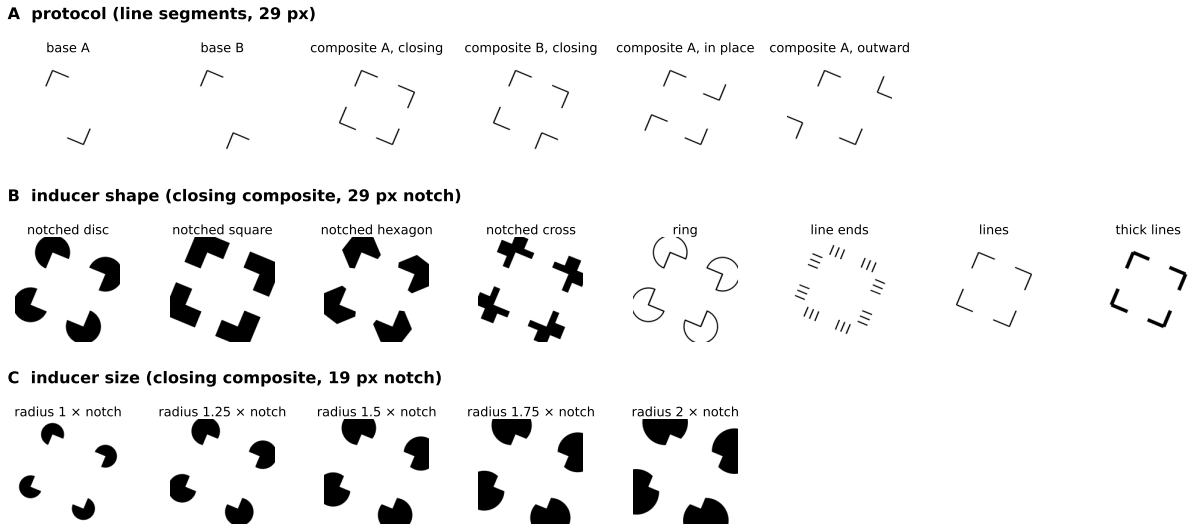


Figure 27: The configural-effect stimuli. (A) The protocol with line segments (edge length 29 px): the base pair differs in the orientation of one corner (flipped about the centroid of its own pixels); the composite pair adds the same two corners, the context, to both. The closing context completes the square; the in-place context rotates each added corner by 90° about the centroid of its own pixels; the outward context rotates it by 180° about its vertex. (B) The eight inducers of the shape audit, each specifying the same corner of the same square with the closing context (29 px notch); the first five have identical notch edges. (C) The size audit: notched discs enlarged at a fixed 19 px notch, so the notch edges and the implied square are unchanged.

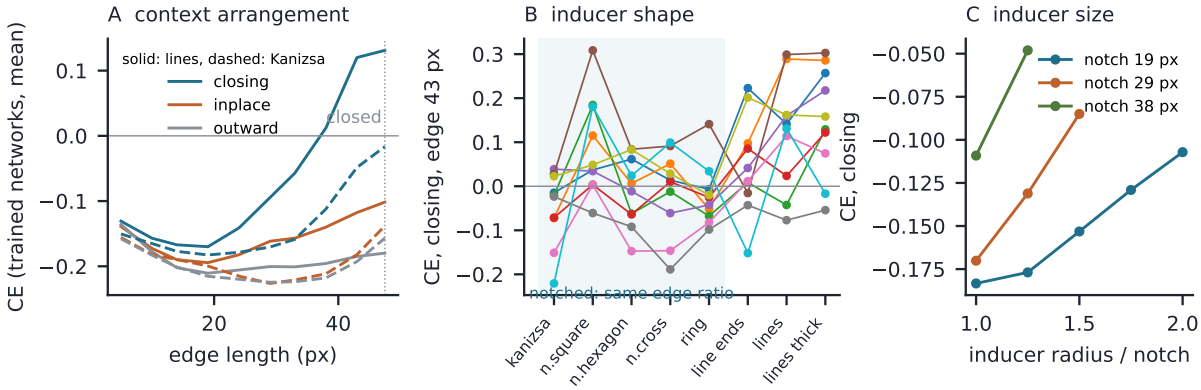


Figure 28: (A) Configural effect for the ten trained networks (nine CNNs and ViT-B/16) with closing, in-place (component-centroid-matched, non-closing) and outward contexts against edge length (solid, line segments; dashed, Kanizsa squares); 47.5 px is a closed square. (B) CE per network across inducer shapes; shaded, notched inducers with identical notch edges (same specified-to-total edge ratio). (C) CE against inducer radius at a fixed notch.

released stimulus code, test polygons are smaller than training polygons (radius 60 versus 67 px), the reported drop at 10% removal mixes size change with removal.

9 Faithfulness of saliency overlays

Overlays are a common argument that a network "sees" a triangle. We scored each map with a deletion test that removes contour pixels in map order or in random order (Petsiuk et al., 2018), and call the resulting score faithfulness, in the sense in which Selvaraju et al. (2017) use the word (Appendix D). Grad-CAM (Selvaraju et al., 2017) and class-specific relevance maps (Chefer et al., 2021a) have a median faithfulness of

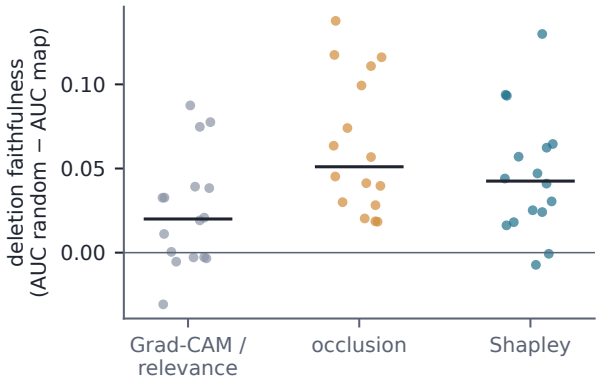


Figure 29: Deletion faithfulness of three attribution methods across networks (one point per network; bar, median).

+0.020 and are near zero ($|\cdot| < 0.02$) for 7 of 16 networks; occlusion (Zeiler & Fergus, 2014) and Shapley values computed over the stimulus's own pieces score +0.051 and +0.043 (Fig. 29; occlusion was run on a subset of 40 images per network). This is separate evidence from the sanity checks of Adebayo et al. (2018), which Grad-CAM passed and which test a different property.

10 Discussion

This paper is meant to settle a question rather than to open one. Across the architectures that have been proposed as the place where closure should appear, feed-forward, looped, horizontally and top-down recurrent, oscillatory and slot-based (Appendix L) and feedback inside frozen transformers, under recognition training, self-supervised pretraining, contour tracing, cued grouping and object-training with object masks, we found no network that completes a contour, and the published signatures of closure did not survive controls that keep the stimuli's low-level statistics. The value of such a result is in its coverage and in what it points to, so we first state what the evidence shows, then where, in our reading, the search should move.

What the evidence shows. Across 24 networks and four published measures, we find at most a weak illusory-contour signal, consistent across seeds only in ViT-B/16 (Kanizsa accuracy at most 0.39 against 1.00 on complete outlines), and the published illusory-brightness and representational-similarity signatures of closure are reproduced by observers that cannot close contours. What networks do show is integration of nearby, consistently oriented contour pieces: fine-tuned transformers use both the positions and the orientations of fragments, and ImageNet CNNs show a configural effect whose advantage for closing arrangements grows as the outline approaches completion. In CNN and recurrent networks the gaps that halve closure are, with one exception (two counting single-iteration controls), smaller than the measured effective receptive field, so the measured receptive field does not appear to be what limits closure. This places current networks closer to the local association-field account of contour integration (Field et al., 1993) than to Gestalt closure: they link pieces that are near and aligned, and do not complete what is absent. In the terms of Roelfsema (2006), the networks perform base grouping, by units tuned to conjunctions of nearby aligned elements, but show no sign of incremental grouping, in which activity spreads along an object through horizontal and feedback connections. Nor do they show the growing reach that incremental grouping predicts (Pooresmaeili & Roelfsema, 2014): the gaps they bridge stay within the receptive field, and neither iterations nor depth extend them. A related dissociation has been reported for vision-language models, which give the expected answer to classic illusions but keep it when the inducers are reversed (Sun et al., 2026); recognizing an illusory configuration and being susceptible to it are different properties, and our measures target the second. Networks trained to trace contours, the task recurrence is thought to be needed for, share this profile: they link aligned pieces a few pixels beyond their training statistics and label almost none of the missing contour as contour. Recurrence does not change this picture. In some fGRU circuits the gap similarity rises over iterations, but classifiers trained on complete outlines have little reason to use it, and single-iteration controls perform as well in all but one case.

The training signal, not recurrence. The feedback networks, like the cued circuits, were always told where to look: the cue at the figure's centre asks what object lies there, so their fills are answers to that question; run without any cue the feedback networks label the same figures, less completely in two of six, and the hGRU and fGRU circuits do as well, while the looped-transformer circuit labels nothing, so the selectivity is the networks' own and the cue names the location or, in one circuit, gates the answer. They were the last place we looked for completion by recurrence, with encoders that had never seen a line drawing and a loop free to build whatever the task rewarded. What they showed is that the label a network puts between four inducers is set by its encoder and by what the training clips ask of it: the video encoders give little or the reverse under every clip set, while two image encoders, SAM 3 (trained with object masks) and DINOv3 (self-distilled), give a selective label at high support ratios once the clips contain outlines and the layer labels are trained, DINOv3 on both seeds and over the scrambled-outline bound on one; the loop prunes the configurations that specify no figure in both and, in one DINOv3 seed, adds to the figure. The loop's measured role is mainly that veto; cutting it raises the Kanizsa label and the controls' labels together in SAM 3 and 2.1 L, lowers it in one DINOv3 seed and raises it in the other, and a network trained without it passes the same battery. The training signal, the encoder's pretraining together with the outline clips and the layer labels, is what moved the result, and it moved it towards labelling a surface bounded by aligned edges, not towards completing a contour: the fill is absent at a support ratio of 0.41, weak at 0.5 and saturated by 0.6, a surface probe on its last block reads a filled region between rotated inducers once they nearly close the figure, and it spills onto the inducers themselves. **Seeing the inducers is not enough.** A network's feature extractors respond where there are features; between the inducers there are none, and nothing we measured puts a response there. Wherever the read-outs work, a contour probe and the most contour-selective units read a real line at 6% contrast almost fully, yet value the illusory edge at no more than 3.1% of a real line's contrast (Section 7.2). This holds at every layer, including late transformer blocks whose tokens attend to every inducer and convolutional and recurrent networks whose receptive fields span the gap, and it holds when the inducers are familiar from training (Appendix G); missing sides stay empty however many others are drawn. A receptive field large enough to contain all the inducers gives a unit access to them, but nothing in training for recognition makes it respond where the image is empty. This is not a necessary property of visual systems: neurons in V1 respond more inside a figure than outside it when the local image is identical (Lamme, 1995), a modulation carried by recurrent and feedback processing (Self et al., 2012; Kirchberger et al., 2021). In incremental grouping, however, the spreading label enhances responses the image already drives (Roelfsema, 2006; Brosch et al., 2015), so even a network that implements it would have nothing to label between the inducers. Circuits trained for such spreading confirm this: reimplementations of the γ-net of Mollard et al. (2026) label the interior of a Kanizsa figure only across gaps of a few pixels, as much after one step as after 40, and recurrence only reduces, without removing, the label in regions that specify no shape, such as those between full discs or rotated inducers (Section 6.5).

Where to look. Four things follow from the pattern of failures, and they say where a completion signal might still be found.

The training signal, not the architecture. Every architecture we trained behaved like its objective. Looping a transformer added depth and no closure; horizontal and top-down recurrence raised the similarity in the gap without any network using it; oscillators and slots reproduced what their feed-forward controls did (Appendix L); feedback inside a frozen encoder produced what the encoder already separated and vetoed the rest. The selective labels came from two image encoders under one recipe, SAM 3, trained with object masks, and DINOv3, self-distilled, and the training changes that mattered inside our own recipe were the outline clips and the layer-label loss, which asks the network to say what is hidden and what occludes. Nothing in training for recognition, for predicting masked video or image features, or for tracing a drawn contour rewards a response where the image is empty, and none of these produced one. More depth, more loops or larger encoders trained on the same objectives are not where completion will appear.

Supervision for amodal structure. What did move the result was supervision about objects and occlusion in the clips, hidden and occluding layers and drawn outlines, on encoders that carry a figure/ground signal (object masks in SAM 3's pretraining; DINOv3's self-distilled features). Both gave a label for a surface bounded by aligned edges, with a sharp support-ratio threshold and a loop that removes the label from configurations that specify no figure. Models trained to recover amodal shape, object permanence under

occlusion, or the layered structure of a scene are the natural next candidates, and this paper's battery, controls and probes apply to them unchanged: the first thing to ask of such a model is whether it labels the Kanizsa surface at a support ratio of 0.5 and below, where humans see the clearest illusory contours and where no network here did.

The constraints from biology that we did not use. Human illusory contours grow in clarity with support ratio (Shipley & Kellman, 1992; Lesher & Mingolla, 1993); in V1, responses to illusory contours appear later than responses to real ones, consistent with feedback from higher areas (Lee & Nguyen, 2001); and the figure–ground modulation of V1 responses is carried by recurrent processing (Lamme, 1995; Self et al., 2012). None of the networks here matched these signatures: the fill, where there was one, switched rather than graded, and it appeared in the first pass and was pruned afterwards rather than arriving late and growing. A model that aims at completion should be held to the graded and late profile, measured on the same displays in people and in the model, rather than to a closure index; we did not collect human data on our own displays, and that comparison is the next experiment.

What would count as closure. Our conclusion is falsifiable with the measures used here. We propose the following as diagnostic signatures, not necessary conditions; a network could complete contours feed-forward without iterations or attention. A network that completes contours would be expected to show a Kanizsa closure index comparable to its fragment closure index and high Kanizsa accuracy when trained only on complete outlines; a critical gap that grows with iterations or loops; empty gap cells whose similarity to the drawn contour is maintained to the readout and grows with iterations at gaps of several patches; attention from gap tokens that prefers aligned contour ends over misaligned pieces at gaps of several patches; and, for a network that labels surfaces from a cue, a fill that grows with support ratio below 0.5 and leaves the inducers' bodies unlabelled. No network meets these criteria jointly; the transformers meet the attention criterion only weakly, and mainly at one-patch gaps; the SAM 3 feedback networks meet the surface criterion only above a support ratio of 0.6. The closest classifier is the γ-net, whose gap similarity grows over iterations and persists to the layer it reads out, yet it gains nothing in recognition over a single iteration and has no Kanizsa effect. Each criterion is a concrete target for models designed to complete contours.

How to test. Closure tests should (i) specify gaps in patch or receptive-field units; (ii) remove corners or otherwise prevent corner reading; (iii) include controls that keep the contour pixels, their pixel growth and their orientations while breaking the arrangement; (iv) report null observers (untrained networks, a filter bank, pixels and a pixel-growth observer) alongside trained ones; (v) for recurrent models, read the time course and a matched single-step control, since in every family here the label of interest was present after the first pass and recurrence only narrowed it; and (vi) for models that label surfaces, vary the support ratio and measure the label on the inducers as well as between them.

Limitations. ViT-B/16 and DINOv2-S were trained with three seeds and ResNet50 with five, whose corner-removed closure index and arrangement effect vary by up to 0.03 and 0.05 (s.d.) and whose mid-gap closure index varies by up to 0.06 (Table 5); the others were trained once, so differences of that size between them remain uncertain without repeated training. Published stimuli were recreated from their papers' descriptions rather than original code. We did not test the implicit-prior account of Toosi & Miller (2024), or abutting-grating illusions, and we did not collect human data on our stimuli. Pathfinder pretraining (Linsley et al., 2018) of our recurrent circuits, and of a ViT-B/16, remained at chance after 1–2 h of training per network; networks from a companion project that were trained to trace contours stand in for it (Fig. 13). Among the feedback networks, the cells with a verdict have two seeds, but the two hybrids were trained once each, with an adapter trained for 4,000 iterations, so their failure to transplant SAM 3's label is a single observation; the earlier objectives on DINOv2 were explored with one to eight seeds per objective and are summarized rather than reported in full; and three chains, the deeper cuts of V-JEPA 2.1 L and SAM 3 and the SAM 3-distilled 2.1 L, were stopped before their results could be read, so whether more looped blocks or a distilled front change the picture is untested. The selectivity of the SAM 3 networks rests on four stage-2 runs, of which one lost its fill; their step-1 label was consistent across the four, their final label was not. DINOv3 under the same recipe rests on two runs that pass the battery with margins of 0.69 and 0.34 but differ on the scrambled-outline gate and in the loop's direction, so its reading is less settled than SAM 3's.

Compute and reproducibility

The feedback networks and the exploratory circuits of the same thread were trained on single 48 GB GPUs; their training logs record 846 GPU-hours over 197 training stages, 459 of them over the 103 stages of the feedback family (median 3.5 h per stage; a stage is 4,000 iterations at batch 24). The classifiers, the recurrent circuits built from the fGRU cell and the cued circuits were trained earlier on the same class of hardware and their wall-clock times were not logged; each classifier trains in under an hour, each cued circuit (21,000 iterations with contractor recurrent back-propagation) in about a day. The probes, layer maps and animations run in minutes per network. Every number in the text is produced by a script from a results file (the macros of the numbers files), and the figures and animations are regenerated from the saved maps; the stimulus generator, training and evaluation code, per-image results and the step-by-step and layer-by-layer maps will be released with the final version.

Statistics

Where a network was trained more than once we report every seed's value, or the range across seeds, rather than a mean with an interval: with two to five trainings an interval would suggest more than it can support. Differences between networks trained once are reported as observed and not tested; the Limitations paragraph gives the seed-to-seed variation that bounds them.

Broader impact

This paper reports what a class of vision models does not do and audits measures that have been used to claim otherwise; its direct use is in evaluation. Claims that a network "perceives" an illusion are sometimes taken as evidence of human-like vision in systems deployed for other purposes; our results argue against reading such claims off saliency overlays or representational-similarity scores without controls, and the battery and null observers are released so that others can run the same checks.

Use of AI tools

Code for stimulus generation, training, evaluation and figures, and drafts of this manuscript, were produced with the assistance of a large language model (Claude, Anthropic). All experimental design decisions, results and text were reviewed by the authors.

Code and data

Stimulus generator, training and evaluation code, and per-image results will be released with the final version.

References

Samira Abnar and Willem Zuidema. Quantifying attention flow in transformers. In *Proceedings of the 58th Annual Meeting of the Association for Computational Linguistics*, pp. 4190–4197, 2020. doi: 10.18653/v1/2020.acl-main.385.

Julius Adebayo, Justin Gilmer, Michael Muelly, Ian Goodfellow, Moritz Hardt, and Been Kim. Sanity checks for saliency maps. In *Advances in Neural Information Processing Systems*, volume 31, 2018.

Guillaume Alain and Yoshua Bengio. Understanding intermediate layers using linear classifier probes. In *International Conference on Learning Representations (ICLR), Workshop Track*, 2017. URL https://openreview.net/forum?id=HJ4-rAVtl.

Mido Assran, Adrien Bardes, David Fan, Quentin Garrido, Russell Howes, Mojtaba Komeili, Matthew Muckley, Ammar Rizvi, Claire Roberts, Koustuv Sinha, Artem Zholus, Sergio Arnaud, Abha Gejji, Ada Martin, Francois Robert Hogan, Daniel Dugas, Piotr Bojanowski, Vasil Khalidov, Patrick Labatut, Francisco Massa,

Marc Szafraniec, Kapil Krishnakumar, Yong Li, Xiaodong Ma, Sarath Chandar, Franziska Meier, Yann LeCun, Michael Rabbat, and Nicolas Ballas. V-JEPA 2: Self-supervised video models enable understanding, prediction and planning, 2025. arXiv:2506.09985.

Nicholas Baker, Gennady Erlikhman, Philip J. Kellman, and Hongjing Lu. Deep convolutional networks do not perceive illusory contours. In *Proceedings of the 40th Annual Conference of the Cognitive Science Society*, 2018.

Johannes Ballé, Valero Laparra, and Eero P. Simoncelli. Density modeling of images using a generalized normalization transformation. In *International Conference on Learning Representations*, 2016.

Irving Biederman. Recognition-by-components: A theory of human image understanding. *Psychological Review*, 94(2):115–147, 1987. doi: 10.1037/0033-295X.94.2.115.

Valerio Biscione and Jeffrey S. Bowers. Mixed evidence for Gestalt grouping in deep neural networks. *Computational Brain & Behavior*, 6(3):438–456, 2023.

Daniel Bolya, Po-Yao Huang, Peize Sun, Jang Hyun Cho, Andrea Madotto, Chen Wei, Tengyu Ma, Jiale Zhi, Jathushan Rajasegaran, Hanoona Rasheed, Junke Wang, Marco Monteiro, Hu Xu, Shiyu Dong, Nikhila Ravi, Daniel Li, Piotr Dollár, and Christoph Feichtenhofer. Perception encoder: The best visual embeddings are not at the output of the network. In *Advances in Neural Information Processing Systems*, volume 38, pp. 67743–67796, 2025. doi: 10.52202/085713-2036. arXiv:2504.13181.

Tobias Brosch, Heiko Neumann, and Pieter R. Roelfsema. Reinforcement learning of linking and tracing contours in recurrent neural networks. *PLOS Computational Biology*, 11(10):e1004489, 2015. doi: 10.1371/journal.pcbi.1004489.

Nicolas Carion, Laura Gustafson, Yuan-Ting Hu, Shoubhik Debnath, Ronghang Hu, Dídac Surís, Chaitanya Ryali, Kalyan Vasudev Alwala, Haitham Khedr, Andrew Huang, Jie Lei, Tengyu Ma, Baishan Guo, Arpit Kalla, Markus Marks, Joseph Greer, Meng Wang, Peize Sun, Roman Rädle, Triantafyllos Afouras, Effrosyni Mavroudi, Katherine Xu, Tsung-Han Wu, Yu Zhou, Liliane Momeni, Rishi Hazra, Shuangrui Ding, Sagar Vaze, Francois Porcher, Feng Li, Siyuan Li, Aishwarya Kamath, Ho Kei Cheng, Piotr Dollár, Nikhila Ravi, Kate Saenko, Pengchuan Zhang, and Christoph Feichtenhofer. SAM 3: Segment anything with concepts. In *International Conference on Learning Representations*, 2026. URL https://openreview.net/forum?id=r35clVtGzw. arXiv:2511.16719.

Hila Chefer, Shir Gur, and Lior Wolf. Generic attention-model explainability for interpreting bi-modal and encoder-decoder transformers. In *2021 IEEE/CVF International Conference on Computer Vision (ICCV)*, pp. 387–396. IEEE, 2021a. doi: 10.1109/ICCV48922.2021.00045.

Hila Chefer, Shir Gur, and Lior Wolf. Transformer interpretability beyond attention visualization. In *IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pp. 782–791, 2021b. doi: 10.1109/CVPR46437.2021.00084.

Mostafa Dehghani, Stephan Gouws, Oriol Vinyals, Jakob Uszkoreit, and Łukasz Kaiser. Universal transformers. In *International Conference on Learning Representations*, 2019.

Alexey Dosovitskiy, Lucas Beyer, Alexander Kolesnikov, Dirk Weissenborn, Xiaohua Zhai, Thomas Unterthiner, Mostafa Dehghani, Matthias Minderer, Georg Heigold, Sylvain Gelly, Jakob Uszkoreit, and Neil Houlsby. An image is worth 16x16 words: Transformers for image recognition at scale. In *International Conference on Learning Representations*, 2021.

Gregor Ehrensperger, Sebastian Stabinger, and Antonio Rodríguez-Sánchez. Evaluating CNNs on the Gestalt principle of closure. In *Artificial Neural Networks and Machine Learning – ICANN 2019: Theoretical Neural Computation*, pp. 296–301, 2019. doi: 10.1007/978-3-030-30487-4_23.

David J. Field, Anthony Hayes, and Robert F. Hess. Contour integration by the human visual system: Evidence for a local "association field". *Vision Research*, 33(2):173–193, 1993. doi: 10.1016/0042-6989(93)90156-Q.

Jonas Geiping, Sean McLeish, Neel Jain, John Kirchenbauer, Siddharth Singh, Brian R. Bartoldson, Bhavya Kailkhura, Abhinav Bhatele, and Tom Goldstein. Scaling up test-time compute with latent reasoning: A recurrent depth approach. In *Advances in Neural Information Processing Systems*, volume 38, pp. 46242–46293, 2025. doi: 10.52202/085713-1380. arXiv:2502.05171.

Robert Geirhos, Patricia Rubisch, Claudio Michaelis, Matthias Bethge, Felix A. Wichmann, and Wieland Brendel. ImageNet-trained CNNs are biased towards texture; increasing shape bias improves accuracy and robustness. In *International Conference on Learning Representations (ICLR)*, 2019. URL https://openreview.net/forum?id=Bygh9j09KX.

Angeliki Giannou, Shashank Rajput, Jy-Yong Sohn, Kangwook Lee, Jason D. Lee, and Dimitris Papailiopoulos. Looped transformers as programmable computers. In Andreas Krause, Emma Brunskill, Kyunghyun Cho, Barbara Engelhardt, Sivan Sabato, and Jonathan Scarlett (eds.), *Proceedings of the 40th International Conference on Machine Learning*, volume 202 of *Proceedings of Machine Learning Research*, pp. 11398–11442. PMLR, 2023.

Michel Grabisch and Marc Roubens. An axiomatic approach to the concept of interaction among players in cooperative games. *International Journal of Game Theory*, 28(4):547–565, 1999. doi: 10.1007/s001820050125.

Stephen Grossberg and Ennio Mingolla. Neural dynamics of form perception: Boundary completion, illusory figures, and neon color spreading. *Psychological Review*, 92(2):173–211, 1985. doi: 10.1037/0033-295X.92.2.173.

Kaiming He, Xinlei Chen, Saining Xie, Yanghao Li, Piotr Dollár, and Ross Girshick. Masked autoencoders are scalable vision learners. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pp. 16000–16009, 2022.

David J. Heeger. Normalization of cell responses in cat striate cortex. *Visual Neuroscience*, 9(2):181–197, 1992. doi: 10.1017/S0952523800009640.

Stefan Heimersheim and Neel Nanda. How to use and interpret activation patching. *arXiv preprint arXiv:2404.15255*, 2024.

Pierre Jolicoeur, Shimon Ullman, and Marilynn Mackay. Curve tracing: A possible basic operation in the perception of spatial relations. *Memory & Cognition*, 14(2):129–140, 1986. doi: 10.3758/BF03198373.

Gaetano Kanizsa. *Organization in Vision: Essays on Gestalt Perception*. Praeger, 1979.

Been Kim, Emily Reif, Martin Wattenberg, Samy Bengio, and Michael C. Mozer. Neural networks trained on natural scenes exhibit gestalt closure. *Computational Brain & Behavior*, 4(3):251–263, 2021. arXiv:1903.01069.

Junkyung Kim, Drew Linsley, Kalpit Thakkar, and Thomas Serre. Disentangling neural mechanisms for perceptual grouping. In *International Conference on Learning Representations*, 2020.

Diederik P. Kingma and Jimmy Ba. Adam: A method for stochastic optimization. In *International Conference on Learning Representations*, 2015. arXiv:1412.6980.

Lisa Kirchberger, Sreedeep Mukherjee, Ulf H. Schnabel, Enny H. van Beest, Areg Barsegyan, Christiaan N. Levelt, J. Alexander Heimel, Jeannette A. M. Lorteije, Chris van der Togt, Matthew W. Self, and Pieter R. Roelfsema. The essential role of recurrent processing for figure-ground perception in mice. *Science Advances*, 7(27):eabe1833, 2021. doi: 10.1126/sciadv.abe1833.

Alexander Kirillov, Ross Girshick, Kaiming He, and Piotr Dollár. Panoptic feature pyramid networks. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pp. 6399–6408, 2019.

Alexander Kirillov, Eric Mintun, Nikhila Ravi, Hanzi Mao, Chloe Rolland, Laura Gustafson, Tete Xiao, Spencer Whitehead, Alexander C. Berg, Wan-Yen Lo, Piotr Dollár, and Ross Girshick. Segment anything. In *Proceedings of the IEEE/CVF International Conference on Computer Vision (ICCV)*, pp. 3992–4003, 2023. doi: 10.1109/ICCV51070.2023.00371.

Lukas Kuhn, Lucas Maes, Giuseppe Serra, Quentin Le Lidec, Yann LeCun, Randall Balestriero, and Florian Buettner. LeVJEPA: Efficient & scalable video pretraining without the heuristics, 2026. arXiv:2608.27395.

Victor A. F. Lamme. The neurophysiology of figure-ground segregation in primary visual cortex. *The Journal of Neuroscience*, 15(2):1605–1615, 1995. doi: 10.1523/JNEUROSCI.15-02-01605.1995.

Tai Sing Lee and My Nguyen. Dynamics of subjective contour formation in the early visual cortex. *Proceedings of the National Academy of Sciences*, 98(4):1907–1911, 2001. doi: 10.1073/pnas.031579998.

Gregory W. Lesher and Ennio Mingolla. The role of edges and line-ends in illusory contour formation. *Vision Research*, 33(16):2253–2270, 1993. doi: 10.1016/0042-6989(93)90104-5.

Tianqin Li, Ziqi Wen, Leiran Song, Jun Liu, Zhi Jing, and Tai Sing Lee. From local cues to global percepts: Emergent Gestalt organization in self-supervised vision models, 2025. arXiv:2506.00718.

Tianqin Li, Isaac Lin, Leiran Song, and Tai Sing Lee. Learning perceptual rules in a fragmented visual world, 2026. arXiv:2610.13988.

Yanghao Li, Hanzi Mao, Ross Girshick, and Kaiming He. Exploring plain vision transformer backbones for object detection. In *Proceedings of the European Conference on Computer Vision*, 2022. doi: 10.1007/978-3-031-20077-9_17.

Tsung-Yi Lin, Michael Maire, Serge Belongie, James Hays, Pietro Perona, Deva Ramanan, Piotr Dollár, and C. Lawrence Zitnick. Microsoft COCO: Common objects in context. In *European Conference on Computer Vision*, pp. 740–755, 2014.

Drew Linsley, Junkyung Kim, Vijay Veerabadran, Charles Windolf, and Thomas Serre. Learning long-range spatial dependencies with horizontal gated recurrent units. In *Advances in Neural Information Processing Systems*, volume 31, 2018.

Drew Linsley, Alekh Karkada Ashok, Lakshmi Narasimhan Govindarajan, Rex Liu, and Thomas Serre. Stable and expressive recurrent vision models. In *Advances in Neural Information Processing Systems*, volume 33, pp. 10456–10467, 2020a.

Drew Linsley, Junkyung Kim, Alekh Ashok, and Thomas Serre. Recurrent neural circuits for contour detection. In *International Conference on Learning Representations*, 2020b.

Wenjie Luo, Yujia Li, Raquel Urtasun, and Richard Zemel. Understanding the effective receptive field in deep convolutional neural networks. In *Advances in Neural Information Processing Systems*, volume 29, 2016.

Kevin Meng, David Bau, Alex Andonian, and Yonatan Belinkov. Locating and editing factual associations in GPT. In *Advances in Neural Information Processing Systems*, volume 35, 2022.

Sami Mollard, Alekh K. Ashok, Lore Goetschalckx, Drew Linsley, Thomas Serre, Sander M. Bohte, and Pieter R. Roelfsema. Recurrent network dynamics explain the time course of perceptual grouping in natural scenes. *arXiv preprint arXiv:2610.05419*, 2026.

Michael C. Mozer, Shoaib Ahmed Siddiqui, Danny Sawyer, Sunny Sanyal, and Rosanne Liu. Recirculation. *arXiv preprint arXiv:2608.17981*, 2026.

Lorenzo Mur-Labadia, Matthew Muckley, Amir Bar, Mido Assran, Koustuv Sinha, Mike Rabbat, Yann LeCun, and Nicolas Ballas. V-JEPA 2.1: Unlocking dense features in video self-supervised learning. In *Computer Vision – ECCV 2026*, Lecture Notes in Computer Science, pp. 671–689, Cham, 2026. Springer Nature Switzerland. doi: 10.1007/978-3-032-37369-4_36. arXiv:2603.14482.

Maxime Oquab, Timothée Darcet, Théo Moutakanni, Huy Vo, Marc Szafraniec, Vasil Khalidov, Pierre Fernandez, Daniel Haziza, Francisco Massa, Alaaeldin El-Nouby, Mahmoud Assran, Nicolas Ballas, Wojciech Galuba, Russell Howes, Po-Yao Huang, Shang-Wen Li, Ishan Misra, Michael Rabbat, Vasu Sharma, Gabriel Synnaeve, Hu Xu, Hervé Jégou, Julien Mairal, Patrick Labatut, Armand Joulin, and Piotr Bojanowski. DINOv2: Learning robust visual features without supervision. *Transactions on Machine Learning Research*, 2024.

Zhaoyang Pang, Callum Biggs O'May, Bhavin Choksi, and Rufin VanRullen. Predictive coding feedback results in perceived illusory contours in a recurrent neural network. *Neural Networks*, 144:164–175, 2021. doi: 10.1016/j.neunet.2021.08.024.

Vitali Petsiuk, Abir Das, and Kate Saenko. RISE: Randomized input sampling for explanation of black-box models. In *British Machine Vision Conference*, 2018.

Arezoo Pooresmaeili and Pieter R. Roelfsema. A growth-cone model for the spread of object-based attention during contour grouping. *Current Biology*, 24(24):2869–2877, 2014. doi: 10.1016/j.cub.2014.10.007.

Nikhila Ravi, Valentin Gabeur, Yuan-Ting Hu, Ronghang Hu, Chaitanya Ryali, Tengyu Ma, Haitham Khedr, Roman Rädle, Chloe Rolland, Laura Gustafson, Eric Mintun, Junting Pan, Kalyan Vasudev Alwala, Nicolas Carion, Chao-Yuan Wu, Ross Girshick, Piotr Dollár, and Christoph Feichtenhofer. SAM 2: Segment anything in images and videos. In *International Conference on Learning Representations*, 2025. URL https://openreview.net/forum?id=Ha6RTeWMd0. arXiv:2408.00714.

Dario L. Ringach and Robert Shapley. Spatial and temporal properties of illusory contours and amodal boundary completion. *Vision Research*, 36(19):3037–3050, 1996. doi: 10.1016/0042-6989(96)00062-4.

Pieter R. Roelfsema. Cortical algorithms for perceptual grouping. *Annual Review of Neuroscience*, 29(1): 203–227, 2006. doi: 10.1146/annurev.neuro.29.051605.112939.

Pieter R. Roelfsema, Victor A. F. Lamme, and Henk Spekreijse. Object-based attention in the primary visual cortex of the macaque monkey. *Nature*, 395(6700):376–381, 1998. doi: 10.1038/26475.

Martin Schrimpf, Jonas Kubilius, Michael J. Lee, N. Apurva Ratan Murty, Robert Ajemian, and James J. DiCarlo. Integrative benchmarking to advance neurally mechanistic models of human intelligence. *Neuron*, 108(3):413–423, 2020. doi: 10.1016/j.neuron.2020.07.040.

Matthew W. Self, Roxana N. Kooijmans, Hans Supèr, Victor A. F. Lamme, and Pieter R. Roelfsema. Different glutamate receptors convey feedforward and recurrent processing in macaque V1. *Proceedings of the National Academy of Sciences*, 109(27):11031–11036, 2012. doi: 10.1073/pnas.1119527109.

Ramprasaath R. Selvaraju, Michael Cogswell, Abhishek Das, Ramakrishna Vedantam, Devi Parikh, and Dhruv Batra. Grad-CAM: Visual explanations from deep networks via gradient-based localization. In *IEEE International Conference on Computer Vision*, pp. 618–626, 2017. doi: 10.1109/ICCV.2017.74.

Lloyd S. Shapley. A value for n-person games. In Harold W. Kuhn and Albert W. Tucker (eds.), *Contributions to the Theory of Games, Volume II*, number 28 in Annals of Mathematics Studies, pp. 307–318. Princeton University Press, Princeton, NJ, 1953. doi: 10.1515/9781400881970-018.

Thomas F. Shipley and Philip J. Kellman. Strength of visual interpolation depends on the ratio of physically specified to total edge length. *Perception & Psychophysics*, 52(1):97–106, 1992. doi: 10.3758/BF03206762.

Oriane Siméoni, Huy V. Vo, Maximilian Seitzer, Federico Baldassarre, Maxime Oquab, Cijo Jose, Vasil Khalidov, Marc Szafraniec, Seungeun Yi, Michaël Ramamonjisoa, Francisco Massa, Daniel Haziza, Luca Wehrstedt, Jianyuan Wang, Timothée Darcet, Théo Moutakanni, Leonel Sentana, Claire Roberts, Andrea Vedaldi, Jamie Tolan, John Brandt, Camille Couprie, Julien Mairal, Hervé Jégou, Patrick Labatut, and Piotr Bojanowski. DINOv3. *Transactions on Machine Learning Research*, 2026. URL https://openreview.net/forum?id=2NlGyqNjns. arXiv:2508.10104.

Xiaoxiao Sun, Mingyang Li, Kun Yuan, Min Woo Sun, Mark Endo, Shengguang Wu, Changlin Li, Yuhui Zhang, Zeyu Wang, and Serena Yeung-Levy. Do VLMs perceive or recall? Probing visual perception vs. memory with classic visual illusions. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pp. 25861–25870, 2026.

Corentin Tallec and Yann Ollivier. Can recurrent neural networks warp time? In *International Conference on Learning Representations (ICLR)*, 2018. URL https://openreview.net/forum?id=SJcKhk-Ab.

Leonard Tang and Dan Ley. Degraded polygons raise fundamental questions of neural network perception. In A. Oh, T. Naumann, A. Globerson, K. Saenko, M. Hardt, and S. Levine (eds.), *Advances in Neural Information Processing Systems*, volume 36, pp. 9695–9706. Curran Associates, Inc., 2023. doi: 10.52202/075280-0425. Datasets and Benchmarks Track.

Tahereh Toosi and Kenneth D. Miller. Emergence of illusory contours in robust deep neural networks by accumulation of implicit priors. *Journal of Vision*, 24(10):639, 2024.

Dmitry Ulyanov, Andrea Vedaldi, and Victor Lempitsky. Instance normalization: The missing ingredient for fast stylization. *arXiv preprint arXiv:1607.08022*, 2016. URL https://arxiv.org/abs/1607.08022.

Rüdiger von der Heydt, Esther Peterhans, and Günter Baumgartner. Illusory contours and cortical neuron responses. *Science*, 224(4654):1260–1262, 1984. doi: 10.1126/science.6539501.

Johan Wagemans, James H. Elder, Michael Kubovy, Stephen E. Palmer, Mary A. Peterson, Manish Singh, and Rüdiger von der Heydt. A century of Gestalt psychology in visual perception: I. perceptual grouping and figure–ground organization. *Psychological Bulletin*, 138(6):1172–1217, 2012. doi: 10.1037/a0029333.

Yuxin Wu and Kaiming He. Group normalization. In Vittorio Ferrari, Martial Hebert, Cristian Sminchisescu, and Yair Weiss (eds.), *Computer Vision – ECCV 2018*, volume 11217 of *Lecture Notes in Computer Science*, pp. 3–19, Cham, 2018. Springer International Publishing. doi: 10.1007/978-3-030-01261-8_1.

Matthew D. Zeiler and Rob Fergus. Visualizing and understanding convolutional networks. In David Fleet, Tomas Pajdla, Bernt Schiele, and Tinne Tuytelaars (eds.), *Computer Vision – ECCV 2014*, volume 8689 of *Lecture Notes in Computer Science*, pp. 818–833, Cham, 2014. Springer International Publishing. doi: 10.1007/978-3-319-10590-1_53.

Yuyan Zhang, Derya Soydaner, Fatemeh Behrad, Lisa Koßmann, and Johan Wagemans. Investigating the Gestalt principle of closure in deep convolutional neural networks. In *European Symposium on Artificial Neural Networks, Computational Intelligence and Machine Learning (ESANN)*, pp. 679–684, 2024. doi: 10.14428/esann/2024.es2024-111. arXiv:2411.00627.

Yuyan Zhang, Derya Soydaner, Lisa Koßmann, Fatemeh Behrad, and Johan Wagemans. Finding closure: A closer look at the Gestalt law of closure in convolutional neural networks. *Computational Brain & Behavior*, 9(1):104–116, 2025. doi: 10.1007/s42113-025-00251-x. Preprint arXiv:2408.12460 (2024).

A Results for every network

Table 3 gives every classifier's accuracies and closure indices, Table 4 the critical gaps against ERF_{50}, and Table 5 the repeated training seeds.

B Gap dynamics for every network

Table 6 gives the net gap similarity of Section 7.1.3 for every network.

C Training details

fGRU cells used 9×9 horizontal kernels at stride 4 and kernels of 7, 5 and 3 at deeper stages of the deep hierarchies. Relevance maps followed Chefer et al. (2021a): for each block, $\bar{A} = \mathbb{E}_h[(\nabla A \odot A)^+]$ with ∇A the gradient of the class logit, and $R \leftarrow R + \bar{A}R$ from $R = I$, the map being the class-token row of R; unlike attention rollout (Abnar & Zuidema, 2020) it is class-specific, and unlike Chefer et al. (2021b) it uses no layer-wise relevance propagation; for looped transformers, loops are unrolled so each executed block contributes once.

Network	complete	frag.	CI frag.	scrambled	conflict: pos.	conflict: ori.	Kanizsa	CI Kan.	Brain-Score*
VGG16 (frozen)	0.99	0.25	+0.14	0.10	0.10	0.00	0.14	+0.02	0.37
ResNet50 (frozen)	0.97	0.41	+0.27	0.12	0.15	0.04	0.20	-0.00	0.38
ViT-B/16 (frozen)	1.00	0.62	+0.43	0.26	0.12	0.16	0.29	+0.10	–
DINOv2-S (frozen)	0.97	0.63	+0.51	0.20	0.02	0.08	0.19	+0.00	0.35
ResNet50	1.00	0.66	+0.53	0.14	0.27	0.15	0.18	+0.04	0.34
ViT-B/16	1.00	0.82	+0.60	0.27	0.05	0.18	0.29	+0.08	–
DINOv2-S	1.00	0.94	+0.79	0.30	0.03	0.33	0.25	+0.10	0.34
Looped ViT-B/16	1.00	0.81	+0.60	0.39	0.04	0.42	0.39	+0.09	–
Looped DINOv2-S	1.00	0.97	+0.80	0.36	0.09	0.30	0.20	+0.03	0.31
fGRU-H	1.00	0.32	+0.13	0.12	0.14	0.12	0.15	-0.00	–
fGRU-H (T=1)	1.00	0.30	+0.18	0.18	0.09	0.11	0.14	+0.00	–
fGRU-H, 1x1 readout	1.00	0.78	+0.63	0.68	0.05	0.70	0.07	-0.00	–
γ-net	1.00	0.58	+0.36	0.41	0.08	0.52	0.04	+0.00	–
γ-net (T=1)	1.00	0.68	+0.50	0.56	0.09	0.53	0.12	+0.04	–
deep fGRU-H D8	1.00	0.54	+0.36	0.11	0.11	0.00	0.16	+0.06	–
deep fGRU-H D8 (T=1)	1.00	0.51	+0.36	0.10	0.10	0.00	0.11	+0.03	–
deep fGRU D4	1.00	0.55	+0.42	0.10	0.16	0.00	0.17	+0.06	–
deep fGRU D4 (T=1)	1.00	0.51	+0.34	0.15	0.10	0.03	0.18	+0.03	–
DN fGRU stack L4	1.00	0.51	+0.29	0.24	0.15	0.12	0.13	-0.01	–
DN fGRU stack L4 (T=1)	1.00	0.49	+0.30	0.24	0.01	0.16	0.09	+0.01	–
DN fGRU stack L8	1.00	0.41	+0.26	0.29	0.12	0.10	0.10	-0.00	–
DN fGRU stack L8 (T=1)	1.00	0.61	+0.40	0.40	0.01	0.21	0.11	-0.00	–
VGG16 (Zhang data)†	0.29	0.19	+0.06	0.12	0.11	0.01	0.17	+0.00	–
ResNet50 (Zhang data)†	0.39	0.28	+0.15	0.13	0.13	0.03	0.10	+0.00	–

Table 3: All networks. complete: accuracy on complete outlines; frag. and CI: accuracy and closure index for corner-removed fragments of the main battery; scrambled and cue conflict: the orientation-control battery (a different set of fragments, so scrambled cannot be subtracted from frag.); Kanizsa: accuracy and closure index; Brain-Score*: mean of public-split V1, V2, V4, IT and behavioural benchmarks run locally (Schrimpf et al., 2020) (not the leaderboard score; computed for a subset of networks, "–" otherwise). †Trained on the stimuli of Zhang et al. (2024) (validation accuracy 0.88 and 0.85 on those stimuli); they transfer poorly to our battery.

D Attribution and faithfulness

Shapley values over pieces were estimated with 16 antithetic permutations over the stimulus's own pieces; the value of a set of pieces is the log probability of the class when only those pieces are drawn, a piece being removed by drawing it in the background colour. Occlusion slides a $P \times P$ background-coloured window at stride $P/2$ and maps the drop in log probability of the true class. The deletion test removes contour pixels (pixels differing from the background by more than 20 grey levels) in ten steps, either in the order of the map or in random order (three random orders), and records the probability of the true label; faithfulness is the area under the random-order curve minus the area under the map-order curve, so positive values mean the map ranks the contour pixels that matter. Only contour pixels are deleted, because deleting background does not change a line drawing.

E Contour-tracing probe

The tracing networks come from a companion project in which hGRUs (with batch or group normalization), an fGRU/γ-net and weight-tied looped local-attention transformers, and an untied 24-block transformer, were trained to segment the dashed contour on which a marker dot sits (128 px images, contours of 6–14 dashes). We drew polygon outlines in the same style, removed a gap from the middle of every edge and placed the marker on one corner piece; in the control every unmarked piece is rotated about its own centre. *Traced* is the share of each unmarked piece's pixels labelled as part of the marked contour (probability above 0.5), averaged over pieces; *gap coverage* is the share of the removed contour labelled as contour, excluding the marker. Linking by continuation predicts traced(fragments) > traced(control); completion predicts gap coverage > 0. The segment-pair probes place a marked and a target segment, aligned or rotated, at increasing extra gaps, straight or along a curve.

Network	family	ERF_{50} or P (px)	g^* (px)	$g^*/(\mathrm{ERF}_{50}$ or $P)$
VGG16 (frozen)	CNN	38	6	0.16
ResNet50	CNN	60	20	0.34
ResNet50 (frozen)	CNN	66	14	0.22
deep fGRU-H D8 (T=1)	T=1 control	16	9	0.58
fGRU-H (T=1)	T=1 control	19	15	0.78
γ-net (T=1)	T=1 control	35	39	1.09
DN fGRU L4 (T=1)	T=1 control	43	11	0.25
DN fGRU L8 (T=1)	T=1 control	60	51	0.84
deep fGRU D4 (T=1)	T=1 control	65	38	0.59
Looped ViT-B/16	ViT (patch)	16	54	3.39
ViT-B/16 (frozen)	ViT (patch)	16	55	3.45
DINOv2-S	ViT (patch)	14	$\geq$56	$\geq$4.00
ViT-B/16	ViT (patch)	16	$\geq$64	$\geq$4.00
DINOv2-S (frozen)	ViT (patch)	14	51	3.64
Looped DINOv2-S	ViT (patch)	14	$\geq$56	$\geq$4.00
deep fGRU-H D8	recurrent	16	10	0.62
fGRU-H, 1×1 readout	recurrent	18	50	2.82
fGRU-H	recurrent	18	15	0.82
γ-net	recurrent	59	29	0.49
deep fGRU D4	recurrent	70	27	0.39
DN fGRU L4	recurrent + DN	58	17	0.30
DN fGRU L8	recurrent + DN	74	44	0.60

Table 4: Critical gap g^* against ERF_{50} (patch size P for transformers) for every network, including the recurrent variants and single-iteration controls not drawn in Fig. 22.

Network	seeds	CI corner	CI mid	arrangement	CI Kanizsa	Kanizsa range
ViT-B/16	3	+0.62 ± 0.03	+0.42 ± 0.04	+0.54 ± 0.05	+0.11 ± 0.03	+0.08 to +0.14
DINOv2-S	3	+0.78 ± 0.01	+0.43 ± 0.01	+0.64 ± 0.02	+0.09 ± 0.07	+0.01 to +0.15
ResNet50	5	+0.52 ± 0.03	+0.23 ± 0.06	+0.51 ± 0.03	+0.02 ± 0.05	-0.05 to +0.07

Table 5: Repeated training seeds: mean ± s.d. of the closure indices and the arrangement effect.

F Recirculation

Following Mozer et al. (2026), after each pass the per-token activation of a deep source block s is rescaled to the norm of a shallow destination block d and added, $z_d \leftarrow z_d + \alpha\, z_s\, \|z_d\|/\|z_s\|$, and the blocks from d upward are run again with frozen weights. For each of the four transformers (fine-tuned and frozen ViT-B/16 and DINOv2-S; 12 blocks, numbered from 0) we swept $(s, d) \in \{(10, 4), (10, 6), (11, 4), (8, 2), (8, 4)\}$, mixture coefficients $\alpha \in \{0.05, 0.15, 0.3, 0.6\}$ and 1, 2 or 4 extra passes (60 settings) on at most 640 main-battery images per condition.

G Turning the inducers

Each display keeps the polygon, inducer radius and position of a Kanizsa figure in the battery; only the inducers turn, each about its own centre. The training outlines already cover every rotation and the size of the shape the inducers specify (circumradius $4.5P$, ±15%), so that shape is familiar to every network. In transformers the Kanizsa advantage over the 180° control falls from +0.15 (median, all radii) to +0.07 at 15° and is +0.08 at 45°; at radius ≥ 1.5 patches, where the inducers nearly touch, it falls from +0.25 to +0.13 and +0.08 (fine-tuned ViT-B/16: +0.47, +0.24, +0.25; Fig. 30). Turned in alternating directions, the transformers' advantage is +0.02 at 15° and +0.01 at 45°. In CNNs and recurrent circuits no turn moves accuracy away from the control by more than 0.11. An advantage that survives a 45° turn in the same direction does not depend on the inducer edges lining up with one another. The contour units on the

	peak, gap (patches)			last layer / iteration			Kanizsa edge	
Network	1	2	4	1	2	4	peak	last
ViT-B/16	+0.24	+0.16	+0.15	-0.02	-0.01	+0.04	+0.09	+0.01
ViT-B/16 (frozen)	+0.18	+0.10	+0.11	-0.05	+0.00	+0.02	+0.07	+0.03
Looped ViT-B/16	+0.26	+0.23	+0.10	+0.07	+0.07	+0.02	+0.05	-0.01
DINOv2-S	+0.43	+0.37	+0.21	+0.08	+0.08	+0.07	+0.13	+0.06
Looped DINOv2-S	+0.48	+0.43	+0.21	+0.11	+0.10	+0.05	+0.20	+0.05
ResNet50	+0.38	+0.10	+0.02	+0.35	+0.04	-0.00	+0.06	+0.06
fGRU-H	+0.09	+0.03	+0.06	+0.09	+0.03	+0.05	+0.05	+0.04
fGRU-H, 1x1 readout	+0.39	+0.20	+0.11	+0.38	+0.19	+0.11	+0.16	+0.16
γ-net	+0.30	+0.21	+0.10	+0.30	+0.21	+0.10	+0.19	+0.19
deep fGRU-H D8	-0.06	+0.04	+0.01	-0.06	+0.04	+0.01	+0.33	+0.32
deep fGRU D4	-0.00	-0.03	-0.01	-0.02	-0.03	-0.01	+0.30	+0.30
DN fGRU L4	+0.30	+0.26	+0.14	+0.28	+0.25	+0.13	+0.27	+0.24
DN fGRU L8	+0.26	+0.25	+0.10	+0.17	+0.12	-0.06	+0.28	+0.13

Table 6: Net gap similarity (Section 7.1.3) for every network: peak over layers (iterations for recurrent circuits) and value at the last layer or iteration, at gaps of 1, 2 and 4 patches; on Kanizsa illusory edges, the largest value over inducer radii. Recurrent hierarchies are read at the classifier's layer (for the deep fGRU-H its 7×7 top layer, where few gaps leave an empty cell: 0, 3, 19, 45, 47 usable triplets at gaps of 0.5, 1, 2, 3 and 4 patches).

Family	epochs	batch	backbone lr	new modules lr	head lr	precision
Frozen backbones (linear probe, Adam)	25	64 / 32	–	–	10^{-3}	bf16 / fp32
Fine-tuned ViT-B/16	15	32×2	10^{-5}	–	10^{-3}	bf16
Fine-tuned DINOv2-S/14	15	64	10^{-5}	–	10^{-3}	bf16
Fine-tuned ResNet50	15	64	10^{-4}	–	10^{-3}	bf16
Looped ViT-B/16	15	16×4	10^{-5}	10^{-4}	10^{-3}	bf16
Looped DINOv2-S	15	32×2	10^{-5}	10^{-4}	10^{-3}	bf16
fGRU-H, fGRU-H 1×1, γ-net	20	32	10^{-4}	10^{-3}	10^{-3}	bf16
Deep fGRU-H D8, deep fGRU D4	20	16×2	10^{-4}	10^{-3}	10^{-3}	bf16
DN fGRU stacks	20	16×2	10^{-4}	3×10^{-4}	10^{-3}	fp32

Table 7: Training hyperparameters. Batch $a \times b$: a images with gradient accumulation over b steps; for the frozen backbones, 64 for feature extraction (bf16 autocast) and 32 for training the head (fp32); the per-layer probes of Fig. 6 train their heads with batches of 64. End-to-end training used AdamW (weight decay 0.05), 5% linear warm-up with cosine decay, label smoothing 0.1 and gradient clipping at norm 1, keeping the checkpoint with the best validation accuracy. "New modules" are the recurrent cells, 1×1 reductions, inter-stage convolutions, top-down projections, readouts and the looped transformer's injection; in the DN stacks the divisive-normalization parameters were trained at the backbone rate. Single-iteration controls use the same settings. The DN stacks were trained and evaluated in full precision: in bf16 the eight-layer stack's accuracy on complete outlines falls from 1.00 to 0.44.

edges fall by a median 0.04 from 0° to 45° (14 networks whose units respond to faint real lines), less than the null observer (0.09). Retrained with inducers in half of the training outlines (aligned or turned 180°, at the battery's radii), the median advantage at radius ≥ 1.5 patches is +0.01 with aligned inducers, and transformers score below the 180° control at 45° (-0.24).

H Shapley interactions over the inducers

Shapley values of image regions cannot test completion: removing an empty region replaces background with background, so its value is zero by construction. What closure predicts instead is an interaction among the inducers: a set of aligned inducers should be worth more than its members alone. Taking the n inducers of a polygon ($n = 3$–6, radii $\frac{1}{2}$–2 patches) as players, we scored every subset by the log probability of the true class (at most 2^6 images per figure, so the values are exact) and compared aligned inducers with the same inducers turned 180° in place. As a positive control we ran the same analysis on the n corner pieces of mid-gap fragments (gaps of 1 and 2 patches), aligned or each turned 180° about its own pixel centroid. In the 16 networks trained on outlines, fragment corners are superadditive relative to their control (median synergy +0.08 nats, positive in 12), Kanizsa inducers are not (-0.02, positive in 7), and fragments exceed Kanizsa

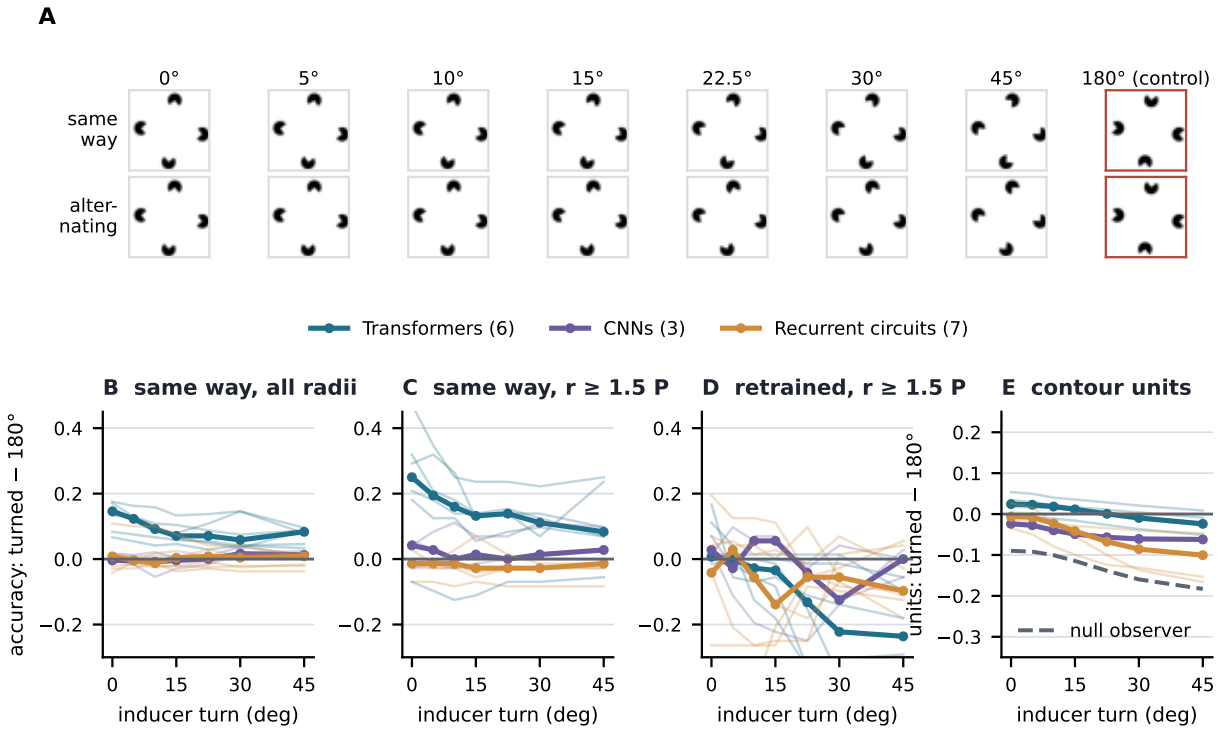


Figure 30: Turning the inducers. (A) Every inducer turned in place by 0–45° (0°: the Kanizsa figure), the same way (top; the fat/thin displays of Ringach & Shapley (1996)) or in alternating directions (bottom), against the 180° control of the main battery (red frame); a square with inducer radius 1 patch is shown. (B–D) Accuracy with turned inducers minus accuracy with the 180° control, inducers turned the same way: all radii (B), radius ≥ 1.5 patches (C), and the same networks retrained with inducers in their training images (D). Bold: median of each family; thin: each network. (E) The 5% most contour-selective units of each network, in the empty cells on the straight edges of the aligned figure, scaled so that an empty cell is 0 and a drawn contour 1; dashed, a null observer (oriented edge energy pooled over half a patch, scaled the same way).

figures in 14 of 16 networks (Wilcoxon $p = 0.009$; Fig. 31A). The exception is the transformers, whose aligned inducers add a little synergy (median +0.08, fragments +0.02), in line with their small Kanizsa advantage, which survives turning the inducers (Appendix G). Pairwise interaction indices (Grabisch & Roubens, 1999) of neighbouring inducers, which share an edge of the shape they specify, do not separate fragments from Kanizsa figures ($p = 0.19$). After retraining with notched discs, fragments no longer exceed Kanizsa figures (6 of 16; Fig. 31B).

I Masked-autoencoder reconstruction on the battery's displays

Before the protocol of Li et al. (2026), we tested the four MAEs on our own displays: Kanizsa squares and hexagons (circumradius 4.5 patches, support ratio 0.50–0.95, four orientations), the same with rotated inducers, and the aligned inducers with each gap closed by a 2 px line. Every 16 px patch crossed by a gap that holds no contour pixel in the Kanizsa or the rotated display was masked, the same patches for all three displays. We read the reconstruction's grey level (0–255) inside the masked patches: the mean 4–8 px from the gap segment minus the mean within 1.5 px of it (a dark line along the illusory edge), and the mean 3–8 px inside the polygon minus 3–8 px outside it (a brighter illusory surface). With aligned inducers the line measure is 0.4–0.8 grey levels in the four MAEs, with rotated inducers −0.3–0.0, and with the drawn line, visible outside the masked patches, 147–154. The surface measure is −0.3–1.6 with aligned and 0.0–0.5 with rotated inducers. For the MAE trained with visible occluders we also covered the gap patches with a visible mid-grey block instead of masking them: the line measure is then 0.8–1.0 with aligned inducers in all four

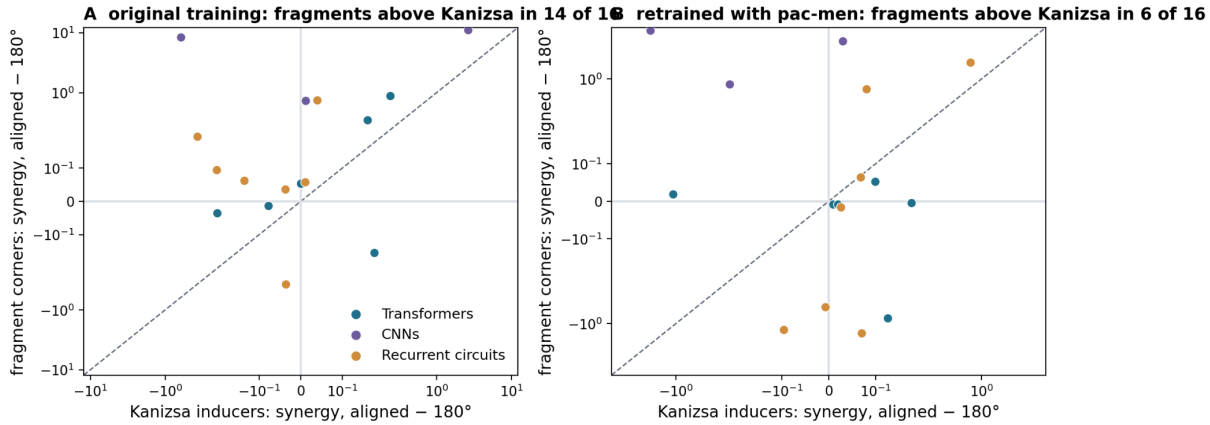


Figure 31: Whole-minus-parts synergy of the inducers, $v(N) - v(\emptyset) - \sum_i \big(v(\{i\}) - v(\emptyset)\big)$ with v the log probability of the true class, aligned minus the 180° control (median over figures; symmetric log axes), for fragment corners (vertical) against Kanizsa inducers (horizontal), one dot per network. (A) Original training. (B) The same networks retrained with inducers in their training images.

MAEs, and the MAE trained this way continues the drawn line (153 grey levels), while the others, for which a grey block is unfamiliar, reach about 50.

J Cued-object segmentation circuits

Architecture. We followed the Methods and Table S1 of Mollard et al. (2026). The backbone is the Panoptic FPN with a ResNet-101 trained on COCO panoptic segmentation (Kirillov et al., 2019) (Detectron2 model-zoo weights), frozen; only its stride-4 map (256 channels) is used, reduced to 64 channels by a 1×1 convolution. Four horizontal hGRU levels at strides 4, 8, 16 and 32 (64, 128, 256 and 512 channels, 5×5 horizontal kernels) are linked bottom-up by blocks of two or four 1×1 convolutions with group normalization and ReLU, each followed by 2×2 max-pooling, and top-down by three hGRUs with 1×1 kernels whose drive is the upsampled higher level (bilinear upsampling, group normalization, two 1×1 convolutions with ReLU) and whose output is mixed with the old state, $H_l = (1 - \sigma(\beta_l))\,\mathrm{hGRU}(H_l, H_{l+1}) + \sigma(\beta_l) H_l$. Each hGRU computes $G_S = \sigma(U_S * H)$, $C_S = \mathrm{GN}(W_S * (H \odot G_S))$, $S = [Z - [(\alpha H + \mu) \odot C_S]_+]_+$, $G_F = \sigma(U_F * S)$, $C_F = \mathrm{GN}(W_F * S)$, $\tilde{H} = [\nu(C_F + S) + \omega(C_F \odot S)]_+$, $H' = (1 - G_F) \odot H + G_F \odot \tilde{H}$, with U 1×1 convolutions, α, μ, ν, ω learnable scalars and $[\cdot]_+$ softplus (Table S1). The network runs for 40 timesteps; a 1×1 convolution to 128 channels, ReLU and a 1×1 convolution on the lowest level read out 14 classes (background and 13 object classes).

Training. Every network is trained to give every pixel of the cued object its class and every other pixel the background class, with per-pixel cross-entropy and the Lipschitz coefficient penalty of C-RBP (Linsley et al., 2020a) (weight 10^{-4}, as in the paper), by C-RBP: 39 steps without gradients, one with, and the gradient at the fixed point from a Neumann series of up to 15 vector–Jacobian products that stops, as in the reference implementation, when a term grows, the sum's norm exceeds 1 or a term becomes negligible; Adam, batch 28, 21,000 iterations, learning rate warmed up to 10^{-4} over 1,000 iterations and multiplied by 0.8 at 10,000 and 20,000; 224×224 images, objects of at least 5,000 pixels, three random cue positions per object, the class *person* subsampled to one in six. Single-step controls are trained by ordinary back-propagation through their one step, without the penalty.

Where the paper is silent or we could not follow it: (i) the cue enters as an extra input, a disc of radius 8 px added after the 1×1 reduction; (ii) we trained on all COCO training images with objects of the 13 most frequent classes after subsampling (person, dining table, chair, car, bowl, couch, cup, truck, cat, bus, bed, motorcycle, train), not on the image set of Adeli et al. that the paper used, and evaluated on 2,000 COCO validation objects rather than on that set; (iii) group normalization uses 8 groups, and the network has about

20 million trainable parameters against about 8 million in the paper; (iv) eq. 4 of the paper has no inner rectification, $S = [Z - (\alpha H + \mu) \odot C_S]_+$; since C_S is group-normalized and can be negative, the suppression then excites, and from some initializations the state grew without bound and training reached non-finite values within 12 iterations, so we rectify the suppressive term as in the hGRU the paper cites (Linsley et al., 2018); (v) initial values (orthogonal kernels, gate biases initialized as in our fGRU, $\alpha = \omega = 0.1$, $\mu = \nu = 0$, $h_0 = 0$) are not given in the paper; (vi) the frozen backbone runs in bfloat16, the recurrence in 32-bit floating point; (vii) we clip gradients at norm 10 and use a guard that restores the last checkpoint and halves the learning rate if the loss is not finite or the penalty exceeds 20 times its recent median (in the first 20 iterations, 20 times its first value and at least 1,000), and skips a batch whose cross-entropy exceeds 25 while the penalty is normal; it fired in two runs (9×9 kernel: iterations 18,630 and 20,756; fGRU on ViTDet: iteration 20,780) and never skipped a batch. We trained one network per setting; the paper trained five.

Variants. (a) Penalty weight 100, the value given in the documentation of the reference C-RBP code. (b) Finest-level horizontal kernels of 3×3, 9×9 and 15×15 (12, 36 and 60 px at stride 4) in place of 5×5; the other levels keep 5×5. In the 9×9 run the penalty rose from about 0.2 at iteration 12,000 to about 18 at 18,000, and the network reaches IoU 0.28 (0.18 on objects under 5% of the image); its checkpoint at iteration 12,000 reaches 0.25 (0.10) and labels 0.02 of the filled hexagon. (c) The fGRU cell of our other circuits (Section 5; group normalization in place of instance normalization) in place of the hGRU, with a gated top-down term added to the lower level in place of the paper's mixing, max-pooling and a 3×3 convolution between levels and a 64-channel read-out. (d) The same fGRU network on the stride-4 map (256 channels) of ViTDet ViT-B, a plain ViT pre-trained with masked autoencoding and trained with Mask R-CNN on COCO (Li et al., 2022) (Detectron2 weights). (e) A looped transformer: the 64-channel drive at stride 4 is embedded in 2×2 patches into 784 tokens of width 512 with a learned position embedding, and one pre-norm transformer block (8 heads, MLP width 2,048) is applied 40 times to the state plus the embedded input, $h_{t+1} = \mathrm{Block}(h_t + e)$, $h_0 = 0$; a layer norm, a linear map to 64 channels, upsampling to stride 4 and the read-out of (c) follow. Attention is written out because fused attention kernels have no double backward, which the penalty needs; the network has 3.7 million parameters. (f) Single-step controls of the hGRU network and of (e), trained and read out after one step.

Displays and measures. For the battery test each display (224×224, the polygon of circumradius $4.5P$ centred, $P = 16$ px, $n = 3$–6, inducer radius 1, 1.5 and 2 patches, four orientations) was shown with the cue at the polygon's centre, and the network's segmentation (argmax over classes other than background, at the read-out step) was compared with the polygon. For the gap test, regular pentagons and hexagons of circumradius 3.75 and 4.5 patches (the main text reports the hexagon of 4.5 patches) were drawn with inducers of radius $r = s\,L/2$ at every vertex, for support ratios $s = 0.50$–0.95 (L the side length, gaps $(1 - s)L$), four orientations. Controls keep the vertices and r: rotated inducers (turned 180° in place), full discs, segments 8 px thick along both sides from every vertex, of length r (corners), the aligned inducers with the stretch of each side between them drawn as a 2 px line, solid or dashed (4 px dashes, 4 px spaces), a blank page and a mid-grey filled polygon. Fill is the share of the polygon's white pixels, beyond one patch from the cue, that the network labels as the cued object; the strip just outside a gap lies 0.25–1 patch outside a side, along the gap's span. Closing the gaps with 8 px bars instead of 2 px lines stops the label in every 40-step network (at most 0.44), so the thin lines serve as the closed-gap control. For reversed contrast the displays' grey levels are inverted before the backbone and the regions are taken from the original display.

Without a cue. With the cue map set to zero on the large hexagon (gaps 4–36 px), the fill at the read-out step is 0.00 on every display for the looped transformer (cued: Kanizsa 0.80, 0.76, 0.72 and 0.49 at 4–14 px; closed line up to 0.82; blank page 0.62), its single-step control (cued: blank page 1.00) and the 15×15 hGRU (cued: Kanizsa 0.24–0.28 up to 11 px). The 3×3 hGRU fills the Kanizsa hexagon uncued at 0.68, 0.56, 0.59 and 0.49 at 4–14 px (cued 0.61, 0.47, 0.53, 0.39), the closed line from 0.70 at 4 px to 0.32 at 29 px (cued 0.62 to 0.18), rotated inducers 0.00 (cued 0.16–0.17 at 4–7 px) and full discs 0.00 (cued 0.58 to 0.28); the single-step hGRU fills the Kanizsa hexagon uncued at 0.96 up to 14 px, then 0.33 and 0.04 (cued 0.97, 0.84, 0.24, 0.09), full discs 1.00 to 0.80 (cued 0.80 to 0.58) and rotated inducers 0.42 falling to 0.01 (cued 0.38 to 0.06); the fGRU with the weighted loss fills the Kanizsa hexagon uncued at 0.72, 0.65, 0.61 and 0.21 (cued 0.63, 0.53, 0.47, 0.23), the closed line 0.73 to 0.18 at 22 px (cued 0.63 to 0.14), full discs 0.30 to 0.04 and rotated inducers 0.00 either way.

Earlier fGRU networks. Two networks with fGRU units on Panoptic FPN features, one with the paper's settings and one with object pixels weighted three times in the cross-entropy, were trained before the corrections above: their Neumann series summed all 15 terms. In the first, the guard fired once, at iteration 19,038, so its last 2,000 iterations used half the learning rate; a run before it, without the guard, diverged at about iteration 16,000. On COCO validation objects they reach IoU 0.43 and 0.50 (0.35 and 0.35 for objects under 5% of the image, 0.50 and 0.71 above 30%; Fig. 32). In the battery's figures (48 figures) both segment the filled surface (mean IoU 0.48 and 0.55) but barely the Kanizsa figure (0.03 and 0.10), little more than with rotated inducers (0.02 and 0.07) or on a blank page (0.01 and 0.03), and over 40 timesteps the cue's effect on the lowest level, $e_t = \text{mean}_c|h_t(\text{cue}) - h_t(\text{no cue})|$, spreads past the polygon, with indistinguishable maps for aligned and rotated inducers (Fig. 32A); a thin real outline is not segmented either (0.01 and 0.08), and a display combining it with aligned inducers reaches 0.10 and 0.17. Pooled over polygons with $n = 3$–6 sides at three sizes (circumradius 3–4.5 patches), neither network fills more than 0.18 and 0.10 of any Kanizsa figure at support ratios up to 0.70; fills of 30% or more begin at 0.75 and 0.80 (Fig. 33B,C), and at support ratios of 0.80 and above full discs fill as much as aligned inducers on average (paper settings 0.47 against 0.34; weighted loss 0.36 against 0.37). These pooled means hide the dependence on size: in the large hexagon the network trained with the weighted loss fills 0.68 of the Kanizsa hexagon with 7 px gaps and 0.19 between full discs (Table 1). Across the five notched inducers of the configural-effect shape audit (Fig. 27B), which specify the same polygon, aligned notched crosses fill nothing (0.00 and 0.00), aligned rings fill less than rings without their notch (0.26 against 0.68; 0.09 against 0.30), and only notched squares and hexagons fill more aligned than without their notch (squares 0.18 against 0.02 and 0.26 against 0.10; support ratio ≥ 0.8, displays whose overlapping inducers seal the interior, 12%, excluded; Fig. 33D,E).

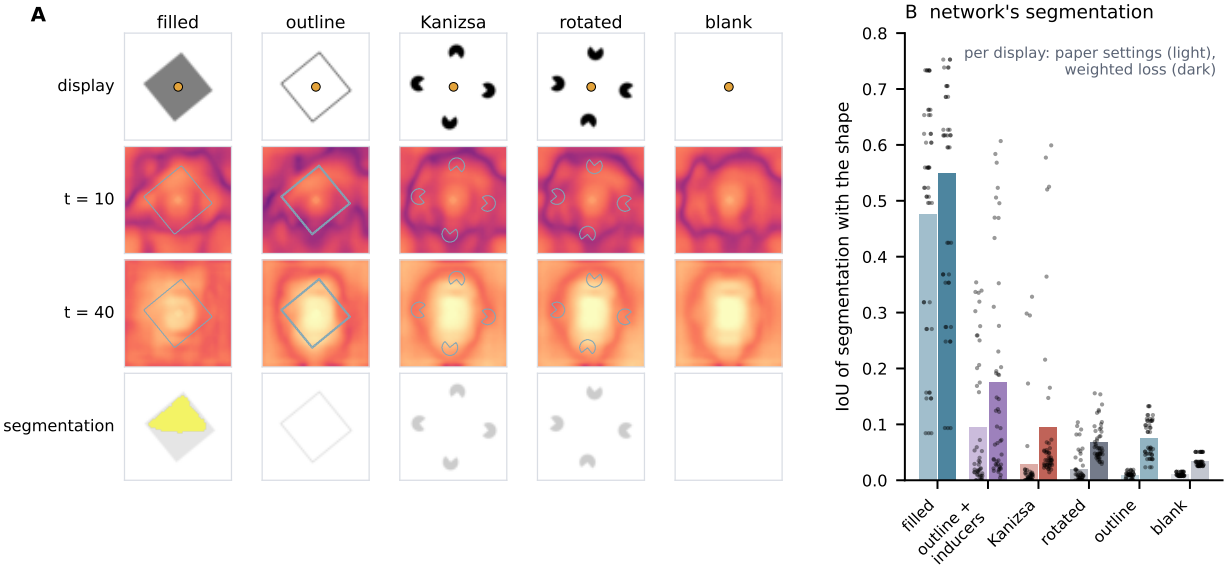


Figure 32: The earlier fGRU networks on the battery, cue (orange) at the centre of a polygon. (A) Paper settings, square with inducer radius 1 patch: displays; the cue's effect on the lowest recurrent level, $|h_t(\text{cue}) - h_t(\text{no cue})|$ averaged over channels, after 10 and 40 timesteps (one logarithmic colour scale for all maps; blue lines, the display's contour pixels); and the region the network labels as the cued object (yellow). (B) Intersection over union of that region with the polygon for each display ($n = 3$–6 sides, inducer radius 1–2 patches, four orientations); bars, mean for the network trained with the paper's settings (light) and with object pixels weighted three times in the loss (dark); dots, single figures.

Table 8: Terms used in this paper.

term	meaning here
patch, P	the square of pixels a vision transformer turns into one token (16 px for ViT-B/16 and the CNN reference, 14 px for DINOv2); every gap, inducer and dash is sized in units of P
inducer	a notched disc of a Kanizsa figure; rotated inducers are the same discs turned by 180° in place
support ratio	the share of a figure's side that is bounded by contour pixels (Shipley & Kellman, 1992); 0.5 means half of each side is specified by the inducers
closure index	accuracy on a stimulus minus accuracy on its rotated control, for the classifiers
rotated control	the same pieces, each turned by a fixed pseudo-random angle about a point inside it, so that the ink and its orientation statistics are kept and the arrangement is broken
closed discs, full discs	the inducers with their notches closed by hairlines; solid discs at the same positions
gap similarity	how much more the empty cells on the missing contour resemble the same cells for the complete outline than they do for the rotated control (cosine similarity)
contour probe	a logistic regression on a layer's feature-map cells, fitted on complete outlines, that labels a cell as holding drawn contour; read on the empty cells where an illusory edge would be
separation probe	one logistic read-out per token of an untouched encoder, fitted on clip frames to mark the object, read on the line drawings as the probability inside the figure minus outside
cue	for the cued circuits and feedback networks: a point in the image (a disc of radius 8 px, placed by us at the figure's centre) at which the network is asked what object lies; it names a location, it does not create the label
fill, interior fill	the share of a figure's interior pixels that a cued network labels as the cued object (0–1); spill, the same on the strip just outside the figure; false fill, the fill of a scrambled contour in the validation set
time course	the fill read at every recurrent step
single-step, single-iteration control	the same architecture trained with one step or iteration instead of 8, 40 or 80
seed	an independently trained copy of the same network from a different random start; "seed 0" and "seed 1" are two such trainings, not different inputs
front, looped blocks	the frozen early blocks of a pretrained encoder, run once; its last blocks, copied and run repeatedly with feedback
layer labels	per-pixel classes of the training clips: visible part of the cued object, hidden part, occluder, background
unity, outline, texture, faded-cover clips	the synthetic training clips of the feedback networks (an object moving behind flat covers); variants with 30% of objects drawn as outlines, filled with texture, or with covers blended into the background
veto	our reading of what the loop does in the feedback networks: the single pass proposes a label on figure and non-figure displays alike, and the loop removes it from the non-figures
null observer	a measurement device that cannot close contours (pixels, a Gabor energy bank, an untrained network, the pixel-growth histogram) used to audit published closure measures

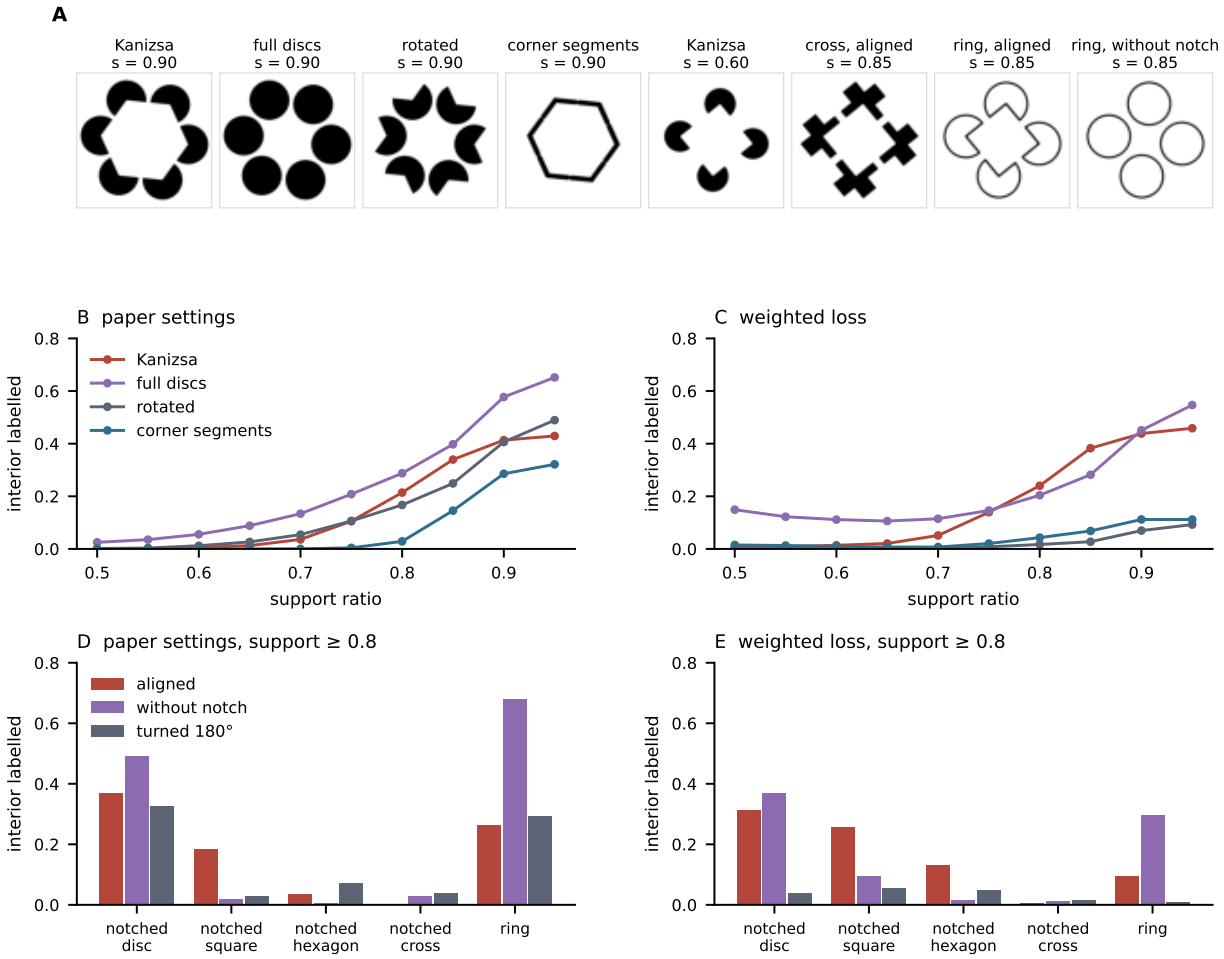


Figure 33: Support ratio and inducer shape in the earlier fGRU networks. (A) Examples: a hexagon at support ratio $s = 0.90$ shown with aligned inducers, full discs, rotated inducers and thick corner segments; a Kanizsa square at $s = 0.60$; aligned notched crosses, aligned rings and rings without their notch at $s = 0.85$. (B, C) Share of the polygon's white pixels labelled as the cued object (fill), against support ratio ($n = 3$–6, circumradius 3–4.5 patches, four orientations), for the network trained with the paper's settings (B) and with the weighted loss (C). (D, E) Fill at $s \geq 0.8$ for five notched inducers that specify the same polygons: aligned, without the notch and turned 180° about the vertex; displays whose overlapping inducers seal the interior are excluded.

K Glossary

L Feedback networks on frozen fronts: details and every run

Architecture. The looped blocks are deep copies of the encoder's last blocks, so that at the first step the network is the pretrained encoder with a cue added; the feedback map (layer norm, linear $D \to D$) is zero-initialized; the two read-outs (object label, layer label) are a layer norm, a linear map to 64 channels and two 1×1 convolutions with a ReLU. The looped blocks, the cue and feedback maps and the read-outs are trained; the front is not, except in the LeVJEPA whole-encoder variant. Resolutions: V-JEPA 2 and 2.1 at 512 px (32×32 tokens), LeVJEPA at 224 px (14×14), DINOv3 at 448 px (28×28), SAM 3 at 336 px (24×24); the hybrids at 336 (SAM 3 front) and 384 px (2.1 front), both 24×24. Inputs are normalized with each encoder's own statistics.

Table 9: Every selected feedback network: interior fill (share of the figure's interior pixels labelled as the cued object, 0–1) at step 80 on the outline (whole square drawn), line (gaps drawn), Kanizsa, rotated-inducer and closed-disc displays; Kanizsa minus closed; drawn corners turned by 180°; full discs; Kanizsa fill with the feedback cut from step 1; the chosen iteration, its validation IoU and false fill. Two seeds per cell where trained; "unity clips" are the earlier clips without outlines (there both the selected checkpoint and the checkpoint at 4,000 iterations are listed where they differ); the no-loop networks are read at their last checkpoint.

family	stage-2 clips	seed	outline	line	Kanizsa	rotated	closed	K−closed	corners rot.	discs	K, FB off	it	val IoU	false fill
SAM 3	outline clips	0	1.00	1.00	0.62	0.00	0.08	0.53	0.00	0.00	0.65	3,500	0.92	0.09
	outline clips	1	1.00	1.00	0.55	0.00	0.22	0.33	0.00	0.00	0.93	3,750	0.92	0.08
	faded covers	0	1.00	1.00	0.54	0.01	0.01	0.53	0.01	0.00	0.66	2,500	0.92	0.10
	faded covers	1	1.00	1.00	0.07	0.00	0.00	0.07	0.00	0.00	0.75	3,750	0.89	0.07
DINOv3 ViT-B/16	outline clips	0	1.00	1.00	0.69	0.00	0.00	0.69	0.00	0.00	0.51	1,750	0.89	0.44
	outline clips	1	0.57	0.94	0.34	0.00	0.00	0.34	0.00	0.00	0.53	3,750	0.71	0.14
V-JEPA 2.1 L	outline clips	0	1.00	0.97	0.33	0.01	0.00	0.33	0.25	0.06	0.29	1,750	0.88	0.12
	outline clips	1	1.00	0.98	0.27	0.00	0.00	0.27	0.81	0.50	0.18	3,000	0.91	0.24
	faded covers	0	0.96	0.92	0.30	0.00	0.00	0.30	0.83	0.00	0.11	1,750	0.89	0.21
	faded covers	1	0.99	0.96	0.33	0.00	0.00	0.33	0.87	0.00	0.13	2,500	0.86	0.12
	no layer labels	0	1.00	0.98	0.09	0.05	0.01	0.07	0.95	0.05	0.01	1,750	0.78	0.70
	no layer labels	1	1.00	1.00	0.69	0.95	0.98	-0.29	0.76	0.91	0.09	1,000	0.71	0.68
	texture clips	0	0.91	0.77	0.22	0.08	0.00	0.21	0.76	0.18	0.36	2,500	0.78	0.14
	texture clips	1	0.98	0.92	0.33	0.06	0.02	0.30	0.93	0.40	0.34	2,750	0.82	0.22
	outline + texture	0	0.99	0.97	0.46	0.01	0.07	0.39	0.74	0.09	0.56	3,500	0.89	0.13
	outline + texture	1	1.00	1.00	0.38	0.01	0.07	0.31	0.72	0.24	0.54	2,750	0.92	0.12
	unity clips	0	0.83	0.72	0.19	0.01	0.00	0.18	0.04	0.06	0.10	1,000	0.74	0.15
	unity clips	0	0.45	0.40	0.00	0.00	0.00	0.00	0.00	0.00	0.03	1,000	0.74	0.15
	unity clips	1	0.99	0.90	0.22	0.13	0.01	0.22	0.00	0.11	0.12	3,500	0.75	0.24
V-JEPA 2 L	outline clips	0	0.95	0.83	0.37	0.00	0.40	-0.02	0.57	0.00	0.63	1,250	0.80	0.24
	outline clips	1	0.34	0.34	0.28	0.00	0.08	0.20	0.00	0.00	0.10	250	0.29	0.06
	texture clips	0	0.99	0.85	0.29	0.00	0.18	0.11	0.00	0.00	0.64	2,750	0.67	0.11
	texture clips	1	0.78	0.94	0.30	0.00	0.03	0.28	0.00	0.00	0.72	1,750	0.75	0.21
	outline + texture	0	0.99	0.97	0.27	0.00	0.25	0.02	0.82	0.00	0.74	2,500	0.86	0.23
	outline + texture	1	0.45	0.15	0.19	0.03	0.05	0.13	0.06	0.02	0.50	500	0.37	0.15
	unity clips	0	0.93	0.78	0.41	0.00	0.26	0.15	0.00	0.00	0.79	1,000	0.73	0.22
	unity clips	0	0.99	0.96	0.40	0.00	0.07	0.33	0.00	0.00	0.86	1,000	0.73	0.22
	unity clips	1	0.96	0.93	0.74	0.00	0.70	0.04	0.00	0.00	0.80	2,000	0.76	0.47
	unity clips	1	0.96	0.92	0.51	0.00	0.44	0.07	0.00	0.00	0.82	2,000	0.76	0.47
	unity clips, no loop	0	0.98	0.93	0.77	0.00	0.54	0.23	0.02	0.00	0.77	–	–	–
	unity clips, no loop	1	0.99	0.98	0.73	0.00	0.42	0.31	0.00	0.00	0.73	–	–	–
LeVJEPA L, whole encoder	outline clips	0	0.98	0.93	0.06	0.00	0.74	-0.67	0.00	0.00	0.07	2,000	0.72	0.25
	outline clips	1	0.96	0.97	0.01	0.00	0.64	-0.64	0.07	0.00	0.06	2,750	0.51	0.17
	outline + texture	0	1.00	0.60	0.00	0.00	0.65	-0.65	0.00	0.00	0.00	3,500	0.72	0.19
	texture clips	0	0.98	0.46	0.00	0.00	0.25	-0.25	0.00	0.00	0.12	3,750	0.51	0.20
LeVJEPA L, frozen front	outline clips	0	0.04	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	3,750	0.09	0.00
	outline clips	1	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	4,000	0.04	0.00
2.1 front + SAM 3 back	outline clips	1	0.85	0.90	0.06	0.00	0.00	0.06	0.00	0.00	0.03	3,250	0.77	0.13
SAM 3 front + 2.1 back	outline clips	0	1.00	1.00	0.00	0.00	0.00	0.00	0.00	0.01	0.18	750	0.86	0.18

Training clips. The unity clips are eight frames of 224 px in which a COCO cut-out (object a) moves behind one or more flat-coloured covers, with a second object as distractor; the targets are the cued object's mask and, per pixel, one of four layer classes (visible part of the cued object, hidden part, occluder, background). The outline clips draw 30% of the objects as their contour only (1–3 px, contrasting with the background in 85% of cases); the texture clips fill the object with its own background's texture taken from another place; the faded-cover clips blend each cover with the background at an opacity drawn uniformly from 0.15 to 0.7 per clip. Earlier clip sets, whose networks are listed in Table 9 under "unity clips", had no outlines.

Checkpoint selection. Checkpoints are saved every 250 iterations in the runs reported here. The validation set has 60 items in four displays (outline, silhouette, outline behind a real grey bar, and a scrambled outline whose half-sides are turned and shifted) for triangles, squares, pentagons, hexagons and irregular quadrilaterals of 48–88 px, off-centre; it is read at step 40. The chosen checkpoint has the best intersection-over-union on the first three displays among those whose fill of the scrambled display (the false fill) is at most 0.25; where no checkpoint meets the bound, the one with the lowest false fill is taken and the network is counted as failing (this happened for V-JEPA 2 L seed 1 with outline clips and for both no-layer-label networks). Selected on validation IoU alone, checkpoints of V-JEPA 2.1 L with IoU 0.73–0.74 scored 0.05–0.19 on the Kanizsa display, which is why the battery, not the training curve, decides.

Support ratio (Fig. 34). At a fixed inducer radius the SAM 3 networks' own label fills the square at step 1 at every support ratio (0.2–0.6 on seed 0, 0.6–0.9 on seed 1) and by step 8 only a trace remains at support ratios of 0.41 and below; at the largest support ratio of this family, 0.58, the fill is 0.35 at step 8 and 0.20 at step 80 on seed 0 and 0.09–0.13 on seed 1; in a second family in which the whole figure is scaled at a support

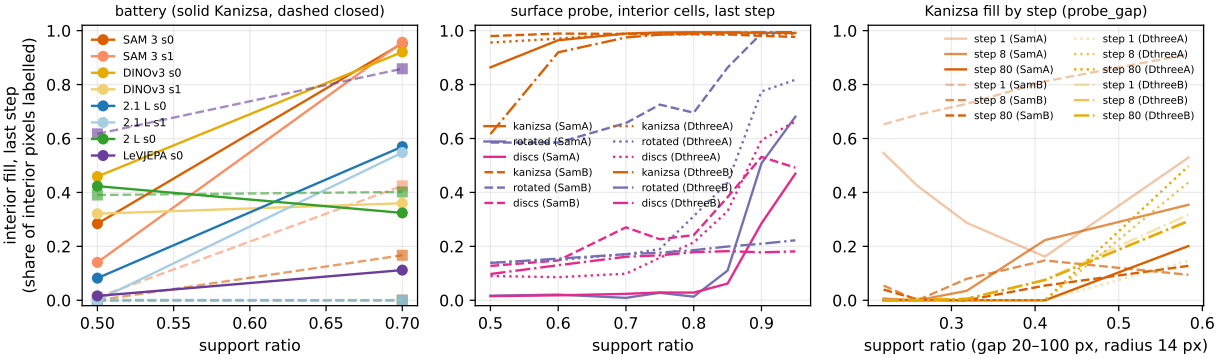


Figure 34: Fill (share of interior pixels labelled, 0–1) against support ratio. Left: the battery's two support ratios, Kanizsa (solid) and closed discs (dashed), per network. Middle: a linear surface probe on the interior cells of the SAM 3 networks' last looped block, fitted on real outlines, for the Kanizsa display, the rotated inducers and full discs, support ratio 0.50–0.95 (regular pentagons and hexagons). Right: the SAM 3 networks' own label on the interior of a Kanizsa square whose inducers keep a fixed radius of 14 px while the gap grows from 20 to 100 px (support ratio 0.58 to 0.22), at steps 1, 8 and 80.

ratio of 0.40 the fill at step 8 is at most 0.33 for any size. The surface probe gives 0.86 at support ratio 0.5, 0.99 from 0.7 (seed 0; seed 1 0.98 and 0.99), stays at or below 0.03 for the rotated inducers up to 0.85 and rises to 0.51 and 0.68 at 0.90 and 0.95 (full discs 0.29 and 0.47); the battery's split (left panel) agrees, with seed 1 part-filling the closed discs at support ratio 0.7 (0.42; seed 0 0.17).

Inducer bodies. From the step-80 maps, each item's square frame is recovered as the orientation whose bounding square of the interior mask is tightest, and the ink is split into an edge band within 6 px of the square's sides and the bodies beyond it; the measure is the mean label on the bodies. Values for every display and network are in Fig. 19; the hybrids, which fill no figure, put part of their label on the inducer bodies of the Kanizsa and rotated displays (0.4–0.6 for the 2.1-front hybrid, 0.2–0.4 for the SAM 3-front one).

Without a cue. The single displays were also run with the cue map set to zero (recurrent_halting/probe_nocue.py), which no training clip ever had. Interior share labelled at step 80, uncued against cued: SAM 3 seed 0 Kanizsa 0.18 against 0.62, closed discs 0.00 against 0.08; seed 1 Kanizsa 0.61 against 0.55, closed 0.20 against 0.22; V-JEPA 2.1 L seed 0 0.32 against 0.33 and seed 1 0.15 against 0.28, closed 0 in both; V-JEPA 2 L seed 0 0.33 against 0.41, closed 0.41 against 0.43; LeVJEPA whole encoder 0.07 against 0.07, closed 0.73 against 0.73; the line square 0.94–1.00 and the rotated inducers 0.00 in every network, cued or not; the outside strip 0.00 and the share of the whole image 0.00–0.08. Uncued, the SAM 3 seed-1 label on the Kanizsa square rises from 0.03 at step 1 to 0.70 at step 4 and settles at 0.61; seed 0 peaks at 0.40 at step 4 and settles at 0.18. DINOv3 under the final recipe: seed 0 Kanizsa 0.52 uncued against 0.69 cued, seed 1 0.32 against 0.34, the line square 1.00 and 0.83 uncued (1.00 and 0.94 cued), rotated inducers and closed discs 0.00 either way; uncued, the seed-0 label grows from 0.25 at step 1 to 0.52 at step 16.

Clutter. Among same-kind clutter (the cued Kanizsa square among stray inducers), the SAM 3 networks keep 0.42–0.45 of the Kanizsa fill and 0.00 for the rotated figure from step 6 on (0.16–0.56 at steps 1–2, vetoed like the single rotated display); V-JEPA 2.1 L keeps 0.15–0.20. In the pair layouts, in which the cued square is shown alone or with a second Kanizsa or rotated figure of a different geometry, the fill on the cued square decays to about 0.1 by step 8 even when it is alone, and the second figure's kind makes no difference; these displays differ in size and position from the single ones, and the decay is unexplained.

Other read-outs. The networks trained on the plain unity clips label a figure under a training-style cover bar as the hidden part of the cued object more for the Kanizsa display (0.22–0.57) than for the rotated or closed displays (0.00–0.14), but a half or whole cover removes the label, and a figure shown and then covered is labelled no more than one that was never shown, so the loop is not used as a memory. Cutting the feedback from step 2, 4 or 8 returns the label to its feed-forward level within one step; permuting the feedback's positions removes most of the selectivity; removing the cue after step 1 changes little. Learning

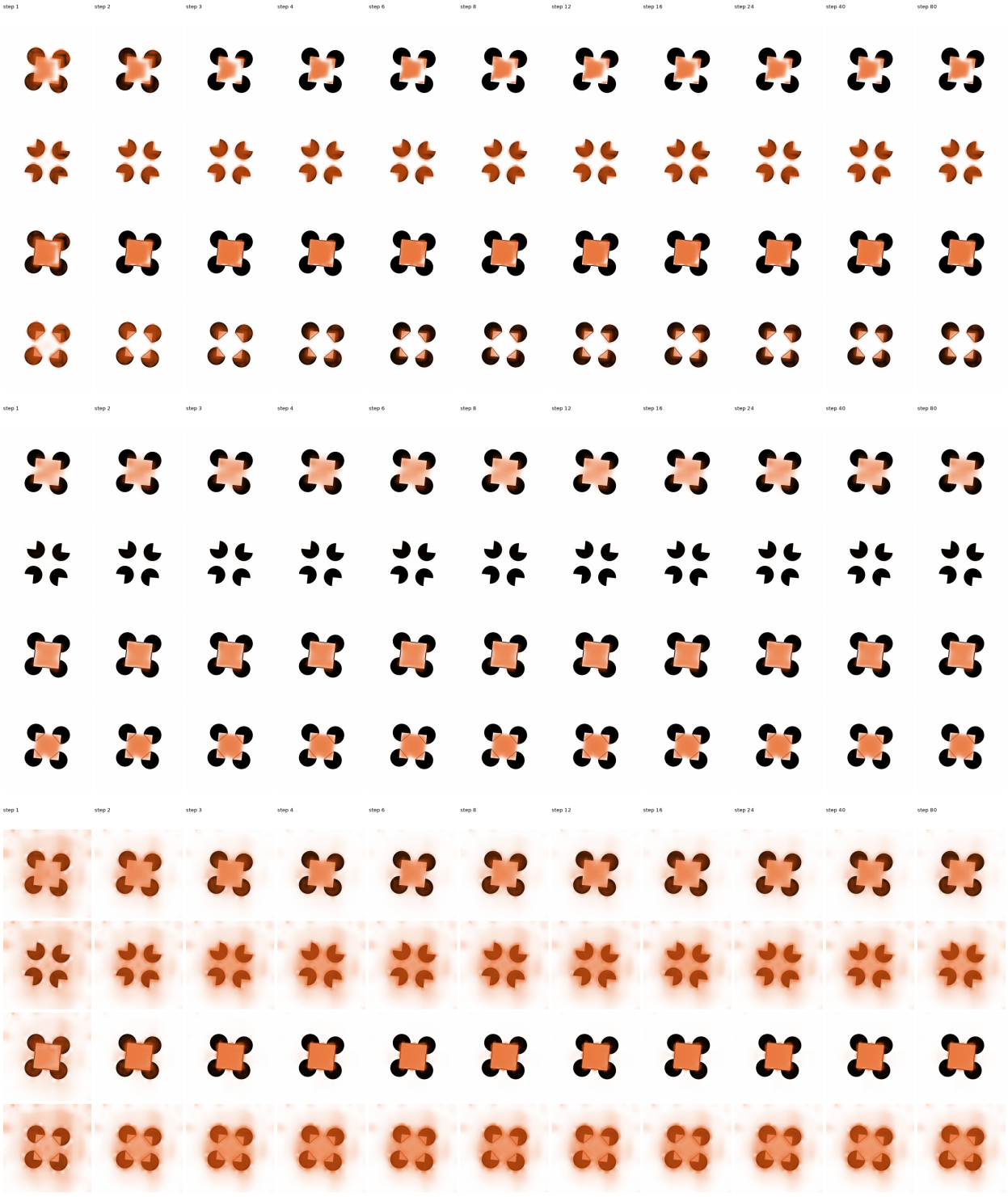


Figure 35: Step-by-step label maps as in Fig. 17 for V-JEPA 2.1 L (top; seed 0), the LeVJEPA whole encoder (middle; seed 0), which labels the closed discs and not the Kanizsa square, and V-JEPA 2.1 L trained without the layer-label loss (bottom; seed 1), whose loop builds a label on every display. Rows: Kanizsa, rotated, line, closed; columns: steps 1 to 80.

the layer labels on the training clips (hidden-part recall and precision 0.80–0.89 for the unity networks) does not predict the Kanizsa battery.

Earlier objectives on DINOv2. With feedback inside a frozen DINOv2 ViT-B/14 (blocks 7–12 looped), we first measured a contour probe on the illusory-edge cells, a far-side deletion test (whether removing one inducer lowers the probe on the far sides, relative to the same deletion with rotated inducers) and a recurrence-off test, with a criterion of three of four seeds. A static layered objective gave a modest Kanizsa-over-rotated contour reading on four of four seeds but failed the recurrence-off test (switching the feedback off from step 1 lowered the Kanizsa and the control alike or not at all); a VGG16 front with a convolutional GRU before the looped blocks passed on one of four seeds; moving-occluder objectives on one of four and zero of four; a slit-reveal objective on three of eight, with the positive seeds losing the effect with further training; one-step training gave zero, confirming that the paired measure is unbiased. One network trained on the plain unity clips built its Kanizsa fill over steps (0.31 at step 1 to 0.82 at step 4), the only such case among the networks of this family, and was not replicated by its sibling seeds. The same recipe on DINOv3 failed in opposite directions after one stage (nothing on the unity clips, everything with the layered objective) and after two stages matched DINOv2 on clutter without filling a plain outline. Oscillator (Kuramoto) and slot-attention circuits warm-started from the fine-tuned DINOv2 looped transformer, and a local-to-global looped attention circuit, reproduced a Kanizsa-over-rotated gap response that matched feed-forward controls trained with one step or without coupling between tokens, and no whole-figure effect that survived a deletion control or replication across seeds.